

HONOURS PROJECT REPORT

An investigation of multi-touch gesture-based gaming

Nina Schiff

Supervised by:

James Gain

	Category	Min	Max	Chosen
1	Software Engineering/System Analysis	0	15	5
2	Theoretical Analysis	0	25	0
3	Experiment Design and Execution	0	20	15
4	System Development and Implementation	0	15	10
5	Results, Findings and Conclusion	10	20	15
6	Aim Formulation and Background Work	10	15	15
7	Quality of Report Writing and Presentation	10		10
8	Adherence to Project Proposal and Quality of Deliverables	10		10
9	Overall General Project Evaluation	0	10	0
Total Marks		80		80

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Abstract

Multi-touch technology allows one to control multiple simultaneous input points through the use of one's fingers. This provides a more natural form of interaction when compared to traditional input mediums, such as a keyboard and mouse. One area that may benefit greatly from this is computer gaming. Games, unlike most other software, are primarily used for their intrinsic enjoyment value. Thus, given the relatively unexplored nature of this area, valuable information may be gained through its exploration.

The effect of multi-touch interaction techniques on game enjoyment was investigated by comparing direct manipulation and gestures, that is the use of symbolic forms in order to relay scope and commands, as well as a combination of the two styles. In addition, the influence of game genre and previous gaming and touch-device experience were also analysed. For the overall project, a system consisting of three games of different genres, as well as a gesture recognition unit, was implemented. This document details the design and implementation of a puzzle game as well as the feature extraction unit necessary to reduce the dimensionality of data for further analysis by the gesture recognition engine.

User experimentation served to assess which interface produced the most flow. Flow, which describes the intrinsic enjoyment present in a task, was assessed using the Flow State Scale. The gestural interface was found to elicit more flow than either of the other two interfaces, while the mixed interface elicited more flow than the direct manipulation version. Genre and previous gaming and touch-device experience were not shown to have a significant effect on interface preference. However, previous gaming experience increased the overall level of flow experienced in all conditions.

The results seem to indicate that for games, the inclusion of gestural controls, where their use is appropriate, when compared to other interaction techniques, may increase the degree of flow, and thus overall game enjoyment, experienced by players.

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Glossary

Affordance Features of the object or interface that suggest how it should be interacted with and what effect this interaction will have.

ANOVA Analysis of Variance. Statistical test used to partition the observed variance into its various sources when considering more than two possible sources (or groups).

Autotelic Experience One of the dimensions used to describe flow. This describes an experience that is intrinsically rewarding and not simply a means to an end.

Between-subjects design The use of distinct and mutually exclusive groups where each group either possesses or is exposed to only one experimental condition.

Bimanual interaction The use of both hands when interacting with a device.

Challenge-Skill Balance One of the dimensions used to describe flow. An appropriate balance between the difficulty presented by a task and the ability or perceived ability of an individual to complete this task.

Clear Goals One of the dimensions used to describe flow. A clarity of intention, aided by goals being set prior to beginning an activity.

Concentration on the Task at Hand One of the dimensions used to describe flow. A state where one has a single-minded focus on the task at hand, while ignoring any potential distractions.

Construct validity The degree to which a test relates to other variables within the same theoretical framework consistently and as expected.

Contrast effect Caveat associated with the use of a repeated measures design. This describes a condition where the response to the second or third condition in an experiment is altered because the two conditions are contrasted with each other.

Counterbalancing A method of controlling for order effects in a repeated measures design by either testing all possible combinations or randomising the order in which experimental conditions are completed.

Curve fitting The construction of a curve or function that best describes or approximates a set of points.

Demand characteristics Any feature of the experiment that may inform participants as to the purpose of the study.

Dimensionality reduction The process of decreasing the number of random variables to be evaluated.

Direct manipulation An interface design philosophy which encourages graphical, manual and incremental interactions over more complex, abstract ones. It is characterised by a perpetual representation of the objects of interest to the user; labelled buttons or physical actions instead of complex syntax; and reversible, incremental operations that provide immediate visual feedback as to their effect.

Direct-touch Directly touching or interacting with the object one wishes to manipulate.

Discriminant validity Measures the degree to which scores on a measure are not related to scores on a second conceptually unrelated measure. This is considered an assessment of construct validity.

Disordinal interaction This describes an interaction that has simple effects that lie in opposite directions. See also: Interaction (statistics).

Double-blind study Experimental design where neither the experimenter nor participants are aware of which experimental conditions the participants have been assigned to.

Ecological validity Refers to how closely the experimental conditions approximate real-life situations and thus the degree to which the findings can be generalised outside of an experimental setting.

EMG Electromyography.

Experimenter bias Also termed experimenter expectancy. Any intentional or unintentional influence the experimenter exerts on participants in order to confirm the hypothesis under investigation.

Face validity Whether on first examination a test appears to be a reasonable measure of some construct.

Fatigue effect Caveat associated with the use of a repeated measures design. This describes a deterioration in performance as the research participant becomes tired, bored or distracted.

Feature A higher-level description of some fundamental unit relevant to computationally solving a specific problem. Often used to describe data in such a way so as to provide dimensionality reduction.

Feature detection The process of locating areas or elements of interest within an image.

Feature extraction A broad topic concerning the process of locating, segmenting and describing features.

Feature segmentation Process associated with partitioning data into a number of segments or features.

Fisher LSD test Fisher Least Significant Difference test. Provides post-hoc testing on the results of an ANOVA by performing pairwise comparisons of the means.

Gesture Symbolic movements used to relay scope and commands. Gestures generally serve two purposes: they are used to draw attention to some point or area and to manipulate or act upon an object.

Gesture recognition The use of mathematical algorithms in order to interpret human gestures.

Greenhouse-Geisser correction Correction used when assumptions of sphericity are violated for repeated measures ANOVAs. In order to produce more accurate significance values, the degrees of freedom of the ANOVA are adjusted. See also: Mauchly's sphericity test, ANOVA, MANOVA, Repeated measures.

Group When three or more blocks of the same colour have at least one face touching, they are considered to be grouped together.

Hawthorne effect See Reactivity.

Immersion A psychological state mediated by the perceived distance between the user and the technological system they are interacting with.

Indirect-touch The use of a pointing device to control and manipulate objects onscreen.

Interaction (statistics) Used to describe a situation where two or more independent variables act on the dependent variable(s) in a non-additive manner. That is the effect of one or more independent variables depends on the level of another.

Internal consistency Refers to the correlations between different items within the same test. This is used to refer to whether all items in the same test or sub-scale measure the same construct.

Interpolation The use of existing data points to determine a function, or similar, which passes through all the given data points. That is, a means of defining a function that takes on specific values at certain points. This function can then be used to define the values of missing points.

Interval measure A rank-order scale where the distance between every element and those adjacent to it is equal.

iOS The mobile operating system developed by Apple for use on the iPhone, iPod Touch, Apple TV and the iPad.

ITC-SOPI Independent Television Commission's Sense of Presence Inventory.

ITQ Immersive Tendencies Questionnaire.

Krippendorff's alpha Measure of inter-rater reliability used to determine the level of agreement between a set of coders on a number of variables.

Least squares Determines the approximate relationship between two or more variables by minimising the sum of squared distances between each data point and the constructed trend line. See also: Ordinary least squares, Total least squares.

Likert scale A rating scale where individuals rate their level of agreement with a statement.

Line of best fit See Trend line.

Linear regression Any approach that models the relationship between an independent variable and one or more dependent variables. See also: Non-linear regression, Least squares.

Loss of Self-Consciousness One of the dimensions used to describe flow. Concern for the self and other everyday issues disappears due to total concentration on the task at hand.

Main effect (statistics) Effect of an independent variable on a dependent variable. If there are other independent variables present in the design, these are averaged across.

MANOVA Multivariate Analysis of Variance. Generalisation of the univariate ANOVA. Used to analyse effects in studies containing more than one dependent variable. See also: ANOVA.

Mauchly's sphericity test Used as a test of the sphericity of data in repeated measures designs. Sphericity refers to the equality of the variances between different levels of the repeated measures independent variable. Sphericity is an assumption of a repeated measures ANOVA and thus deviations may cause invalid results. See also: ANOVA, MANOVA, Repeated measures.

Merging of Action and Awareness One of the dimensions used to describe flow. This refers to a state of deep immersion where an individual feels at one with their actions.

Multi-touch The ability to simultaneously control two or more input points or touches on a touch-device.

Multivariate contrast analysis This is a procedure that provides a linear combination of two or more factor level means, that is, it allows one to collapse these means into one dimension, in order to compare specific elements of an experimental design.

Multivariate regression Regression performed in order to fit a trend line to more than one dependent variable. See also: Total least squares.

Nominal measure A level of measurement where elements are classified into distinct categories which cannot be logically ranked or manipulated mathematically.

Non-linear regression Regression where data is modelled by a non-linear function that depends on more than one independent variable. See also: Linear regression.

Non-parametric regression A method that derives the relationship between two or more variables as well as the model used to describe this relationship from the data available. See also: Parametric regression.

Occlusion This refers to hiding or partial obstructing the view of an object while interacting with it.

OpenGL ES OpenGL for Embedded Systems. This is a subset of OpenGL specifically designed for use with PDAs, mobile phones and other such embedded systems.

Order effect Caveat associated with the use of a repeated measures design. The dependent variable is affected by the order in which conditions are presented.

Ordinal interaction This describes an interaction that has simple effects that lie in the same direction. See also: Interaction (statistics).

Ordinary least squares Least squares procedure where the vertical offsets between each point and the trend line are minimised. Thus the trend line is fitted based on only the dependent variable. See also: Least squares, Total least squares.

Orthographic projection A means of representing a three-dimensional object in two dimensions using projection lines that are orthogonal to the projection plane.

Parametric regression A method that attempts to describe the relationship between two or more variables according to a specified model. This model may take the form of an equation or polynomial.

PQ Presence Questionnaire.

Practice effect Caveat associated with the use of a repeated measures design. This describes a condition where performance on a task improves as a result of repeated practice with the task.

Presence A psychological state where a user believes that the environment presented to him or her through some medium is reality.

Quota sampling The non-probabilistic selection of participants based on specific subject characteristics deemed important to the topic under investigation.

Reactivity Phenomenon that occurs when participants alter their behaviour or performance as a result of the knowledge that they are being observed. Also known as the Hawthorne effect.

Regression Statistical technique used to ascertain the relationship between two or more variables. This is done through the construction of a trend line or line of best fit.

Reliability The degree to which a test generates consistent results both across time and with other equivalent tests.

Repeated measures design Also known as a Within-subjects design. The use of a single group exposed to multiple experimental conditions in order to assess their effect on the dependent variable(s).

Sense of Control One of the dimensions used to describe flow. A feeling of total mastery over one's environment and actions.

Shape This refers to the larger arrangement of all blocks visible on the screen.

Single-blind study Experimental design where participants are unaware of which experimental condition they have been assigned to. However, the experimenter has full knowledge of this information.

Smoothing In the context of curve fitting, smoothing describes the process of determining a function that approximates the most important points or features of the data.

Splines Used to define curves, or more complex shapes, in a piecewise manner through the use of polynomials of low degree.

Surface normal Vector perpendicular to the flat surface.

SUS Slater, Usch and Steed questionnaire. Used to assess presence.

Total least squares Also known as geometric least squares. Least squares procedure where the perpendicular offsets between each point and the trend line are minimised. Thus the trend line is fitted based on both the dependent and independent variables. See also: Least squares, Ordinary least squares, Multivariate regression.

Transformation of Time One of the dimensions used to describe flow. This describes a perceived distortion in the normal passage of time.

Trend line Line or other function used to describe the relationship between two or more variables. Constructed through the use of regression.

Tukey's HSD test Tukey's Honestly Significant Difference Test. Used to determine which means are significantly different from each other through the use of multiple pairwise comparisons.

Unambiguous Feedback One of the dimensions used to describe flow. This refers to feedback that provides a clear awareness of how well one is progressing towards one's goals.

Unimanual interaction The use of one hand only.

Validity The degree to which a test measures what it purports to measure across a range of different situations.

Vertex normal Normalised average of the surface normals of all faces that contain the vertex.

Wilks' lambda distribution Multivariate generalisation of the F -distribution. Used by the MANOVA procedure in order to determine significance values. See also: MANOVA.

Within-subjects design See Repeated measures.

Chapter 1

Introduction

Computing devices are becoming more and more prevalent in everyday life. Yet the way in which we interact with these devices is often given little thought. With the recent surge in accessibility, in terms of both cost and diversity, of multi-touch computing devices [Moscovich 2007], one can begin to consider alternative input techniques to the more traditional mouse and keyboard. Multi-touch refers to the ability to control two or more points of input simultaneously through the use of one's fingers [Jaimes and Sebe 2007]. This makes an understanding of the appropriateness of their use paramount, as these devices represent a possible shift in the way commands and information are relayed to computing devices.

However, even when considering such emerging technology, the direct manipulation paradigm, that is, a design philosophy which discourages complex, abstract interaction in favour of manual, graphical and incremental techniques [Shneiderman 1982], is ubiquitous to the degree that few alternatives are even considered. While direct manipulation is regarded as providing the most appropriate means of interaction for many systems utilising a mouse and keyboard for input, the same may not be true for touch-devices.

With this said, it must be noted that an interaction technique should, in most cases, be matched to its application. That is, broad statements regarding which technique is most appropriate for all applications often ignore the fundamental differences present in various pieces of software [Frohlich 1993]. An interesting example of this is computer games. Unlike much conventional software, games are often used for their intrinsic enjoyment value alone. That is, there is often no added incentive, apart from enjoyment, in their use [Sweetser and Wyeth 2005]. Thus the choice of input technique should not detract from this, and if possible, should add to the potential amusement derived from the game.

Gestures, that is, symbolic movements or drawn forms used to relay scope and commands, provide an alternative to traditional direct manipulation techniques [Rubine 1991]. While their use has been explored in a number of domains, such as Virtual Environments and Human-Computer Interaction, very little research has been performed to assess their impact on gaming [Kratz et al. 2007]. In addition, the majority of this research relates to three-dimensional gestures detected using Computer Vision or accelerometer-based techniques [Kang et al. 2004]. Furthermore, it is often easier to entice users to use a novel or unfamiliar interaction technique within a gaming context. This is due to the fact that games are often disassociated, to some degree, from more traditional applications due to the enjoyment present in them and users are thus less hesitant to make use of novel techniques [Starner et al. 2000].

This becomes especially relevant when considering the relatively unexplored field of direct manipulation alternatives for touch-based computing [Kin et al. 2009]. Furthermore, as multi-touch technology has been shown to be more intuitive and naturalistic [Forlines et al. 2007] and experiments on the use of spatial gestures in gaming have suggested the same [Payne et al. 2006], the combination of gesture-based interaction with multi-touch technology may greatly improve the overall gaming experience.

The combination of gestures, as an interaction technique, with multi-touch devices, as an input medium, allows us to investigate the effects of two relatively unconventional elements within the context of computer gaming. Therefore, the findings of this research will provide information on how game enjoyment is affected by the use of interaction technique. This thus provides novel findings within both the fields of computer gaming and multi-touch

gestural interaction.

1.1 Research question

The research question that will be investigated within this document is whether multi-touch gestures improve gaming experience when compared to other interaction techniques. In order to test this hypothesis, gestural interaction will be contrasted with direct manipulation. In addition, an interface that uses a combination of the two will also be created so that their combined interaction can be evaluated. The creation of games is therefore necessary. This provides an opportunity to assess whether game genre affects the appropriateness of gesture-based gaming. In addition, the effects of previous gaming and touch-device experience will also be investigated in relation to the enjoyment experienced through multi-touch gesture-based gaming.

Thus, the research question and sub-questions under investigation are:

- The effect of interaction technique on the enjoyment experienced during multi-touch gaming.
 - Whether game genre affects the appropriateness of interaction techniques within multi-touch gaming.
 - The degree to which previous gaming or touch-device experience influences the enjoyment experienced through the use of different interaction techniques within multi-touch gaming.

In order to assess which interaction technique provides the most enjoyable gaming experience, a measure of flow will be used. Flow is defined as an emotional state, where an individual is immersed within some task that produces feelings of strong gratification [Ermi and Mayra 2005]. This emotional state is so powerful that an individual is often “willing to do it for its own sake with little concern for what they will get out of it” [Csikzentmihalyi 1990]. This thus provides an ideal measure for gaming experience, as games are played for their intrinsic enjoyment value.

The evaluation of each game will therefore require user experiments. In order to investigate the effects of genre, three games of different genres (Puzzle, Strategy and Fighting/Duelling) will be developed. In addition, participants will be recruited so that the subject characteristics of gaming and touch-device experience can also be evaluated. Each participant will play a game, from one genre, three times. The interaction technique in each of these sessions will be varied so that the levels of flow produced by each interface can be determined. Flow will be assessed through the use of the *Flow State Scale* developed by Jackson et al. [2001]. This consists of a 36 question survey administered after the task being evaluated has been completed.

The key success factors will thus be an understanding of the effects of interaction technique on the level of enjoyment experienced during gaming. The additional data concerning game genre and subject characteristics will also mediate the interpretation of these results. These results will be obtained through the measurement of flow and subsequent statistical testing.

1.2 System overview

In order to test the above research questions, a number of components need to be developed. These, and the work allocation amongst them, are summarised in Figure 1.1. The developed system will thus contain two distinct but related components. The first of these will be the gesture engine. This includes a feature extraction unit which will be used to transform raw touch input into a simpler form for use in the gesture recognition. The actual recognition of gestures will be performed by the gesture recognition engine. Two gesture recognition units will be developed in parallel, one making use of a Hidden Markov Model for recognition, with the second utilising simpler techniques.

The gesture engine will be integrated with three games. Each game developed will be of a different genre (Puzzle, Strategy or Fighting/Duelling). The games will be developed with a limited scope, however, they should still prove sufficient in eliciting a flow state. Each game will contain three interface versions. One of these interfaces will contain direct manipulation controls, another will be purely gesture-based with the third containing a combination of the previous two. Finally, experiments need to be designed in order to evaluate the research hypotheses.

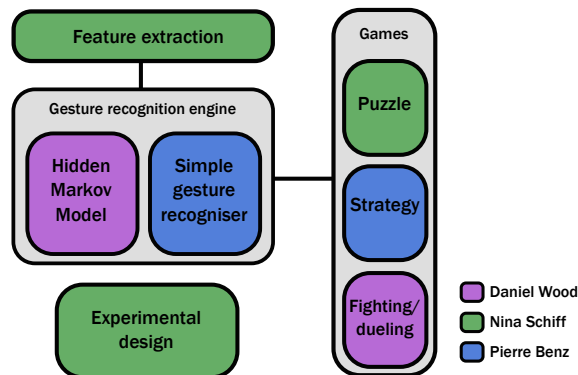


Figure 1.1: Overview of system components developed or designed in order to assess the effect of interaction style on game enjoyment. These components will also be used in order to investigate whether game genre and previous gaming or touch-device experience impact the way in which interaction styles are perceived. The work allocation for the project is illustrated through the use of colour.

The system will be developed for the Apple iPad. This device was chosen due to its multi-touch capabilities and relatively large screen size when compared to other similarly priced commercial products. The choice of device necessitates the environment within which the system will be created. Thus, iOS will serve as the platform for development and the Objective C language will be used to create both the gesture engine and games. OpenGL ES will be used for any three-dimensional graphical requirements, while interface elements will be handled by the iOS SDK [Apple 2010a].

The feature extraction and Puzzle game as well as the experimental design are described in more detail within this document.

1.3 Ethical considerations

As user testing will be conducted, in order to assess the research questions, Ethics Clearance was obtained from the Science Faculty Ethics Clearance for all planned user experiments. This ensures that participants will not be adversely affected by the experimental procedure.

With regard to legal and copyright issues surrounding use of libraries and other components, a number of factors must be considered. Development for the Apple iPad requires a Developer Licence. As an education licence was obtained, submitting one's code to the Apple App Store is disallowed. In addition, the use of external libraries or software, not approved by Apple, is also not permitted. For this reason, development was done using components provided within the iOS SDK.

The *Flow State Scale*, used to assess flow, is also copyrighted. As the purchase of a licence, linked to the number of participants it would be administered to, was required, the experimental design cannot make use of more than 50 participants. In addition, due to these copyright restrictions, only a limited portion of the scale was able to be reproduced as an appendix within this document.

Due to the use of restricted materials and frameworks, care was taken to not violate the above provisions. In addition, ethical considerations were carefully considered in the design of the user experiments to ensure that there was no undue harm to participants.

1.4 Outline

This document outlines the development and testing of the feature extraction unit and a puzzle game. It also contains the sections necessary for evaluating the research hypotheses. Chapter 2 provides background information on multi-touch interaction, gestures, the evaluation of games as well as the limited existing research on gesture-

based gaming. The design, implementation and evaluation of the puzzle game, *Hexahedron*, is contained in Chapter 3. Chapter 4 details the design and implementation of three feature extraction techniques. It also contains an evaluation of these methods. Background information on feature extraction has also been provided in this chapter. The design of the user experiments is provided in Chapter 5, while the results and discussion of the data obtained from these experiments is included in Chapter 6. Finally, Chapter 7 contains the overall findings and conclusions of this work as well as a discussion of the limitations of the present study and suggestions for future work.

Chapter 2

Previous work

2.1 Introduction

The topic of gesture-based gaming on a touch-device spans a number of distinct research areas. This chapter briefly covers the most relevant of these fields. This includes multi-touch interaction where the various ways of controlling a touch-device are discussed. The use of gestures, in contrast to other more conventional means of input, is also considered. As these techniques will be applied to gaming, a discussion of the ways in which gaming experience can be evaluated is also included. Finally, the limited research on gesture-based gaming is reviewed.

2.2 Multi-touch

2.2.1 Introduction

The way in which we interact with computers has changed very little over the last few decades. That is, the keyboard and mouse remain the primary means through which individuals interact with graphical user interfaces [Kin et al. 2009]. However, these means of interaction greatly limit the number of simultaneous input points and consequently the types of interfaces that can be developed [Moscovich and Hughes 2008]. In addition, these devices distance the user by limiting the degree and magnitude with which they can physically interact with the system [Moscovich 2006]. Multi-touch technology, however, offers an alternative to the traditional computing paradigm. Multi-touch, in relation to touch-screen devices, refers to the ability to simultaneously control two or more input points or touches [Jaimes and Sebe 2007]. These systems allow users to physically interact with the computing device with the option to use two or more input points. This in turn, increases the degrees of freedom that the user is able to manipulate, which has implications for the type of applications that are possible and the way in which interaction itself is performed [Moscovich and Hughes 2008].

2.2.2 Types of interaction

At present, there is no standard hardware with which multi-touch interfaces are developed. Some devices utilise capacitive technology, while others use cameras and Computer Vision techniques in order to provide touch capabilities [Wigdor et al. 2009]. This results in there being a number of variations in the input techniques in use. These techniques include direct-touch versus indirect-touch, single-hand versus bimanual, as well as the use of differing numbers of fingers [Kin et al. 2009]. There are also techniques where the shape and orientation of the user's whole hand is tracked and identified [Wu and Balakrishnan 2003]. These mechanisms will be discussed in further detail below.

Direct-touch versus indirect-touch

Indirect-touch refers to the use of a pointing device to control and manipulate objects onscreen [Moscovich and Hughes 2008]. Users are required to visually track this pointer or cursor. Direct touch, in contrast, refers to directly touching or interacting with the object one wishes to manipulate [Kin et al. 2009]. Thus one tracks the finger or fingers interacting with the object rather than an onscreen cursor [Moscovich and Hughes 2006].

Perhaps the most well known example of indirect-touch is a mouse. However, research has also been conducted on using multiple fingers in order to control a number of control points in an indirect manner [Moscovich and Hughes 2006, 2008]. In this scenario one manipulates multiple onscreen control points through the use of more than one finger. However, this manipulation is done on a device connected to a separate output screen rather than on the display screen itself [Moscovich and Hughes 2006].

Direct-touch has been shown to be faster than using single-point, indirect-touch methods, such as a mouse [Kin et al. 2009]. However, no comparison for speed has yet been performed between direct-touch and indirect, multi-touch methods. Direct-touch, however, suffers from issues of occlusion. That is, when interacting with an object, one often hides or blocks portions of the object [Kin et al. 2009]. This is of concern for interfaces with small buttons and other such elements. In addition, due to the size of a finger relative to that of a mouse cursor, both direct and indirect touch-based methods suffer from decreased precision of selection [Kin et al. 2009]. However, as mentioned above, the nature of touch-based interaction allows for a greater number of control points, and thus degrees of freedom within an interaction, when compared to conventional mouse-based pointing devices.

Multi-finger techniques

Many touch-devices support multiple points of input. This increases the degrees of freedom that a user has when manipulating or controlling an object or interface [Moscovich and Hughes 2008].

However, research has found that using multiple fingers on the same hand often produces a minimal increase in the degrees of freedom the user is able to control [Moscovich and Hughes 2008]. This is due to the fact that fingers on the same hand are linked physically and neurologically [Moscovich 2006]. Thus the movements produced by multiple fingers are often highly similar. Therefore, while multiple fingers on the same hand may increase input bandwidth on tasks that require coordinated control, such as using two fingers to scale an object, they do not actually offer a large increase in the degrees of freedom available to a user [Kin et al. 2009]. They may, in fact, hinder a task as they increase the chance of simultaneously selecting multiple targets. Finally, research has suggested that users often do not attempt to use more than two fingers simultaneously unless prompted to do so [Kin et al. 2009]. This suggests that while many devices offer recognition of multiple simultaneous input points, this multiplicity is often not used or required in unimanual applications.

Unimanual versus bimanual interaction

Devices that are able to detect multiple simultaneous inputs also present the opportunity for interaction with more than one hand. This too can increase the degrees of freedom available to a user [Kin et al. 2009].

Bimanual interaction does not suffer from the same limitations as unimanual, multi-finger interaction due to the decoupled nature of the input points [Moscovich and Hughes 2008]. That is, as the multiple input points come from two hands, their motion is far more separable. However, this decoupling of movement also means that it is more difficult to perform coordinated tasks using two hands instead of one [Moscovich and Hughes 2008]. Bimanual interactions have also been shown to have a higher miss rate than unimanual conditions [Kin et al. 2009]. Moscovich and Hughes [2008] showed that bimanual interaction is preferable when performing separable tasks but that unimanual (multi-finger) interaction is preferred when attempting to perform more precise, coordinated tasks. A further constraint on bimanual manipulation is that, without a clear correspondence between each hand and the control points, users often become confused. This may be due to the increased cognitive load of having to control and coordinate both hands simultaneously [Kin et al. 2009]. With this said, however, bimanual interaction shows a speed advantage over unimanual interaction in decoupled tasks that do not require coordinated movement [Kin et al. 2009].

The differences in the degrees of freedom, accuracy and control presented by each method suggests that these variants are not interchangeable. Thus it appears that while both unimanual and bimanual interaction have

their benefits, the type of application and level of control required dictates which of the two methods is more suitable.

2.2.3 Benefits

As is apparent, there are a number of ways in which touch-devices can be utilised. Furthermore, there does not appear to be one universally ideal technique. Instead it seems that the type of interaction style that should be used is highly application dependant. With this said, multi-touch technology does offer a number of distinct benefits over traditional interaction means such as the mouse and keyboard.

One such advantage is that, in many cases, applications that utilise touch-devices are often easier for uninitiated users to learn. As shown by Moscovich [2006], all that is usually needed is the instruction to touch the device. This in turn makes applications that use touch interaction far more accessible. This may be because multi-touch interfaces have been shown to be more naturalistic, which perhaps aids learning of a novel system [Forlines et al. 2007].

With regard to speed, it has been shown that the fastest multi-touch interaction (bimanual, direct-touch) is roughly twice as fast as using a mouse pointer. Furthermore, the use of multiple fingers or hands enables task parallelism and thus increases task performance [Kin et al. 2009]. The increase in the number of input points and degrees of freedom also means that input points can be used simultaneously to control more than just current position. One example is the use of multi-touch to combine translation and rotation of an object into one operation by utilising multiple fingers [Moscovich 2006].

Perhaps the most important benefit of multi-touch devices, and the interaction techniques they make possible, is that they allow for a whole range of novel applications and ways in which one can interact with a computing device.

2.2.4 Limitations

As with any interaction technique, multi-touch has certain limitations when compared to other input mechanisms. These limitations are primarily caused by the nature of touch computing, that is, users use their fingers, rather than a more precise pointer (such as a mouse cursor), to interact with objects on screen [Moscovich 2006].

Direct-touch suffers from issues of occlusion not present in indirect methods [Wu and Balakrishnan 2003]. This has implications for the design of applications that utilise touch technology. While some work has been done on mediating these issues by making use of techniques such as using a second finger to manipulate the scale or coordinate system of the display, occlusion remains a problem within touch interface design [Moscovich and Hughes 2008].

Another issue related to the size of a finger when compared to other pointing devices, is that of accuracy. Touch-based interaction has been shown to have higher miss rates when compared to mouse-based pointers [Moscovich and Hughes 2008]. In addition, the precision of movements available is greatly reduced when compared to a mouse or pen device. This reduced accuracy is also not fully eliminated by the use of indirect-touch methods [Moscovich and Hughes 2006]. Furthermore, due to the nature of interaction, there are limitations on the proximity and size of the display [Moscovich and Hughes 2008]. The display also fails to provide any form of haptic feedback. For example, touch typing would be considerably more difficult, as one is unable to feel the relation of different keys to each other. However, this is often also lacking in other interaction techniques [Wobbrock et al. 2009].

Finally, the use of more than one pointing device can also, in some cases, cause the user to become disoriented and confused due to the increased cognitive load [Moscovich and Hughes 2008]. The same has been shown to be true for multi-finger interaction when this movement is not coordinated [Kin et al. 2009].

2.2.5 Conclusion

Multi-touch technology presents the opportunity for creating applications that utilise two or more control points. This has been shown to increase task parallelism and selection speed as well as changing the way in which people are able to manipulate and control digital objects and interfaces. However, despite these benefits, touch technology

also introduces issues of occlusion and inaccuracy. With regard to cognitive performance, while touch technology has been shown to be highly intuitive to the uninitiated, problems of increased cognitive load are introduced when using multiple fingers or hands. Therefore, it appears that for multi-touch technology to be truly effective, the application itself must suit this style of interaction. Thus touch interaction should not be viewed as a panacea. Instead one should see it as a technique that, within the right framework, can greatly improve the quality of interaction of an application.

2.3 Gestures

2.3.1 Introduction

Hand gestures are an integral additional component when considering interpersonal communication. They allow for further nuances and meaning to be added to conventional communication [Sato et al. 2001]. In addition, they are often able to provide a multitude of symbolic meaning in a very simple and brief manner. Due to their permeation of traditional communication, they are also an intuitive means of providing commands as their purpose and nature is familiar [Kettebekov and Sharma 2000]. With the advent of Computer Vision and touch-based systems, along with the increased penetration of computing devices in everyday life, gestures have begun to be explored as a means for interacting with computers [Rubine 1991]. The following review will discuss hand gestures in this context by exploring their benefits and limitations.

2.3.2 Gestures versus direct manipulation

The definition of gestures is often highly dependant on their application. However, an overarching theme is that they are symbolic movements used to relay scope and commands [Rubine 1991]. According to Huang and Pavlovic [1995], “Every gesture is the physical expression of a mental concept”. In the context of Human-Computer Interaction, gestures generally serve two purposes: they are used to draw attention to some point or area and to manipulate or act upon an object. These two functions are often linked within the same gesture, highlighting the context-sensitive nature of most gestural commands [Kettebekov and Sharma 2000]. Thus a gesture is composed of the physical context in which it is created along with its drawn gestural form.

More importantly, perhaps, is the distinction between gestures and direct manipulation techniques. Direct manipulation refers to an interface design philosophy which encourages graphical, manual and incremental interactions over more complex, abstract ones. The main premise of this is that it reduces the cognitive demands placed on users by the interface [Frohlich 1993]. Shneiderman [1982] provided three properties which can be used to characterise a direct manipulation system. These are a perpetual representation of the objects of interest to the user; labelled buttons or physical actions (for example, drag and drop) instead of complex syntax and “rapid, incremental reversible operations whose impact on the object of interest is immediately visible” [Frohlich 1993]. This can be further explained as an attempt to reduce the perceived distance between the computerised representation of a subject and the way in which a user might normally think about it. This is often accomplished through the use of real-world metaphors which the user is able to manipulate graphically rather than through a series of complex commands [Hutchins et al. 1985]. Thus direct manipulation interfaces can be seen to reduce the cognitive load required of users. However, unlike gestural interaction, these interfaces do not provide the ability to rapidly execute complex and abstract commands due to their incremental nature [Frohlich 1993].

2.3.3 Benefits and limitations of gestural interaction

When considering novel interaction techniques, the benefits and limitations of such interactions should be carefully studied. This is of added importance given the highly abstract nature of gestures.

As discussed above, gestures are often used to encapsulate symbolic meaning. This implies that gestures can be used as a form of shortcut, where multiple commands are specified through the use of a single gesture [Nielsen et al. 2004]. This, in turn, increases communication bandwidth as one is able to complete a number of commands far more quickly. In addition, the context-sensitive nature of gestures means that they can be used to create alternative communication channels within different situations [Jaimes and Sebe 2007].

The main concern regarding the use of gestures is that the purpose or form of gestures may not always be immediately apparent. That is, users may not be easily able to identify what task a gesture performs in relation to its form [Wobbrock et al. 2009]. This is further complicated by the context-sensitivity of gestures. That is, the same gesture may be used for multiple scenarios with its action being determined by the context within which it is performed [Wobbrock et al. 2009]. The reuse of gestures is directly related to a second concern regarding their integration. That is, the amount of learning required of the user. It may not be possible to provide the same affordances, that is, features of the object or interface that suggest how it should be interacted with, in gestural interfaces as are available in direct manipulation designs [Nielsen et al. 2004]. This implies that users will be required to learn and remember a fair number of gestural commands. This, in turn, increases the cognitive demands placed upon the user and may ultimately decrease the usability of the system [Nielsen et al. 2004].

Thus, while gestures have the potential to provide a means for executing a number of commands rapidly and with a high level of abstraction, care must be taken when selecting gestures for use. Gestures should be easily memorable and provide as many affordances as possible in order to lessen the cognitive requirements placed on the user.

2.3.4 Guidelines for gesture selection

In light of the issues identified above, a number of guidelines have been identified by Nielsen et al. [2004] in order to attempt to make gestural interfaces more usable. These are briefly discussed below.

The first suggestion is that gestures should be memorable and easy to perform [Nielsen et al. 2004]. Related to this, gestures should also be ergonomic, so that unnecessary physical stress is not placed on the user [Wobbrock et al. 2009]. Gestures should also be intuitive and should be used to represent logical metaphorical and iconic links with their proposed functionality. Finally, the system should be able to unambiguously identify all gestures in use [Jaimes and Sebe 2007].

These guidelines primarily serve to decrease the learning and memory demands placed upon users of such interfaces. This in turn, should make gestural interfaces more usable and thus more useful.

2.3.5 Conclusion

Gestural interfaces provide an alternative to the well-established direct manipulation form. However, as is the case for any interface, gestures are not appropriate for all applications. Instead, one needs to consider the specific requirements of the application under development. Thus, while they should not be considered as a replacement for direct manipulation, there are cases where the inclusion of gestural controls can indeed make the manipulation of interface elements far more effective.

2.4 Game evaluation techniques

2.4.1 Introduction

The computer game industry has been rapidly growing over the last few decades with revenues exceeding those made in the film industry in many areas [Mandryk et al. 2006]. Despite this, the factors necessary for game success are not well understood. In addition, games are fundamentally different to many other forms of software as they serve the purpose of entertainment rather than aiding the user in attaining some goal [Sweetser and Wyeth 2005]. That is, when evaluating games one has to keep in mind issues, such as enjoyment and fun, which are not normally present in other types of software [Mandryk et al. 2006].

This introduces a number of other topics of investigation apart from just standard usability. It also highlights the need to draw on findings and research from other fields such as virtual reality and psychology [Sweetser and Wyeth 2005]. In particular, a focus on topics that allow one to assess enjoyment and emotional effect is needed. These topics include the fields of immersion, presence and flow which have previously been strongly linked to investigations of virtual environments and their related psychological correlates. This is not to say that traditional usability measures should be ignored, but rather that they should serve as a secondary, rather than primary, means

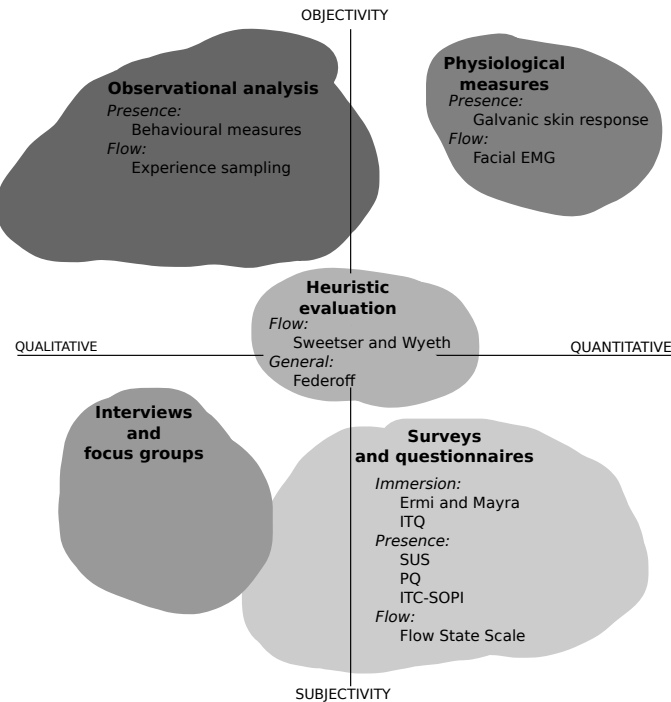


Figure 2.1: A summary of current research methods for games plotted according to how objective/subjective their data collection is as well as whether the data returned is qualitative or quantitative.

for evaluating games [Hazlett 2006]. However, with this said, it should be noted that conventional usability criteria need to be adapted to suit the gaming environment itself [Federoff 2002].

The current research in this field outlines a number of methods for collecting data on the playability and enjoyment produced by games. This includes surveys, questionnaires and interviews, heuristic methods, observational analysis as well as physiological means of evaluating player response. These methods range in both their quantitative and qualitative nature as well as how subjective or objective their means of data collection are [Mandryk et al. 2006]. Figure 2.1 summarises the various available methods according to these criteria [Mandryk et al. 2006].

This paper will provide a summary and comparison of these potential methodologies and how they can be applied to evaluating concepts such as immersion, presence and flow. In addition, ways of evaluating general usability issues will be considered. However, given that the single most important feature of any game is arguably the degree to which it provides enjoyment and entertainment for its players, this paper will focus more on how to measure and assess issues affecting this rather than on pure usability in the conventional sense.

2.4.2 Computer games

In order to better analyse the factors that lead to a game's success, one must first understand what differentiates games from other software. This is necessary as the metrics used to assess computer games must take into account these differences. The traditional concept of usability is defined as being composed of three independent measures: effectiveness, efficiency and satisfaction [Korhonen and Koivisto 2006]. However, within a game environment the first two measures appear to be far less important than the third [Federoff 2002].

This is perhaps because the most apparent difference is that games are generally designed purely to entertain users rather than assist them in completing some goal [Pinelle et al. 2008]. In addition, individuals learn to play games without the use of manuals as well as developing their own techniques and strategies in order to increase their performance [Federoff 2002]. Furthermore, whether or not a game succeeds at eliciting positive emotions is

less dependant on the outcome and more on the actual process of playing [Mandryk et al. 2006]. That is, unlike in other applications, players are unaware of what will happen next and often actively construct their potential experience [Korhonen and Koivisto 2006]. This is an important point as it means that much of the analysis should focus on how and whether a game elicits enjoyment and fun rather than just focusing on usability [Hazlett 2006]. This also raises the question of how the process of game-play can be adequately assessed.

Finally, games do not provoke only positive emotions within players [Hazlett 2006]. Rather, many games, through the use of challenges, appear to first develop negative emotions, such as frustration or forms of stress, which are then followed by positive emotions when the challenge is overcome [Ermi and Mayra 2005]. Thus the affective aspect of games should also be considered as this may form a large part of what makes certain games more successful than others.

Overall, games differ significantly from non-leisure software applications as their primary goal is to entertain. This means that many existing techniques cannot be used in their current forms in order to assess computer games. Rather, an approach that attempts to integrate affective components with those of usability is needed to better gauge how and why users respond in particular ways to various games.

2.4.3 Immersion

One aspect that is thought to influence game enjoyment is immersion. This is defined by McMahan et al. [2006] as “the objective level of fidelity of the sensory stimuli produced by a technological system”. This can be further described as a psychological state mediated by an interface of sorts wherein the subject feels included in and constantly interacting with an environment through the means of a stream of experiences and stimuli [Witmer and Singer 1998].

The degree to which a user becomes immersed appears to lie along a continuum. That is, there are a number of preliminary stages a user proceeds through before reaching total immersion [Brown and Cairns 2004]. These initial states have been termed different things in the literature, from absorption [Ermi and Mayra 2005] to engagement [Brown and Cairns 2004] to involvement [Witmer and Singer 1998]. However, all these terms appear to describe a state where attentional resources are directed towards an experience. This is distinct from full immersion as when a user becomes immersed they feel as if they are completely involved in a separate reality and thus direct all their attention and perceptual tools towards this environment [Ermi and Mayra 2005]. There are thus parallels between full immersion and presence [Brown and Cairns 2004]. However, it should be noted that full immersion is often very difficult to achieve [Witmer and Singer 1998].

This may be due to the fact that, in a sense, immersion requires the user to forget or overlook the fact that they are interacting with an environment through the means of some medium [Federoff 2002]. This is often not possible in settings other than virtual reality. However, being able to gauge and measure immersion is of particular importance as immersion can be considered a necessary state in order for presence to be experienced [Brown and Cairns 2004].

Measuring immersion

Assessing the degree of immersion has proved to be a difficult problem to solve [McMahan et al. 2006]. While there is much theoretical discussion of what immersion entails, there are very few methods of quantifying the level of immersion experienced by participants.

Ermi and Mayra [2005] identified three aspects of immersion within the game-play experience. These are *Sensory immersion*, which relates to the audiovisual environment of the game; *Challenge-based immersion*, which refers to the match between a user’s abilities and the challenge presented by the game, and *Imaginative immersion*, which consists of the narrative dimension of the game experience. Using these three dimensions, a questionnaire was constructed to allow immersion to be assessed. However, while they did test their questionnaire on a number of different games, no statistical tests of internal consistency, validity or reliability were performed. Thus the validity of their findings and the usefulness of their survey remains questionable.

Another test of immersion, developed by Witmer and Singer [1998], measures the propensity an individual has to become immersed in an environment. The Immersive Tendencies Questionnaire [ITQ] uses three dimensions in order to do this. These are *Involvement*, which assesses an individual’s inclination to become passively immersed

in some medium; *Focus*, which refers to the degree a participant is able to concentrate on one task while blocking out other distractions, and *Games*, which considers how often the individual plays games and how involved the individual becomes in the games when doing so [Witmer and Singer 1998]. The test has been thoroughly evaluated to show it has a high internal consistency ($\alpha = 0.81$) as well as displaying reliability and validity (within each sub-scale) [Witmer and Singer 1998].

While this questionnaire does not directly measure the amount of immersion experienced in a game environment, it is meant to provide an indication of how likely an individual is to experience presence [Witmer and Singer 1998]. However, this correlation was not shown to be present in all tests conducted by the authors, despite an overall significant positive correlation ($r = 0.24, p < 0.01$) [Witmer and Singer 1998].

Overall it appears that there are no concrete methods of evaluating the immersive effect of a game. This is potentially due to the very nature of immersion. However, as full immersion has been equated with presence, it may be more appropriate to attempt to quantify this construct instead [Brown and Cairns 2004].

2.4.4 Presence

Presence is defined as the “perceptual illusion of nonmediation” [Ravaja et al. 2004]. That is, a state where a user believes the environment presented through some medium is real [Ivory and Kalyanaraman 2007]. Presence is of particular importance within a game environment as high levels are thought to increase the amount of enjoyment and fun experienced by players [Ravaja et al. 2004]. Some research also suggests that presence has a positive influence on the ability to complete tasks [Ivory and Kalyanaraman 2007]. While immersion and presence may appear to touch on a number of similar elements, they are in fact distinct concepts. Immersion is specifically concerned with the physical interface and its effects while presence refers to the psychological state produced by immersion and a myriad of other factors [Brown and Cairns 2004].

A number of different things that are thought to increase the possibility of presence have been identified in the literature. These can loosely be grouped into player and media characteristics [Lessiter et al. 2001]. Media characteristics refers to elements such as the narrative and overall story as well as the audiovisual and sensory characteristics of the game. For example, highly realistic and naturalistic displays and interaction techniques have been shown to positively contribute to the experience of presence [Lessiter et al. 2001].

Player characteristics include elements such as an individual’s perceptual, motor and cognitive abilities as well as transient factors such as mood [Lessiter et al. 2001]. Furthermore, some research suggests that focus or selective attention, which refers to an individual directing their attention towards a particular stimuli while excluding others, can play an important role in eliciting and explaining presence [Witmer and Singer 1998]. That is, with an increase in game difficulty, players are forced to expend more cognitive resources on interacting with the game and, by proxy, less attention is given to factors suggesting the game environment is not real [Ravaja et al. 2004]. However, while this suggests that some form of attentional resource threshold exists in order for presence to be experienced, it is likely to be only one of many contributing factors [Witmer and Singer 1998].

Thus one needs to consider a variety of factors when assessing presence, both in terms of the simulated environment and in relation to the users themselves.

Measuring presence

A number of methodologies have been used when attempting to measure presence. Due to the difficulty in both eliciting and measuring presence in situ, no well-accepted method has yet been developed [Mandryk et al. 2006]. The following section will review the different techniques proposed in order to quantify presence.

A number of introspective techniques have been developed in order to assess presence. Perhaps the most widely used is the Slater, Usoh and Steed [SUS] questionnaire [Blake 2010]. This questionnaire, administered after experiencing the game environment, consists of six items that require the participant to assess their level of presence [Ivory and Kalyanaraman 2007]. While this scale has been shown to have face validity, it falls prey to many of the problems inherent in introspective measures. That is, due to its highly subjective nature, it suffers from issues of internal consistency and validity [Blake 2010].

Introspection, however, does have its uses as a number of introspective, psychometrically validated studies exist. One of these, developed by Witmer and Singer [1998], consists of three sub-scales: *Involved/Control*,

which addresses issues of perceived control and responsiveness within the game; *Natural*, which measures the extent to which the game was consistent with reality as well as how naturalistic elements such as interactions were, and *Interface Quality*, which measures the extent to which participants were able to concentrate on tasks by examining whether control devices and displays distract or interfere with the task at hand [Witmer and Singer 1998]. This presence scale is intended to be used in conjunction with the ITQ mentioned above. Overall, the internal consistency of this measure is high ($\alpha = 0.88$), however, issues of construct validity remain suggesting the need for further research [Witmer and Singer 1998].

The Independent Television Commission's Sense of Presence Inventory [ITC-SOPI] is another introspective questionnaire [Blake 2010]. This attempts to quantify presence through the use of four independent sub-scales. These are *Sense of Physical Space*, which refers to a feeling of being a part of the mediated environment; *Engagement*, which includes emotional aspects related to fun and feeling psychologically involved with the environment; *Ecological Validity*, which relates to believing elements in the environment were real, and *Negative Effects*, which describes any adverse physiological effects that might lessen the sense of presence [Lessiter et al. 2001]. Cronbach's alpha was very high for all four scales, ranging between 0.77 and 0.94. In addition, the scales displayed sensitivity to cross-media comparisons as well as significant factor inter-correlations [Lessiter et al. 2001]. At present, the authors suggest that the four factors should be analysed separately in order to detect differences that may develop according to the type of media being assessed [Lessiter et al. 2001]. Overall, however, this scale provides a fairly reliable measure of presence [Blake 2010].

Apart from introspective techniques, a number of behavioural and physiological methods also exist. Behavioural measures, however, are difficult to implement and may be subject to extraneous variables such as reactivity [Blake 2010]. Physiological assessment techniques rely on using involuntary physiological responses in order to gauge emotional responses. Some examples of these include monitoring heart rate or galvanic skin response. While some correlation has been shown between certain patterns of physiological response and results obtained using introspective presence questionnaires [Ravaja et al. 2004], there is still doubt as to whether certain physiological patterns can be unambiguously linked to presence [Blake 2010]. Furthermore, physiological measures often require expensive and specialised technology, meaning that they are not an appropriate measure for all circumstances [Blake 2010].

Overall, while a number of different means for measuring presence exist, no one method has been shown to be superior to all others. With this said, however, the ITC-SOPI appears to be the most reliable and psychometrically sound of the techniques discussed above. However, while the importance of assessing presence should not be overlooked, one must also consider other aspects of the game process, such as the ability to elicit enjoyment and fun from users.

2.4.5 Flow

Flow describes a game state where there is a successful balance between a player's ability and the perceived challenge or difficulty of the game [Ermi and Mayra 2005]. However, what is important about this balance is the emotional reaction it produces within the player [Johnson and Wiles 2003]. Csikzentmihalyi [1990] describes the experience of flow as being "so gratifying that people are willing to do it for its own sake with little concern for what they will get out of it, even when it is difficult or dangerous". This description highlights the importance of attempting to create flow within a game environment.

Flow is thought to consist of eight interrelated elements. The first of these is a task that can be completed [Sweetser and Wyeth 2005]. The ability to concentrate on this task is also required. This concentration should be possible because the task has clear goals and/or because the task provides immediate feedback [Johnson and Wiles 2003]. The feeling of being able to exercise control over one's actions is also needed. All this leads to a deep involvement that removes awareness of everyday life, which in turn leads to one's concern for self disappearing [Sweetser and Wyeth 2005]. However, one's sense of self emerges stronger than before once the flow state is completed [Sweetser and Wyeth 2005]. Finally, the perceived duration of time is also altered during this state [Johnson and Wiles 2003].

The application of flow to games would provide a powerful way of understanding the ways in which games manage to elicit enjoyment and fun despite, in general, serving no other intrinsic purpose.

Measuring flow

Flow is particularly difficult to measure as the only way to determine its presence during game-play is to disrupt it [Mandryk et al. 2006]. That is, direct measures of flow end up disturbing the flow state itself. This problem has been tackled in a number of different ways which will be discussed below.

The first of these is to analyse a game according to a fixed set of heuristics that have been adapted to measure flow or potential elements that may disrupt flow. Such a set of heuristics, which draws upon the definition of flow as well as conventional usability standards, was developed by Sweetser and Wyeth [2005]. Their heuristics were found to have varying degrees of applicability depending on game genre [Sweetser and Wyeth 2005]. However, the applicability of heuristics to a topic such as enjoyment remains questionable. While heuristics are very useful for identifying potential problems in the game process, their use becomes less clear when attempting to understand the emotional effect of a scenario or environment on users [Mandryk et al. 2006].

Another method which may have some application to measuring flow is experience sampling. Experience sampling attempts to measure random instances of an individual's experience [Hormuth 2006]. In essence this method involves interrupting a user and then assessing whether one has in fact interrupted some sort of flow state [Marsh et al. 2005]. While this technique may seem counter-intuitive, it allows one to get glimpses into a user's current mental state [Hormuth 2006]. However, as this method relies on a participant's introspective analysis, it is subject to issues of reliability [Blake 2010].

The Flow State Scale, developed by Jackson et al. [2001], is a questionnaire used to assess flow along the nine dimensions identified by Csikzentmihalyi [1990] as well as to provide an overall flow state score based on a higher order factor. The questionnaire comes in a number of variations that range in length, as well as the recommended time of administration (during the event versus after it) [Vlachopoulos et al. 2000]. The long version of the scale has acceptable reliability ($\alpha = 0.83$) [Jackson et al. 2001] and high internal consistency, with Cronbach's alpha equalling 0.93 [Tenenbaum et al. 1999]. A Rasch analysis conducted on the nine factors also suggests high consistency between and across samples [Tenenbaum et al. 1999]. Finally, the questionnaire has also been shown to have acceptable discriminant and construct validity [Marsh and Jackson 1999]. Despite the high reliability and validity of this instrument, care should be taken in its use as it has primarily been evaluated in the context of sports and exercise-based activities [Jackson et al. 2001; Vlachopoulos et al. 2000].

A fourth technique is to use physiological methods in order to assess a player's emotional state. Through the use of techniques such as facial electromyography [EMG], one is able to assess minute changes in facial expression and thus, by proxy, emotion [Hazlett 2006]. Facial EMG thus allows the valence of a player's emotions to be measured [Mandryk et al. 2006]. While there are other methods for measuring how positive or negative a player's emotions are, such as galvanic skin response and cardiovascular measures, facial EMG appears to be the most favoured by the research community and as such will be the only such measure discussed [Mandryk et al. 2006].

The real benefit of such a technique is that it does not require interrupting game-play [Mandryk et al. 2006]. In addition it allows for complex emotional profiles of game-play to be compiled. However, facial EMG is not without its caveats. To begin with, due to the sensitivity of the measure, it is easily influenced by external (and mentally internal) extraneous factors [Hazlett 2006]. Furthermore, the need for special equipment means that cost may be a prohibiting factor [Blake 2010]. Finally, the use of such a method does not allow one to specifically identify which elements produce a favourable response, only that such a response is occurring [Hazlett 2006].

Overall, due to its nature, flow appears to be difficult to measure. While heuristics may prove useful they do not allow one to assess the actual flow process. Physiological measures, on the other hand, allow for this but may not identify specific aspects. Experience sampling allows for an intrusive combination of the two. The Flow State Scale seems to provide the best alternative, but is in need of further validation outside of a sports context. Which method is most appropriate may ultimately depend on the task being assessed and the resources available.

2.4.6 General considerations

In addition to research focusing specifically on issues such as immersion, presence and flow, much research has been done on general usability principles in games [Sweetser and Wyeth 2005]. The majority of these take the form of sets of heuristics that have been adapted from standard usability measures in order to assess

games [Federoff 2002]. These heuristics have proven to be of particular use in the design and development stages of a game [Desurvire et al. 2004]. However, as mentioned above, a heuristic evaluation of elements other than usability and mechanics may prove difficult and unsatisfactory [Mandryk et al. 2006]. Thus while heuristics should perhaps be considered in the design stages, they should be used only when appropriate [Sweetser and Wyeth 2005].

2.4.7 Conclusion

Game enjoyment is a multifaceted issue which appears to depend on a large variety of determinants. These components, however, are interconnected and dependant. In addition to this, a shared difficulty in quantifying the degree to which each is elicited within a game environment appears to be a common factor.

Immersion, which describes a state where one feels one is a part of an environment, despite the technological system used to create said environment, appears to lack any concrete means of quantification. While means of doing so, such as the questionnaire developed by Ermi and Mayra [2005], do exist, the validity of such instruments remains uncertain. Given the overlapping nature of full immersion in relation to presence, it may be more appropriate to measure presence instead. In addition, the use of the ITQ questionnaire in conjunction with presence studies may provide valuable information on how dependant presence is on a person's innate immersive characteristics.

As mentioned, presence appears to be, at the very least, strongly dependant on an immersive state. Presence is defined as the perceptual illusion where an individual feels that an environment is real despite its virtual nature. While there is far more research on methods of evaluating presence in comparison to immersion, much of this research lacks a psychometrically sound basis. For this reason, should an in-depth study of the degree of presence elicited by a game be needed, the ITC-SOPI questionnaire appears to be the most appropriate instrument. However, due to the short nature of other surveys, such as the SUS questionnaire, these surveys may be more useful when one wants to determine a more general level of presence in order to investigate something else.

Flow, which also appears to depend on immersion to some degree, can be thought of as a state where the intrinsic process of playing the game elicits enjoyment and euphoric feelings. Physiological measures, such as facial EMG, allow one to assess emotional valence during game-play thus avoiding interrupting the flow state. However, the need for special equipment may make such methods inappropriate. Experience sampling, despite its disruptive nature, allows for a simple means of determining whether the player subjectively felt they were experiencing flow. The Flow State Scale, on the other hand, provides a measure of flow without requiring the disruption of the current task. Due to the difficulty and cost associated with physiological measures and the disruptive nature of experience sampling, the Flow State Scale appears to be the most appropriate choice.

While heuristics do not appear to be relevant to measures of immersion, presence or flow, they should not be completely overlooked as they can serve a valuable purpose during the game development process. Since many such heuristics have been developed, a combination of a number of them may make for the most comprehensive coverage of the usability issues at hand.

Overall the topic of why people enjoy playing games is complex and multifaceted. However, since a fundamental component of the majority of games is to provide entertainment, this concept is of the utmost importance when evaluating games. Through the use of a variety of techniques, one can begin to assess the core issues at hand as well as to gain some insight into what may be done to improve the enjoyment experienced by players.

2.5 Gesture-based gaming

2.5.1 Introduction

The use of gestures as a mode of interaction has been explored in areas such as Human-Computer Interaction and Virtual Environments. However, very little work has been done to investigate their potential uses within gaming [Kratz et al. 2007]. Games, unlike much other software, are used for their intrinsic enjoyment value. This means that it is often easier to entice users to use novel technology or modalities of interaction within a gaming context [Starner et al. 2000]. While there has been growing interest in technologies that would enable such gesture-based gaming, such as Sony's EyeToy, Microsoft's Kinect and Nintendo's Wiimote, very little has been done to assess how the inclusion of gestures will affect gaming [Wilson et al. 2009]. This section will discuss the limited literature available on this topic.

2.5.2 Previous research on gesture-based gaming

Previous research in this area has primarily used Computer Vision or accelerometer-based means for detecting gestures [Kang et al. 2004; Payne et al. 2006; Wilson et al. 2009]. Thus the discussion below primarily pertains to spatial three-dimensional gestures.

Studies done in this area have generally focused on how to implement gesture-based gaming rather than considering whether this interaction style increases the enjoyment and immersion a player feels during game-play [Kang et al. 2004; Kratz et al. 2007]. With this said, a number of factors have been identified that influence the effectiveness of gesture-based gaming. The first of these is the intuitiveness of the gestures used. In this regard, gestures should seem natural within the context of the game so that the user does not feel limited by their use [Kang et al. 2004]. Another important factor is appropriate and immediate feedback so that the user feels that they have control over their environment. In addition, this allows users to calibrate the way in which they perform the various game-play gestures [Payne et al. 2006]. This is related to the fact that gestural interfaces do not provide the same affordances as those present in traditional games [Kang et al. 2004]. For this reason, the gestures in use should be simple and easy to learn [Payne et al. 2006]. Finally, simple reminders or graphical hints should also be provided to aid recall and a consistent design should be used so that related commands share a gestural structure [Payne et al. 2006].

Payne et al. [2006] found that previous gaming experience had a significant effect on how the gestural interface was received. Experienced gamers expected greater precision in interaction whereas inexperienced gamers felt that the gestural interface was easier to use and more reactive as it required less memorisation of key combinations [Payne et al. 2006]. The differences in perception in this case may suggest that gesture-based gaming reduces the learning curve present in many games by making it easier for inexperienced players to become immersed in the environment [Payne et al. 2006].

Thus it appears that the appropriateness of gestures in gaming is dependant on the characteristics of the game and gestural interface themselves as well as the expectations of the player.

2.5.3 Conclusion

Gestures in games may serve as a novel interaction technique that is able to more rapidly immerse novices within a game-play environment. However, the cognitive demands surrounding learning and memorisation of gestures need to be carefully considered and affordances provided where possible. The choice of gestures should also be consciously examined within the context of the specific game being developed. Overall, however, further research is needed on this topic in order to fully understand the implications of the use of gestures in games.

Chapter 3

Game design and implementation

3.1 Introduction

In order to investigate the research question, the development of a game was needed. A number of limitations were placed on the game, however. To begin with, the game had to be able to incorporate both gestures and direct manipulation in a way that did not adversely effect the playability of either version. It also had to present the opportunity for creating an interface that used elements of both. Another important consideration was that the game itself had to be of a relatively limited scope given the time constraints for development. However, despite this, the game should still be sufficiently involving so as to create a flow state within the player. This is necessary as the goal of much of the experimentation is to measure this flow state for each interface type. The following sections describe the design and implementation of a three-dimensional puzzle game where players attempt to match blocks of the same colour in order to score points.

3.2 Game design of *Hexahedron*

3.2.1 Brief overview

Players will arrange different coloured blocks, which are initially grouped in a three-dimensional shape, so that blocks of the same colour are next to each other. These blocks can then be removed, with the number of blocks removed at a time being exponentially related to one's score. Three different interface versions of *Hexahedron* will be implemented in order to test the hypothesis under study. These include a gestural, direct manipulation and mixed interface. The gestural interface will use touch-based gestures in order to control block manipulations, while the direct manipulation version will employ buttons. The mixed interface will use a combination of the previous two. A mixed interface was added so that one can determine whether gestures or direct manipulation are appropriate for some controls, but not for all. *Hexahedron* will be specifically implemented for the Apple iPad.

3.2.2 Style and genre

Hexahedron aims to expand upon existing matching-style puzzle games by introducing a third dimension. That is, while it will bear some similarity to games such as *Same* and *Bejewelled*, it will expand on the basic premise of these games by allowing players to also arrange elements in the third dimension. The game will consist of a three-dimensional block made up of a number of small coloured cubes. Each cube will be consistently coloured, that is, all of its faces will share the same colour. This third dimension creates an additional element within game-play, as cubes can now be grouped together in more than just two dimensions. This allows for a more challenging experience that will encourage and help develop the spatial thinking skills of its players. The touch nature of the interface also allows for controls to be implemented through the use of gestures, which will hopefully increase flow and the ease with which the game can be played. In addition, it may give players the feeling that they are physically manipulating the three-dimensional object.

3.2.3 Core game-play

The aim of *Hexahedron* is to remove as many blocks in as large a group as possible. As the score received is related to the number of blocks removed, players should attempt to remove as many blocks at one time as possible. Players are able to rearrange blocks by using the predefined movement techniques discussed below. The aim of the game is to get as high a score as possible. The previous best score will be displayed underneath the current game's score.

Scoring system

When removing blocks from the screen, scores will be determined as follows, where S is the final score and n is the number of blocks removed:

$$S = S + 2 \times n$$

3.2.4 View controls

Rotation

Rotation controls will be implemented so that the shape can be rotated in one of the four cardinal directions. That is, a player will be able to rotate the shape by 90 degrees in either the North, South, East or West directions. A simple 90 degree rotation was chosen as it ensures that there is no ambiguity with regard to movement controls, as the shape is snapped to a number of predefined positions.

Rotation controls for each of the three interface versions are described below:

- **Gestural interface:** Swiping with two fingers on the screen will rotate the block. The direction of rotation is determined by the normal of the line created between start and end points of the swipe. This is then snapped to one of the four cardinal directions.
- **Direct manipulation version:** Buttons will be used to control rotation. That is, the player will be able to select the direction they want the block to rotate in through the use of a menu interface.
- **Mixed interface:** The mixed interface will use the same rotational controls as the gestural version. That is, a two-finger swipe will be used to control rotation.

A short animation will be used to give the illusion of movement. In addition, the block will be rotated by 5 degrees along the x - and y -axes in its static state in order to better illustrate its three-dimensional nature. The gestural controls were chosen so that they would appear intuitive to the user. That is, research has shown that gestures of a particular class (in this case, rotation), should be consistent so as not to place unnecessary memory stress on the user [Payne et al. 2006].

3.2.5 Game-play controls

Term definitions

In order to describe movement and orientation, a number of terms are used. This section provides a brief summary of their intended meaning.

- **Row:** All blocks which share a common y value. In addition, they should be equidistant from the z -axis. This ensures that only blocks on the same 'layer' are moved by controls affecting a single row. The active layer is always the outermost one (i.e. where at least one face of every cube in that row is visible).
- **Column:** All blocks which share a common x value. In addition, they should be equidistant from the z -axis. This ensures that only blocks on the same 'layer' are moved by controls affecting a single column. Figure 3.1 illustrates what is meant by a 'layer'. The active layer is always the outermost one (i.e. where at least one face of every cube in that column is visible).

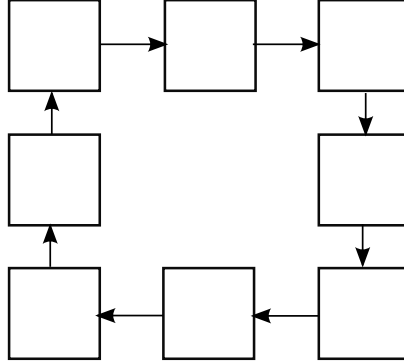


Figure 3.1: The rotation of one column shown apart from the shape. The active elements are considered to be part of the same layer.

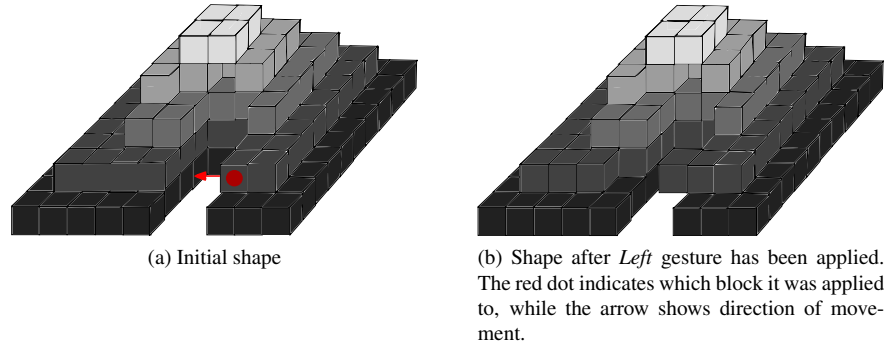


Figure 3.2: Illustration of *Left* gesture on the square pyramid with blocks missing.

- **x-axis:** This axis is considered to run horizontally across the screen.
- **y-axis:** This axis is considered to run vertically along the screen.
- **z-axis:** This axis is considered to run perpendicular to the *x*- and *y*-axis. That is, it lies perpendicular to the surface of the screen, with the positive *z* values coming out towards the user.
- **Origin:** The origin is defined as follows in relation to the overall shape:
 $Centre_x = \text{Maximum length of shape in } x \div 2$
 $Centre_y = \text{Maximum length of shape in } y \div 2$
 $Centre_z = \text{Length of shape in } z \text{ along the line that goes through point } (Centre_x, Centre_y) \div 2$
Thus, the origin is defined as $(Centre_x, Centre_y, Centre_z)$.

Controls

Cubes will be able to be arranged by means of controls that displace blocks with those around them in much the same way as is used in a *Rubik's Cube*. That is, all cubes in a single row (blocks along the *x*-axis with a constant value on the *y*-axis) or column can be rotated around the relevant axis. An example of such rotations around the *y*- and *x*-axes are shown in Figure 3.2 and 3.3. In addition, cubes will be able to be swapped with the cubes below or above them (on the *z*-axis) as well. Finally, controls for removing any gaps that may have been caused by block removal as well as slightly rocking the shape (so as to give an impression of three-dimensionality) will also be included. These controls were decided upon as they provide a maximum amount of control while only requiring

minimal learning from the player. That is, as many of the controls are just variations of the same idea, but in different directions, a player only has to grasp and remember a small number of movement techniques. This, in turn, reduces the cognitive load placed on the player [Nielsen et al. 2004]. A *New game* button will also be included in case the user wants to begin a new game or can no longer execute any significant movement commands.

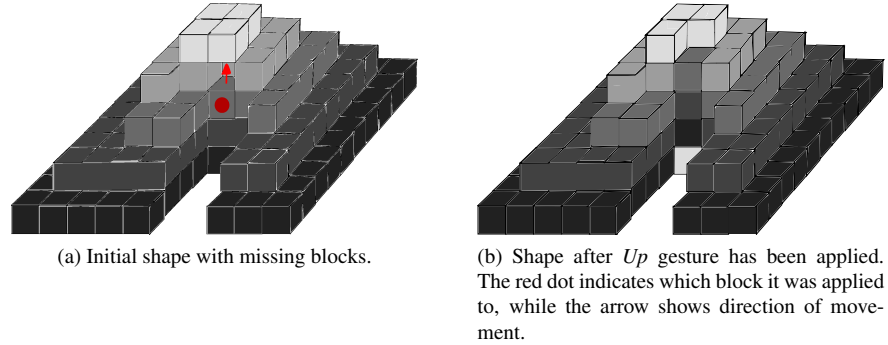


Figure 3.3: Illustration of *Up* gesture on the square pyramid with blocks missing. The red dot indicates which block it was applied to, while the arrow shows direction of movement.

Gestural interface Gestures will be used to control block manipulations. In order to apply a specific action to cube, the player will need to begin that gesture on the required cube. This removes the need for extra taps in order to determine the target of each gesture. It also utilises the context-sensitivity possible with gestural interaction [Wobbrock et al. 2009]. Gestures were chosen so that they provided some form of affordance to the player. That is to say that their form could logically be linked with their action. This is of importance as gestures do not provide the same affordances as more traditional, button-based interfaces [Nielsen et al. 2004]. This in turn may mean that users are required to memorise a far greater number of commands. By making gestures logically relateable to their purpose, one can lessen this memory load [Jaimes and Sebe 2007]. In addition, gestures that applied the same movement technique, but in different directions, were created from variations of a single model. This too was done to reduce cognitive requirements placed on the player [Nielsen et al. 2004].

A full list of possible moves as well as their descriptions are provided in Table 3.1. A dot indicates the beginning of the gesture in order to show directionality. Block removals will be done by tapping the required block. Gaps left by removing blocks will remain until the user applies a *Fill in gaps* command.

Direct manipulation version The direct manipulation version will have the same functionality as the gestural game. However, all controls will take the form of buttons displayed on the side panel which displays hints in the

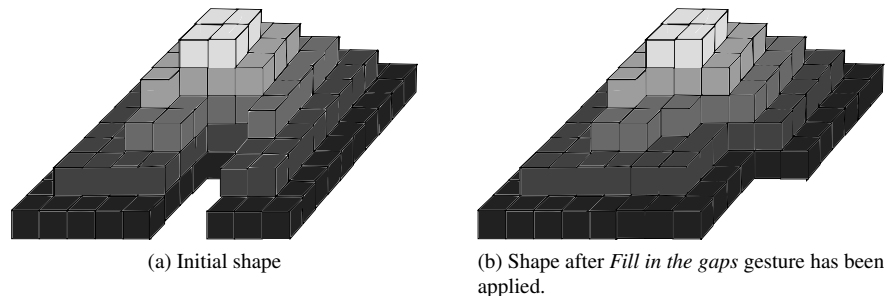


Figure 3.4: Illustration of *Fill in the gaps* gesture on the square pyramid with blocks missing.

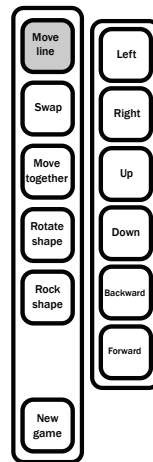


Figure 3.5: An example of the direct manipulation menu structure. As can be seen, when selecting an option with multiple variants, a second sub-menu is displayed.

gestural version. Rotation will also be implemented by means of buttons in order to maintain consistency and not confuse the user. In order to use the controls that relate to specific blocks, a player will be required to select the appropriate control. They are then able to apply the movement by selecting blocks. When a movement control has a number of variants (for example, *Left*, *Right* and so on) a second sub-menu will be shown on the side of the main menu. This sub-menu will display the variants so as not to clutter the main menu. The selected control will remain selected until the user either deselects it or selects a different control. This means that any interaction with the shape will be interpreted as applying the selected movement control. This persistence of a selected command was chosen so that users who wish to apply a movement technique multiple times do not become frustrated with the number of menu selections they need to perform. An example of what such a menu will look like is provided in Figure 3.5

Mixed interface A combination of the two previous interface versions will be used in the mixed interface. That is, controls will be implemented through the use of both gestures and buttons. Gestural controls will be used for rotating rows and columns, moving blocks forwards or backwards along the z-axis, swapping blocks and rotating the shape. These commands were selected for gestural interpretation as they hold a strong logical correspondence to their action. In addition, all the commands represent only three underlying movement techniques (that of rotating the shape, swapping two blocks or moving a number of physically related blocks). The inclusion of both of these elements in a gestural form should reduce cognitive and memory requirements placed on the user. This, in turn, is expected to increase the effectiveness of their use [Nielsen et al. 2004]. Buttons will be used to move blocks together in order to fill in the gaps as well as to rock the shape. These commands and their related gestures do not bear as much of a logical connection and thus were selected for button implementation as it may be found that this is more appropriate. Two disconnected menus will be placed on the left-hand side of the screen. The first will contain the buttons for manipulating the shape while the second will hold the appropriate subsection of the gesture hint panel.

Removing blocks

Cubes can be removed from the game by means of tapping a group. If there are more than three neighbouring cubes of the same colour, they will be removed. Points are scored depending on how many blocks are removed at one time. Blocks can be linked together in any of the three dimensions, so long as at least one side is touching another block of the same colour.

When a group removal has caused there to be spaces between a block and the other blocks comprising the

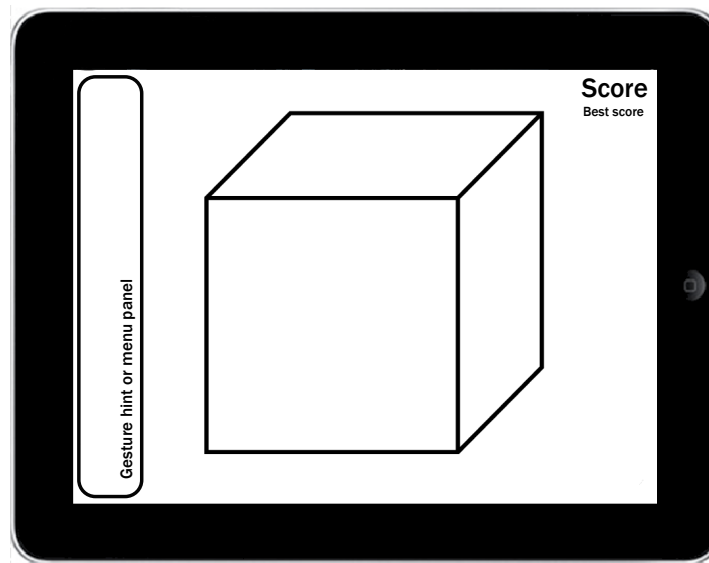


Figure 3.6: An example of possible interface design.

shape, the block will not be automatically moved to fill these gaps. Instead a gesture has been provided so that the player can choose whether to apply this effect. This is done so that players can strategically arrange blocks without having to take into account movement caused by removed blocks as this may be difficult to predict due to the three-dimensional nature of the shape.

3.2.6 Target audience

The demographic characteristics of the target market are fairly broad, as anyone who enjoys spatial thinking or puzzle-type games would be able to play. In addition, the puzzle market is not generally geared towards a specific age group or gender, allowing for a more diverse audience. The platform that will be used, however, does limit the target audience. As *Hexahedron* will be developed for the iPad, the primary target market of the game will thus be individuals who own such a device and enjoy puzzle games.

3.2.7 Design

A very simple, clean design will be used throughout *Hexahedron*. That is, there will be minimal interface elements, as the majority of the move related commands will be executed by means of gestures or a single menu structure. This was chosen so that users are able to better focus on the core game-play without being distracted by extraneous elements. This is of importance as the purpose of the game is to elicit a flow state through game-play. In order to provide the three-dimensionality required by the game, shading will be used when colouring the cubes. In addition, a light-source will be positioned so that there is an illusion of depth. Finally, the overall shape will also be rotated slightly so that the other faces are also slightly visible.

Four different colours will be used. These include red, blue, green and orange. Block movements will be accompanied by a brief animation which shows the movement of affected blocks before they reach their final positions.

A gesture hint panel will be provided on the left-hand side of the screen as initially there will be a large array of gestures to learn. This should provide a form of affordance to new players. Tapping one of the displayed icons on this bar will display the associated gesture and its meaning. In the direct manipulation version, this panel will serve as the block manipulation menu. This panel will also have the ability to be hidden. In order to hide the panel a three-finger swipe starting on the panel and ending at the edge of the screen will be used. Users will be

able to redisplay the panel by using a three-finger swipe that starts at the edge of the screen and moves inwards. Apart from this, the only other interface elements visible will be the score, best previous score and a *New game* button (which will be placed on the left-hand menu). A basic mock-up of the described interface can be seen in Figure 3.6.

3.2.8 Summary

While *Hexahedron* itself is fairly simple, its correspondence to a number of equally simple puzzle games, along with the addition of a third dimension should prove sufficiently compelling to induce a flow state. Furthermore, as gestures have been carefully selected to maximise the use of inherent affordances provided by their structure and relation to other gestures, this should serve to lessen the cognitive demands placed on the user. This is necessary as if users are unable to remember gestures they may feel as if they are not in control of the game [Wobbrock et al. 2009]. This would, in turn, reduce the likelihood of achieving a flow state [Sweetser and Wyeth 2005]. Finally, due to the choice of movement controls, it should prove possible to implement both a direct manipulation and mixed interface version of the game without compromising the fundamental game-play and thus, gaming experience. Therefore this game should serve as an appropriate flow-inducing task in all three conditions. This allows the effect of interface type to be investigated within the context of touch-based gaming.

3.3 Game implementation

3.3.1 System overview

The game can be logically divided into a number of components. These are illustrated in Figure 3.7. A Model-View-Controller design pattern is used for the game implementation [Savitch 2007]. The model consists of a three-dimensional array which is used to store block information, while the view elements are handled by the *OpenGL Renderer* class that translates this block information into a visual representation. Finally, the *Game controller* coordinates interaction between these and all other game components. These other components include input handling, which interfaces with the *Feature extraction* and *Gesture recognition* units, as well as a *Game mechanics* module. The *Game mechanics* unit is responsible for manipulating block data in order to implement movement controls. Finally, the *Game version controller* determines which game version should be played and enables and disables the requisite components for each. This unit also handles the tutorial screens and logging of information necessary for later experimental analysis. The visual elements of the game are created using *OpenGL ES*. The menu system and touch events are handled using standard iOS SDK components. The rest of this chapter will discuss these components in more detail. It should be noted, however, that a full discussion of how the gesture engine and in particular, the feature extraction works, has been included in Chapter 4 and as such will not be expanded on here.

3.3.2 Input events

Input is received via touch events using the standard touch event handling provided by the iOS SDK. The event handling is broken up into `touchesBegan`, `touchesMoved` and `touchesEnded`. Touch events are handled differently depending on the game version that is being played. The following section provides a brief overview of the input handling and how it is used by each game version.

Single finger

`touchesBegan` When a single touch is initiated, the (x, y) position of the touch is sent to the Game Mechanics module. At this point it is mapped from screen coordinates into the coordinate system used by *OpenGL* drawing (and the block placement array). If the gestural or mixed version is active, this point is also sent to the feature extraction class.

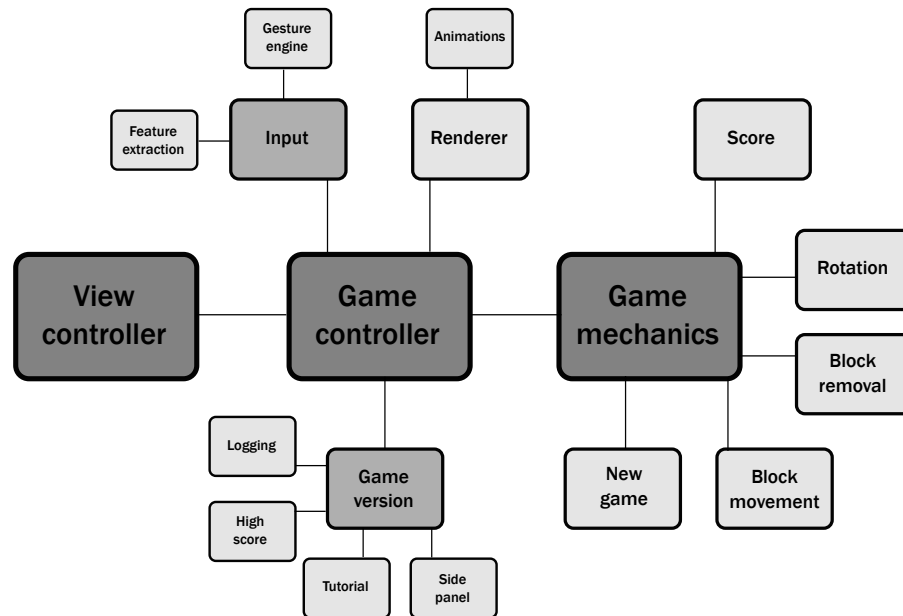


Figure 3.7: Overview of game components and their relation to each other. The use of colour indicates the importance and scope of each component. Dark gray elements are responsible for coordinating a number of other components. Lighter gray items are responsible for the integration of a number of elements within a specific area, while the components in the lightest gray perform a single function and are not responsible for coordinating any of the other named items.

touchesMoved In the gestural and mixed version, these points are sent to the feature extraction class. The direct manipulation version ignores these points (as there is no gestural component associated with the game in this case).

touchesEnded In all versions a check is done to determine whether the distance between the start and end point is less than 10 pixels. If it is, the movement is considered to be a tap and handled accordingly. The direct manipulation game considers all touch interactions to be taps and thus ignores all points except the start position.

In the gestural and mixed versions, the point is sent to the feature extraction endpoint method. The gesture engine is then passed this data and either returns which gesture is recognised or `null` if no gesture is found.

Two fingers

Multi-touch is disabled in the direct manipulation version as all controls are handled via buttons, therefore this section refers to the gestural and mixed versions only.

The start and end point of the touch interaction are recorded and sent to Game Mechanics where the angle of rotation is determined. This angle is returned and set as the angle of rotation for the renderer class (in order to visually rotate the shape).

Three fingers

As mentioned above, multi-touch is disabled in the direct manipulation version. In the other two game versions, the touches are ignored unless they fall within the 100 leftmost pixels of the screen. In this case, the direction of the touch interaction is determined and passed to the method that handles the hiding and redisplay of the gesture hint panel (See Section 3.3.7).

3.3.3 Data organisation

The primary data handled within the game is the information associated with each block. While touch information is also stored, the description of how this is done is included in the Gesture Engine design chapter. The following section describes how block data is stored.

Block objects

Each on screen block has an associated block object. This object stores the following information:

- (x, y, z) position
- Colour
- (x, y, z) position and increment for translation

Block placement array

Pointers to these block objects are stored in a three-dimensional array that mimics their onscreen arrangement. The array is of size $11 \times 11 \times 11$. As the centremost block on screen is located, in visual coordinates, at position (0, 0, 0), this block is positioned at index (5, 5, 5) within the array. Thus, in order to convert array indices to visual coordinates, -5 is added to the array coordinates. When blocks are removed, their position in the array is set to `null`.

Drawing array

Pointers to all visible blocks are also added to a second linear array. A pointer to this array is passed to the renderer class so that each block can be drawn.

3.3.4 File formats

Logging information is recorded during game-play for later analysis. In addition, the top high score is also recorded. The format of these files is described below.

High score

The current score is checked against the high score at the end of each game, as well as when a new game is started manually by the player. If the new score is greater than the previous high score, this new score is written out to a file as a string. This overwrites the previous high score.

Logging information

Information on the order in which the games are played, as well as which controls were used and how often, is recorded for each gaming session. Each session appends this data to the end of a file in the following format:

```
Date and time
Game version number 1
Control data
Game version number 2
Control data
Game version number 3
Control data
```

Game version number refers to a number between 0 and 2 which specifies which game was played. The numbers represent the following game conditions:

0 - Direct manipulation interface 1 - Gestural interface 2 - Mixed interface

Number	0	1	2	3	4	5	6
Control	Row left	Row right	Column up	Column down	Forward	Backward	Swap left
Number	7	8	9	10	11	12	
Control	Swap right	Swap up	Swap down	Move together	Rock shape	Rotate	

Table 3.2: Control numbers and their corresponding movement controls

Control data refers to the information on the number of times each game control was used. This is organised as follows:

Control number: Number of times used

The control numbers range from 0 to 12 and correspond to the movement techniques as shown in Table 3.2.

3.3.5 Game mechanics

This module is responsible for determining block movements and removal. The implementation of these controls is described in more detail below.

Block identification

As all block information is stored in a three-dimensional array that mirrors placement on screen, a block's screen position can be used to determine its position within the array. However, as the contents of the array are not rearranged when a rotation of the shape takes place, the currently visible face may not correspond to the frontmost elements in the array. For this reason, the correct coordinates need to be calculated from the screen coordinates. This is done using rotational matrices corresponding to the x -, y - and z -axes. A description of how the coordinates of these axes are calculated is included in a subsection below.

The mapped input coordinates are first checked to ensure they fall within the onscreen range of the shape. The following matrix multiplication is then applied:

$$\begin{bmatrix} X' \\ Y' \\ Z' \end{bmatrix} = \begin{bmatrix} x_0 & x_1 & x_2 \\ y_0 & y_1 & y_2 \\ z_0 & z_1 & z_2 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} + 5$$

where X , Y and Z are the input x and y coordinates and Z is set to 5 (as this is the frontmost block at these coordinates). The 3×3 matrix stores the current orientation of the axes. 5 is added to the result as view coordinates range from -5 to 5, while array indices begin at 0. This entire operation maps the view screen coordinates to array indices.

If the block at that position does not exist because the user has removed it from screen, and is thus attempting to interact with the currently visible block beneath it, the values are recomputed with Z set to 4. The value of Z continues to be decremented until a valid block is found or until the Z value is outside the valid range of block positions. If a block is found, a pointer to it is returned, otherwise `null` is returned.

Movement of rows and columns

For the purpose of this explanation, a row and column are defined as follows:

- **Row:** All blocks which share a common y value. In addition, they should be equidistant from the y -axis.
- **Column:** All blocks which share a common x value. In addition, they should be equidistant from the x -axis.

The manipulation of columns and rows is fairly similar, except that in the case of rows the y coordinate is held constant, while with columns the x coordinate remains the same. For this reason, the description below refers to the implementation of *Left*, *Right*, *Up* and *Down* operations. The following equation describes this operation

mathematically:

$$V(Rot(P_{x,y,z})) = V(P_{x,y,z}) \text{ where}$$

$$\begin{aligned}
Rot_{Left}(P_{x,y,z}) &= \begin{cases} P_{x-1,y,z'} & \text{if } P_x > -z' \\ P_{-z',y,z-1} & \text{if } P_z > -z' \\ P_{x+1,y,-z'} & \text{if } P_x < z' \\ P_{z',y,z+1} & \text{if } P_z < z' \end{cases} & Rot_{Right}(P_{x,y,z}) &= \begin{cases} P_{x+1,y,z'} & \text{if } P_x < z' \\ P_{z',y,z-1} & \text{if } P_z > -z' \\ P_{x-1,y,-z'} & \text{if } P_x > -z' \\ P_{-z',y,z+1} & \text{if } P_z < z' \end{cases} \\
Rot_{Down}(P_{x,y,z}) &= \begin{cases} P_{x,y-1,z'} & \text{if } P_y > -z' \\ P_{x,-z',z-1} & \text{if } P_z > -z' \\ P_{x,y+1,-z'} & \text{if } P_y < z' \\ P_{x,z',z+1} & \text{if } P_z < z' \end{cases} & Rot_{Up}(P_{x,y,z}) &= \begin{cases} P_{x,y+1,z'} & \text{if } P_y < z' \\ P_{x,z',z-1} & \text{if } P_z > -z' \\ P_{x,y-1,-z'} & \text{if } P_y > -z' \\ P_{x,-z',z+1} & \text{if } P_z < z' \end{cases}
\end{aligned}$$

where V is the view screen to array coordinates transform described above, P is the three-dimensional position of a cube in view screen coordinates, Rot describes the operations used to rotate around the specific layer in the required direction and z' is the maximum extent in the z direction.

Thus, the coordinates of the block touched by the user are passed to the block identification method described above. This checks whether this is a valid block and returns a pointer to this block in the array. It also sets the current z value for use with the later manipulations as this defines which layer within the shape the movement is being applied to. As the application of one of these commands may actually require rotation around some other axis than the one provided by view screen coordinates (due to rotation of the shape), visual coordinates, rather than array coordinates, are used to iterate through the row. That is, the visual coordinates are manipulated and then mapped back to array coordinates (using the block identification method) in order to adjust the position of block objects.

The coordinates stored within each block object as well as their position in the array are updated using this method. Positions with a `null` value are also translated in this manner.

Movement along the z -axis

The implementation of the *Forward* and *Backward* commands is similar to that for rows and columns. In this case, the visual x and y coordinates are held constant, while the z coordinate is changed. These visual coordinates are mapped to array coordinates and the block objects moved accordingly. This is illustrated mathematically below:

$$\text{Forward: } V(P_{x,y,(z+1)\%z'}) = V(P_{x,y,z}) \quad x = x_c, y = y_c$$

$$\text{Backward: } V(P_{x,y,(z-1)\%z'}) = V(P_{x,y,z}) \quad x = x_c, y = y_c$$

where V is the view screen to array coordinates transform described above, P is the three-dimensional position of a cube in view screen coordinates, x_c and y_c are constants and z' is the maximum extent in the z direction.

Swapping of blocks

In order to determine which blocks are to be swapped, the array coordinates for the selected block are calculated. Visual coordinates of this block are then manipulated in order to determine the neighbour with which the block should be swapped. These are then mapped to array coordinates and the position of the two blocks is updated.

Moving blocks together

When a *Move together* command is executed the following three steps are completed in order to move all blocks in towards the view screen origin, thus removing any gaps between them caused by the removal of blocks. The first step involves checking the z array coordinate of a block and the block behind it. The z coordinate of the previous block is determined as follows:

$$z_{prev} = \begin{cases} z - 1 & \text{if } z > 0 \\ z + 1 & \text{if } z < 0 \end{cases}$$

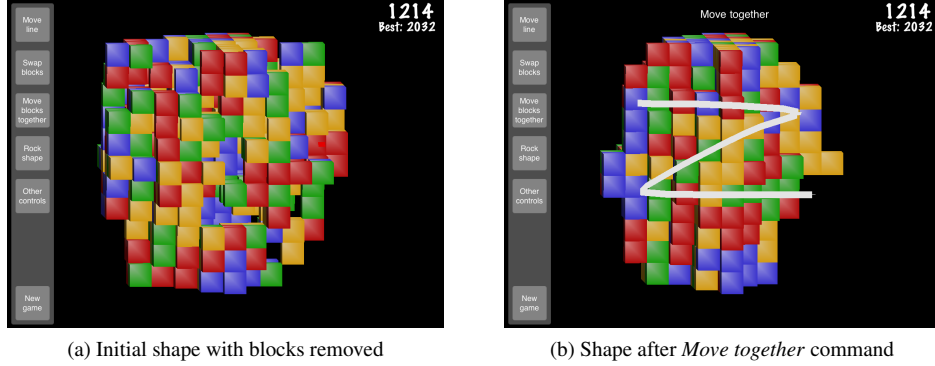


Figure 3.8: Screenshots of the game illustrating the *Move together* command.

If the current block has a z coordinate of 0, it is ignored. The same procedure is then followed for the x - and y -axes. This process is applied from the origin outward to ensure that all blocks are moved as far in as possible. This serves to move all blocks in towards the origin, compacting them in the process. Figure 3.8 illustrates the effects of this command.

Block removal

When a block is set to be removed, it recursively calls a method on its neighbours in order to determine whether they are of the same colour. Blocks are considered to be neighbours if they have at least one face touching. If a block is the same colour, it sets a check variable (so that it is not removed more than once), adds itself to a list of blocks to be removed and calls the removal method on its neighbours. This continues until no more blocks of the same colour can be found. If the removal list has a length of 3 or more, the blocks in this list are removed from both the block position array and the drawing array.

Shape rotation

Rotations are primarily implemented in the *OpenGL* renderer, however, the game mechanics module is also required to keep track of rotations in order to correctly identify blocks within the placement array.

In order to correctly map visual coordinates to array coordinates, it is necessary to keep track of the orientation of each axis. This is done by recording the orientation of each axis as a column vector and then multiplying them with the appropriate rotational matrix every time a rotation is needed. As rotation is only possible in one direction at a time, this is a fairly simple procedure. The rotational matrices for rotation around the x - and y -axes are shown below:

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$

$$R_y(\theta) = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$$

New game

When the *New game* button is selected a number of steps are taken to return the game to its original state. This requires resetting the block array as well as the rotational coordinates of the axes in both the renderer and the game mechanics module.

The block placement array is iterated through. In order to prevent unnecessary reallocation of memory, any positions which already hold block objects simply have the block object passed a new randomly generated colour. Indexes with a `null` value are assigned a new block object. The drawing array is cleared and these new objects added. Finally, the rotational matrix in the renderer and the axes in game mechanics are reset. The score is also checked to determine whether it is better than the previous highest score. If it is, the high score value is changed. The score is then reset to 0.

3.3.6 Game rendering

Rendering of the blocks that compose the overall shape as well as touch events created by the user is done using *OpenGL ES*. An orthographic projection was used to display the blocks. Simple lighting was also implemented in order to aid the illusion of three-dimensionality. In order for the lighting to best illustrate the three-dimensional nature of each cube, surface, rather than vertex normals are required. For this reason, cubes are drawn using `GL_TRIANGLES` rather than `GL_TRIANGLE_STRIP`. The shape is also rotated by 5 degrees in the x and y directions so that additional shape faces are visible to provide depth cues. The matrix used to represent the orientation of a block is translated to the position of each block (specified in the array containing block objects). This is then drawn with the colour listed in each block object.

A pointer to an array of all the current touch points is also passed to the renderer. These are then linked and drawn in white. When the touch ends, this array is cleared. When a gesture is not recognised, a flag is set, and the renderer draws this line in red.

Animations

Rotation Since applying an *OpenGL* rotation also rotates the coordinate system in use, a separate transform matrix is kept. Rotations are then applied to this matrix which is then loaded as the current matrix. In order to do this, an instance of the built-in type, `CATransform3D`, is used to record the current orientation of a cube. The required rotation is then incrementally applied to this matrix. This is done separately along the x - and y -axes. The incremental nature of the rotation is used in order to produce a rotational animation effect.

Translation Most block movement operations require some form of animation. A simple translation is used to achieve this effect. Blocks are incrementally moved from their original position to their new position. Each block object contains additional variables used to determine their position during this gradual translation. When a block is marked for translation, the new position is passed in and the increment determined. These blocks are then removed from the main drawing array and placed in a separate translation array. The renderer then calls the translation method within each block until it has reached its new position. At this point, the block is removed from the translation array and re-inserted into the main drawing array. In this way, it is possible to translate multiple blocks in different directions simultaneously.

3.3.7 Interface elements

Each game version has an associated side panel. The function of this panel, however, depends on the interface in use. The side panel can be hidden by completing a three-finger swipe towards the left edge of the screen. A swipe starting at the left edge and moving inward redisplay this panel. An animation is used when moving the panel in and out. Figure 3.9 shows all three interface versions, as well as a sample side panel for each case.

Gesture hint panel

This panel is used to display information on the possible gestures. Selecting a button displays a second panel which illustrates the gesture as well as its function. This panel is visible in the gestural and mixed game versions.

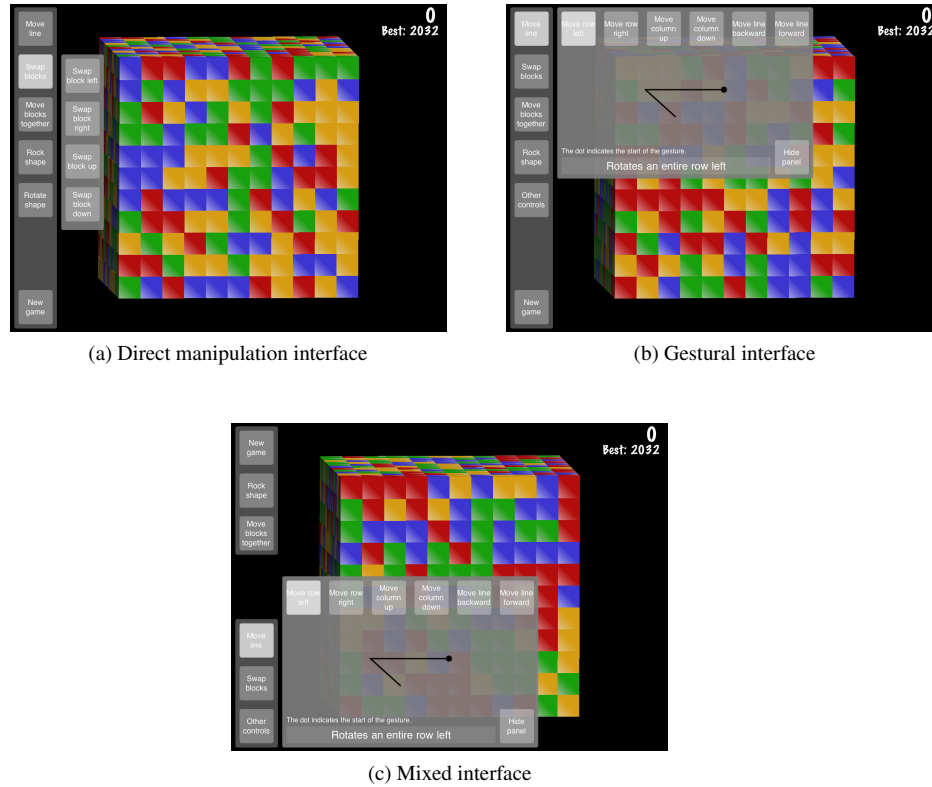


Figure 3.9: Screenshots of the three interface versions. A side menu has been activated in each case.

Control button panel

This panel is used in the direct manipulation and mixed game versions. It contains buttons that can be used to control block manipulation. Once a button has been selected it remains selected until either a new button is chosen or it is unselected by the user. If a button is selected, all interaction with the shape is interpreted as applications of the selected control.

3.3.8 Game version controller and sequence

As there are multiple game versions, a mechanism is needed to transition between the interfaces of each. In addition, various components need to be enabled or disabled depending on the active game version. Game-play consists of a tutorial session followed by the three game versions. This is handled as described below.

Tutorial

During the tutorial the player is shown an image of the gesture they are to practice. They are then asked to draw this gesture ten times. Once they have successfully completed ten gestures (that is, the gesture they have drawn has been recognised correctly ten times), the next gesture is displayed and the process repeated. This is done for all gestures used in the gestural and mixed interface versions of the game.

The tutorial screen drawing is handled by the renderer. During this stage the block drawing array is empty so only the user's touches are drawn. These touches are passed to the gesture recognition engine and handled accordingly. Once all gestures have been tested in this manner, control is passed to the game version controller.

Interface version	Gestures enabled	Hint panel	Button panel
<i>Direct manipulation</i>	None	Hidden	Visible
<i>Gestural</i>	All enabled	Visible	Hidden
<i>Mixed</i>	All enabled apart from <i>Rock shape</i> and <i>Move together</i>	Visible but without buttons for <i>Rock shape</i> and <i>Move together</i>	Visible but only contains the <i>Rock shape</i> and <i>Move together</i> buttons along with the <i>New game</i> button

Table 3.3: Overview of the interface and interaction elements available in each game version

Game choice

Once the tutorial section is complete, a game description screen is displayed. This contains information on what the game entails. Control is then passed to a method which determines which game version should be played. This is done by selecting a random number between 0 and 2. The numbers represent the following game conditions:

0 - Direct manipulation interface 1 - Gestural interface 2 - Mixed interface

This number is checked against a `BOOL` array of size 3. If the element at this index is set to `True`, the process is repeated until a number is chosen that corresponds to a game version that has not yet been played. The array is then updated and the setup for that game version is completed. This entails enabling and disabling menu or hint panels and the accepted gestures for that game version. Table 3.3 illustrates the components used for each game version.

Once the game has been setup, the user plays it for 7 minutes. At this point a timer interrupts game-play and the user is presented with the instructions for the next game version. Logging information is also written to file and the timer is reset. The logging information is described in more detail above. Once all three games have been played, a game end screen is displayed.

3.4 Game evaluation

In order to evaluate the game, a number of techniques were used. These include play testing the actual game and evaluating the interface versions and gestures used within the game. These methods are described in more detail below.

3.4.1 Play testing

In order to uncover any bugs in the actual game mechanics, play testing was conducted. This was done throughout the game development process as well as after the game had been completed.

After each movement technique had been implemented, the game was played for half an hour to ensure that this technique and all other previously implemented mechanics worked as expected. Any errors found were corrected and the game tested again until no more game-play discrepancies were discovered.

Once the game was completed, each group member play tested the game for 20 minutes. Between each session, any discovered bugs were fixed. Once each group member had played the game once and all errors had been corrected, the game was tested for a further half an hour. This was done to ensure that all mechanics worked as specified so that the elicited flow state during experimentation would not be interrupted due to discrepancies in game-play.

3.4.2 Interface evaluation

The design of each interface went through a number of iterations. That is, the final design was successively refined. This was done by asking individuals within the Honours class to play the game with each interface version with

the task of identifying any issues that might detract from game-play. An example of this is the rotational controls in the direct manipulation version. Initially these were going to be implemented via a two-finger swipe, as in the gestural and mixed interface versions. However, play testing revealed that this was not the expected interaction style and represented a break in the continuity of the interface. That is, as all other controls were implemented through button use, including a more gestural technique for rotation was felt to be inconsistent with the remainder of the interface.

3.4.3 Gesture evaluation

The choice of gestures was also evaluated. This was a two-part process whereby group members were first asked to draw gestures they felt would be appropriate for each movement technique. These gestures were then refined into a set of gestural controls for the interface. The gestures were then shown to three members of the Honours class. They were given a list of possible movement techniques and asked to identify what they thought the function of each gesture was. After half an hour, they were then instructed to draw the gestures for each movement command. This was done to test the intuitiveness and memorability of each gesture. Subsequent refinements were made using the data from this procedure in order to determine the final gestures, which can be seen in Table 3.1.

Chapter 4

Feature extraction

4.1 Introduction

As the project aims to investigate the effects of gestures on touch-based gaming, a recognition unit to recognise these gestures was needed. Two gesture engines have been developed for this purpose. One of these uses a Hidden Markov Model, while the other uses an approach based on state machines. Both these engines, however, share a common feature extraction unit. This unit is responsible for reducing the dimensionality of the input data by segmenting it into what are known as features. Each feature then describes this decomposed data using some implementation specific metric. The following sections will describe and discuss what feature extraction is in further detail, as well as the specific design limitations placed on an implementation for the iPad. As a number of different techniques were tested, the implementation of each as well as a comparison of their robustness and consistency is also included.

4.2 Background

The fundamental idea behind feature-based image processing techniques is that of data or dimensionality reduction. That is, complex data is reduced to a simpler and more compact representation [Guyon and Elisseeff 2006]. As the nature of both the representation and the process of reduction is highly dependant on the problem at hand, this section will provide only a high-level introduction to these concepts before discussing techniques specific to feature extraction for touch-based gesture recognition.

4.2.1 Features and their detection, extraction and segmentation

There is no universal definition of what constitutes a feature. Rather, they can be thought of as a fundamental unit with a higher level interpretation relevant in the context of computationally solving a specific problem [Erdas 2008]. Thus a feature could be a single point, line, texture, shape or other appropriate sub-image description [Marais 2010]. The type of feature required also determines the method of feature extraction needed.

Feature extraction is a very broad topic primarily concerned with locating and describing features within the domain of image processing [Erdas 2008]. Feature detection and segmentation are two other related and often overlapping topics in this field. Feature detection refers to locating areas or elements of interest within an image [Guyon and Elisseeff 2006]. This is of importance in images that contain much data. Segmentation pertains to the process of partitioning information into a number of segments or features [Nixon and Aguado 2008]. In general, both detection and segmentation are used when extracting features. Finally, features need to be related to each other in some way in order to fully describe the sub-image they represent [Marais 2010]. The way in which this is done is also highly dependant on the nature of the features and the underlying object they represent. Thus, when considering these techniques, it is vital that the specific characteristics and requirements of the data in question be fully understood.

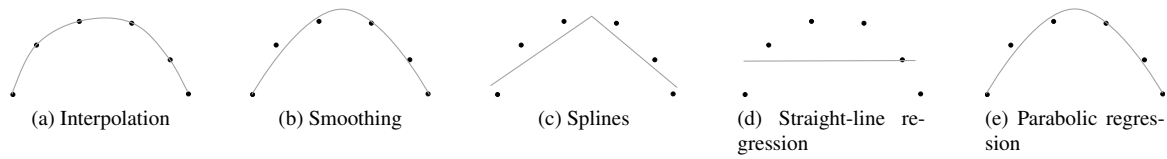


Figure 4.1: Illustration of various curve fitting techniques. The black dots represent the data to be fitted, with the gray line showing the fitted curve.

4.2.2 Curve fitting

Curve fitting refers to constructing a curve or function that best describes a set of points [Akima 1970]. Thus it can be seen as a form of dimensionality reduction. There are a number of ways of achieving this effect. However, since this is a topic in its own right, only an overview of the basic premises will be presented. Figure 4.1 provides a graphical overview of some of the techniques discussed below.

One of these concepts is interpolation. This refers to using existing data points to determine a function, or similar, which passes through all the given data points as well as defining the values of those not present [Moler 2004]. Thus it can be thought of as a means of defining a function that takes on specific values at certain points. Therefore interpolation is different to other curve fitting methods as it requires that the function pass exactly through all the data points provided [Akima 1970]. Another curve fitting technique, with less stringent requirements, is that of smoothing. Smoothing is used to create a function that approximates the most important points or features of the data [Savitzky and Golay 1964]. Thus, unlike interpolation, smoothing does not attempt to create a function that passes through every existing data point, but rather provides an approximation that highlights the most important features of the data.

Splines are a technique that can be used by both interpolation and smoothing. They are used to define curves in a piecewise manner through the use of polynomials of low degree [Marais 2010]. Through the use of this method, a curve would be decomposed into a number of splines which together describe the most important features of the data. Splines are used because they allow one to reduce the complexity of data by approximating complex curves in a way that is simple to construct and evaluate [Dierckx 1993].

4.2.3 Regression

Another method that uses elements of both interpolation and smoothing is that of regression. Regression is a statistical technique used to ascertain the relationship between two or more variables [Sykes 2000]. This is done by constructing a trend line or line of best fit. In general, this line is described by some function with the regression serving to determine the coefficients of this equation [Sykes 2000]. As regression can be applied in a number of different ways an overview of the differences in accuracy and computational complexity will be discussed.

The main premise behind regression is that of constructing some simplified representation of the data that encapsulates the relationship between the variables [Sykes 2000]. One of the simplest techniques to determine this relationship, is that of least squares. Least squares approximates the relationship between the variables by minimising the sum of squared distances between each data point and the trend line, thus providing a line of best fit [Tredoux and Durrheim 2002]. The method of minimising the sum of squared distance can be replaced with the minimisation of some other norm, however, this is the most commonly used technique in simple regression [Lawson and Hanson 1995]. This minimisation can either be done for only the dependant variables (ordinary least squares), or for all variables (total or geometric least squares). Thus, ordinary least squares almost always minimises the vertical offsets while total least squares minimises the perpendicular offsets in relation to the line of best fit [Lawson and Hanson 1995]. While total least squares offers a better line of fit for general data (as opposed to data investigating the relationship between independent and dependant variables), there is a significant performance cost associated with computing the geometric fit due to the addition of a second minimisation parameter. In addition, the complexity of the problem as well as computational cost increases considerably when attempting to fit multivariate equations [Lawson and Hanson 1995]. These constraints are true of multivariate regression in

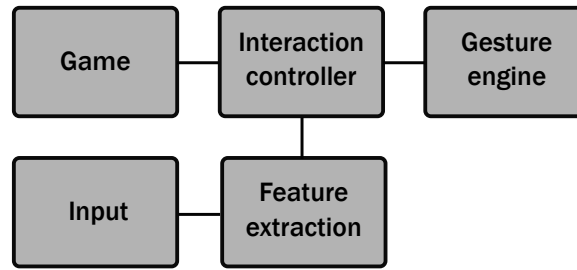


Figure 4.2: An overview of how the various gesture recognition components are arranged in relation to each other

general [Sykes 2000].

A similar problem is faced when considering non-linear regression. Unlike its linear counterpart, non-linear regression attempts to fit data using one or more independent variables and a non-linear combination of parameters [Sykes 2000]. Computing non-linear least squares is done by using a linear approximation and refining the parameters through successive iterations. The need for multiple iterations over the data greatly increases the performance cost of such a technique [Lawson and Hanson 1995].

The techniques discussed above have referred to parametric regression. That is, regression where the unknown parameters of a specific function are calculated from the data. This requires some previous knowledge of what form the data takes, as the use of an incorrect fitting function can produce highly inaccurate or spurious results. Non-parametric regression does not suffer from this problem to the same degree, as it determines both the model or function and the coefficients from the data [Sykes 2000]. While the ability to fit data to a family of functions makes non-parametric regression extremely powerful, it also greatly increases the number of data points required. This is due to the fact that both the parameters and the model itself need to be calculated from the data. In addition, the computational cost of deriving the model must be considered [Corder and Foreman 2009].

Thus while regression techniques can be extremely powerful in reducing the dimensionality of data, the computational and data requirements of each approach need to be carefully considered. That is, where possible, simpler techniques should be used to avoid the computational overhead associated with increased statistical complexity.

4.3 Design

4.3.1 Introduction

A gesture recognition unit is highly dependant on the quality of input it receives, as erroneous data can drastically skew results. For this reason care needs to be taken in designing the feature extraction unit so that it provides the necessary information required by both gesture engines. The feature extraction is required to be robust as well as consistent. In addition, the specific limitations placed on the system need to be considered. The following section discusses these design considerations as well as providing an overview of the system and the way in which features are described.

4.3.2 System overview

The feature extraction module is required to take raw touch data and reduce this to a form that the gesture engines will then interpret. This is needed for single as well as multi-touch input. Thus the components involved in the recognition of gestures include the feature extraction unit, input handling, the gesture engine as well as a controller to mediate interactions between the preceding three units. Figure 4.2 illustrates how the feature extraction relates to the gesture engines and other components.

All touch data must pass through the feature extraction module so that relevant features can be constructed and then passed to the gesture recognition unit. This needs to be done for each currently active touch. This is

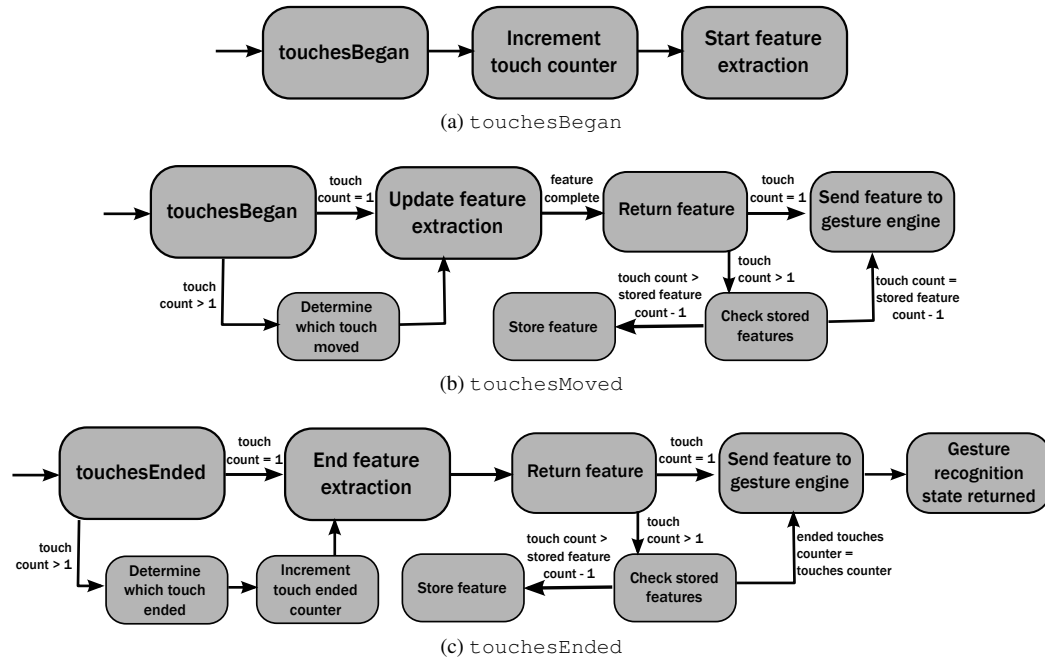


Figure 4.3: Illustration of the flow of control as each event handling method is called

handled by creating a feature extraction unit for each simultaneously active and distinct touch point. Separate feature extraction is chosen as it allows for modular design that can be scaled according to the requirements of each game. In addition, it provides a consistent interface to the gesture engine, regardless of the number of active touches. This is of importance as the feature extraction will be used by two separate and distinct gesture engines.

Three touch event handling methods are provided by the iOS SDK. These event handlers are called under the following conditions:

- `touchesBegan`: New input point is detected
- `touchesMoved`: Position of an existing touch point changes
- `touchesEnded`: Finger is lifted off the device, effectively ending touch interaction

When a touch event is detected, the active touch it corresponds to is determined. Once this is done, the touch data is passed to the relevant feature extraction unit. The feature extraction then determines whether this point constitutes the beginning of a new feature. If it does, this feature is returned. When multiple touches are active, the control unit waits until every touch has returned a completed feature before passing it to the gesture recognition engine. This is done so that the gesture engine does not have to determine or handle the temporal locality of multiple touches, thus simplifying the recognition process.

When a touch ends, if there is only one active touch, the gesture engine is polled to determine whether a gesture has been recognised. If there are multiple active touches, the controller waits until all touches have ended before sending the final features of each touch to the gesture engine. Figure 4.3 illustrates the process for each type of touch event for both single and multi-touch input.

4.3.3 Feature description

The choice of feature description was strongly mediated by the requirements of the two gesture engines. That is, features were designed so that they encompassed the data required by both gesture units.

When a feature is detected, its start point will be set to (0, 0) and the end point adjusted accordingly. This is done to simplify and normalise feature descriptions. The angle the tangent to the feature at the start point makes

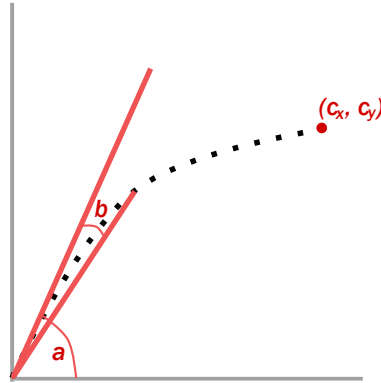


Figure 4.4: An illustration of the elements used to describe a feature. The feature is shown as the black dotted line. a is the angle made by the tangent at the start point and the x -axis; b is the angle between a and the line from the origin to the midpoint of the feature; c_x and c_y are the adjusted coordinates of the end point.

with the positive x -axis is then calculated. In addition, the angle between this tangent and the line made when connecting the start point to the midpoint of the feature is recorded. This is done to provide a more coarse-grained approximation of curvature, as the actual curvature did not show enough deviation between different features to be of use to the Hidden Markov Model gesture engine. The scale of the feature is also recorded. This is calculated by assigning the first feature a scale factor of 1. All subsequent features then use the length of the first feature to determine their scale. This allows gestures to be scale invariant. In order to calculate the length of a feature, the distance between each point within that feature is calculated and summed. Thus the data used to describe each feature is as follows:

- Adjusted end point
- Scale
- Start tangent angle
- Angle between start tangent and midpoint line

Figure 4.4 provides a graphical representation of these factors.

4.3.4 Design considerations

The nature of the system under development places a number of limitations on possible feature extraction designs. This is due to the nature of interaction as well as the fact that the feature extraction is to be implemented for the Apple iPad. A discussion of these considerations is contained below.

Device limitations

As the Apple iPad has a 1GHz processor and only 256MB of RAM [Apple 2010b], it has limited computing capabilities when compared to a conventional desktop computer. Thus care must be taken in selecting a computationally inexpensive feature extraction technique. This is compounded by the fact that the feature extraction is expected to be completed in real-time while a game is in progress. In addition, as the actual gesture recognition can only occur once features have been identified, a design that does not place unnecessary memory or computation demands on the device is required.

Finally, the rate at which touch events are detected should also be considered. When a swipe is made, touch events are fired at a rate of 60 Hz. While this provides considerable data when gestures are drawn slowly, only very

limited data is collected when they are drawn rapidly. This adds the additional constraint that the feature detection should be able to work with sparse datasets as well as dense ones.

Thus, due to device constraints, a computationally inexpensive method that is able to handle sparse data quickly and efficiently is needed.

Nature of problem

Given the variety of feature descriptions and techniques for extracting these features, a thorough understanding of the specific problem at hand is required. As the feature description is fairly simple and feature detection is unnecessary (as the input points do not need to be extracted), in this case, the problem is one of feature segmentation, rather than extraction.

That is, a method to segment the input into features in a consistent and robust manner is needed. So the crux of the problem becomes one of deciding when to end a feature and begin a new one. With this said, the method used to do this needs to be efficient, requiring minimal computation. Furthermore, as each touch point will be received separately, the solution should also be able to be computed incrementally. That is, the addition of new data points should not require recomputation of the entire dataset. This immediately disqualifies non-parametric regression techniques, as these are computationally expensive and would require much recomputation. In addition, they require a large number of input points, which is not possible with the sparse data provided. Multi-variate or total least squares and non-linear regression also present problems of computational speed. Interpolation, on the other hand, solves the wrong problem. That is, it is used to describe features, rather than segment the data. Thus, if one were to use interpolation, some additional function would be required to determine when a new feature should be started. Furthermore, as interpolation provides a function that passes through all the data points, the addition of new data would require this function to be recomputed. The use of splines would also require a secondary evaluation function to determine when to end each feature. However, as fitting splines of a low degree is not very computationally expensive, this may be a possible solution.

Thus it appears that an ordinary least squares regression or some form of spline approach would be most appropriate for this problem.

4.3.5 Feature extraction approach

As discussed above, ordinary least squares or splines appear to be the most feasible approaches for this specific problem when considering the constraints. In order to investigate which of these methods proves most effective, three feature extraction modules were developed. These include parabolic and circular least squares regression with the degree of error from the line of best fit used to determine when to start a new feature. Parabolic and circular regression have been chosen as they are more robust than straight-line fitting. In addition, they are able to handle and describe curves. A spline approach was also attempted. This uses straight-lines as splines with segmentation determined according to changes in direction.

All three of these methods allow for incremental computation. That is, the addition of new touch points will not require recomputation. In addition, the amount of computation required at each step is minimal. Thus the general design for all three methods is as follows: Touch points are sent to the feature extraction unit; Each point is then manipulated and checked against the current model and if the error between the value provided by the model and the actual point grows too large, the feature is ended and a new one started.

The details of how this works in each case have been included in the implementation section with a comparison between the various techniques described in the testing section. Overall, however, all should provide a means of implementing feature extraction and segmentation in a way that is not computationally taxing.

4.4 Implementation

4.4.1 Introduction

As the success of the gesture recognition unit is highly dependant on the quality of the data provided by the feature extraction module, three different techniques were implemented in order to determine the type of extraction that

would be used. These include parabolic and circular least squares regression as well as an approach that uses changes in direction to determine feature segmentation. These are discussed in more detail below.

4.4.2 Parabolic least squares regression

The first method attempted was parabolic ordinary least squares regression. In this case, a set of linear equations is solved in order to obtain the parameters a , b and c for the following equation: $y = a + bx + cx^2$. This equation is then used to describe a feature and its requisite information. The set of linear equations, as described by eFunda [2010], where x_i and y_i refer to the x and y coordinates of each data point, n to the number of data points, and a , b and c the parameters for the equation of a parabola, is shown below:

$$\begin{aligned}\sum_{i=1}^n y_i &= a \sum_{i=1}^n 1 + b \sum_{i=1}^n x_i + c \sum_{i=1}^n x_i^2 \\ \sum_{i=1}^n x_i y_i &= a \sum_{i=1}^n x_i + b \sum_{i=1}^n x_i^2 + c \sum_{i=1}^n x_i^3 \\ \sum_{i=1}^n x_i^2 y_i &= a \sum_{i=1}^n x_i^2 + b \sum_{i=1}^n x_i^3 + c \sum_{i=1}^n x_i^4\end{aligned}$$

These equations are solved so that a , b and c can be calculated [eFunda 2010]. Since in order to calculate all three parameters, various combinations of summated data points are used, the regression can be calculated incrementally for each new data point. That is, recalculation is not necessary when adding new data. This is done by storing the values for each summation combination. When a new data point is inserted, its values are added to the current summation value for each combination. Thus the addition of a data point is accomplished by updating the summation values and then simply inserting these into the regression equations in order to obtain the new a , b and c values.

In order to determine whether a new feature should be started, a rolling error value, with a windows size of 5 points, is kept. The size of the window was chosen as a compromise between the robustness of the measure (that is, a single outlier should not cause a new feature to be started) and the distortion caused at the end of each feature due to erroneous values being included. That is, when using a rolling error, a number of points have to have a large enough error for the overall value to be sufficient to end the feature. If the size of the window is too large, this can cause distortions in the feature description. This value is updated with the error associated with each new touch point. The error for each touch point is calculated as follows, where (x_i, y_i) is the current touch point:

$$\begin{aligned}y_{new} &= c + bx_i + ax_i^2 \\ y_{error} &= y_i - y_{new}\end{aligned}$$

When this error exceeds 5, the feature is ended and a new one started. This involves calculating the information required for each feature and resetting the summation values. The value of 5 was selected through testing the system to determine a value that would maximise the length of a feature, while minimising erroneous data at the feature's end. This value was found to be the most appropriate.

Limitations

One issue related to the use of parabolic fitting, is the inability to recognise vertical lines due to the nature of the function. That is, vertical lines produce parameter values that tend towards infinity. In order to compensate for this, these values are set to `FLT_MAX`. However, this does not adequately address the problem, resulting in features being ended too early due to widely different results caused by these inaccurate parameter values. In addition, the shape of a parabola may not be appropriate for describing all features. This leads to premature feature termination, which increases the number of features within a gesture.

In order to compensate for this, a more robust model, that of a circle, is attempted.

4.4.3 Circular least squares regression

Unlike parabolic regression, a circular model allows for vertical features to be more easily and accurately described. In order to perform a regression to fit data to the equation of a circle ($R^2 = (x - x_c)^2 + (y - y_c)^2$), the following steps are performed [Bullock 2006]. Coordinates are first transformed from (x, y) to (u, v) coordinates as follows:

$$\begin{aligned} u_i &= x_i - \bar{x} \\ v_i &= y_i - \bar{y} \end{aligned}$$

where

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad \text{and} \quad \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

The circle to be fitted has centre (u_c, v_c) , with radius R . For the purpose of the regression, α is set to equal R^2 . The linear equations to be solved to determine the centre point are:

$$\begin{aligned} u_c \sum_{i=1}^n u_i^2 + v_c \sum_{i=1}^n u_i v_i &= \frac{1}{2} \left(\sum_{i=1}^n u_i^3 + \sum_{i=1}^n u_i v_i^2 \right) \\ u_c \sum_{i=1}^n u_i v_i + v_c \sum_{i=1}^n v_i^2 &= \frac{1}{2} \left(\sum_{i=1}^n v_i^3 + \sum_{i=1}^n v_i u_i^2 \right) \end{aligned}$$

α is determined as follows:

$$\alpha = u_c^2 + v_c^2 + \frac{\sum_{i=1}^n u_i^2 + \sum_{i=1}^n v_i^2}{n}$$

Thus, once the regression has been completed, the centre point in (x, y) coordinates and the radius, R , can be calculated as shown below:

$$\begin{aligned} (x_c, y_c) &= (u_c, v_c) + (\bar{x}, \bar{y}) \\ R &= \sqrt{\alpha} \end{aligned}$$

As described in the parabolic least squares section, due to the fact that solving these equations relies on summated values, they can be calculated incrementally for each new touch point, thus avoiding recalculation when adding values. $\bar{x}$ and $\bar{y}$ are calculated anew for each touch point, however. This was done by dividing the summated values for x and y by n . Once again a rolling error with a windows size of 5 is used in order to determine whether to start a new feature. However, due to the fact that the circular regression is far more sensitive to slight deviations, the error bound was set to 15, rather than 5.

Limitations

Circular regression proved to be far more sensitive to small deviations than the parabolic regression. This meant that the parameter values could change greatly with the inclusion of one or two outliers. Thus the error bound had to be increased. However, this also meant that the end of each feature contained more erroneous points than in the parabolic version. Attempts were made to handle this by setting the end point of the feature to an earlier position, so that these erroneous points were included in the next feature, rather than the previous one. However, while this did improve the results, the sensitive nature of the regression decreases the robustness of this implementation.

4.4.4 Direction-based segmentation

As very little descriptive information is required from each feature, the third method uses changes in direction as a segmentation technique, with each feature being interpreted as a straight-line. This is done by calculating the

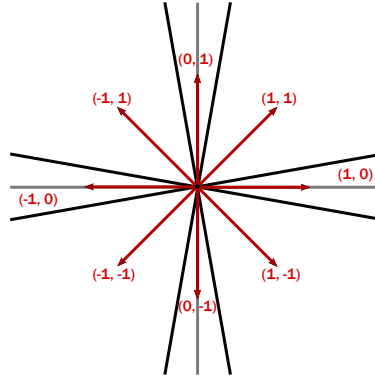


Figure 4.5: Demonstration of the eight possible directions used for segmentation. The black lines delimitate the boundaries of each direction with the red numbers indicating the assigned values in each condition.



Figure 4.6: Illustration of the technique used to combine similar sequential features. The touch points are shown in black. The red dots illustrate the starting points of the two respective features. a and b are the angles between each feature (red line) and the positive x -axis. If the difference between a and b is less than 20 degrees, they will be combined into one feature.

change in x and y between the current touch point and the one immediately preceding it. These values are then used to determine in which direction, of eight, the difference lies. Figure 4.5 illustrates the possible eight directions.

Two values are stored in order to record the current direction in x and y . Both the x and y variables can take on one of three values; 1, 0 or -1, depending on the current direction. The possible combinations are shown in Figure 4.5. In order to better capture horizontal or vertical lines, if the distance between two points in one direction is between 10 and -10, the segment is considered to lie along one of the axes.

When a new direction value is calculated, it is compared against the previous direction values. If the values are the same, the touch point is set as the current end point of the feature. If the values differ and the length of the current feature is greater than 100 pixels, the current feature is ended, and a new one started. The distance check was added so that outliers do not cause very short (and thus non-informative) features to be created.

A number of other attempts were made to help counteract the effect of outliers. One of these techniques is to extend the previous feature. When the end of a feature is found, the parameters of this feature are checked against those of the previous feature. If they are found to be similar enough, these two features are combined. This is done by checking the angle made by the line connecting the start and end points of the previous feature with the angle formed by the line from the current feature's start point to the point immediately preceding the most recent divergent touch point. If the absolute difference between these angles is less than 20 degrees, the previous feature is extended. Figure 4.6 illustrates this concept. If the difference is greater than 20 degrees, however, the current feature is ended and a new one started with the most recent direction as the current direction.

This technique allows for a simplification in the feature extraction process. In addition, due to the limited number of possible features, it provides a more robust approach, as well as simplifying the computation needed within the gesture engines.

Limitations

Due to the course-grained approach described above, this feature extraction technique is unable to differentiate between subtle changes in direction. This means that gestures have to be composed of well-defined lines and curves. The same problem is present when considering small gestures as the features are not large enough to be

consistently recognised. Thus, the use of this method places limitations on the range of possible gestures.

4.5 Evaluation

In order to determine which feature extraction technique provides the most appropriate results, an evaluation of the three was conducted. This involved tests of the reliability and consistency of each method. The following sections describe the test cases, experimental design as well as the results of testing.

4.5.1 Test cases

The assessment of the various feature extraction techniques requires choosing a number of test cases for further experimentation. These test cases should be as representative as possible of all likely gestures. For this reason, a *Z*, *U* and *Circle* gesture were chosen. The *Z* gesture contains only straight-lines, while the *Circle* is composed entirely of curves. This allows the feature extraction units to be tested on each of the basic elements that could potentially compose a gesture. In addition, since these two gestures are composed entirely of single element types, it allows for a comparison between the straight-line and curve finding capabilities of each extraction technique. The *U* gesture, on the other hand, contains a combination of curves and straight-lines. This gesture has been included so that the feature extraction units can be assessed on how well they extract a combination of elements. The use of these three gestures, thus hopefully manages to include the most important features present in the majority of gestures.

4.5.2 Experimental design

Two primary aspects, the consistency and robustness of the techniques, were assessed in our experiments. Consistency was evaluated since, for successful gesture recognition, the input into the gesture recognition engine has to be as consistent as possible for variations of the same gesture. That is, if the features extracted for the same gesture are widely different, there is very little that can be done by the gesture engine to accommodate. This is related to the second factor under evaluation. As people may draw gestures in very different ways, the feature extraction should be robust enough to extract variations of the same gesture in a similar way. The combination of a robust and consistent feature extraction unit should provide input that greatly aids the process of gesture recognition. The testing of these attributes is discussed below.

Consistency

In order to determine how consistently a gesture was recognised by each unit, each of the three test cases was drawn 75 times for each feature extraction unit. Of these 75 samples, the final 50 were selected for analysis. This was done so that the initial samples, which may have been inconsistent due to an unfamiliarity with the particular gesture, did not bias the overall results. As this experiment aims to see how consistently the feature extraction technique recognises a gesture, attempts were made to draw each test case as regularly as possible. This was aided by a sample gesture drawn on the screen over which the test cases could be traced.

Robustness

In addition to consistency, the feature extraction is also required to be robust. That is, variations of a gesture should still be recognised as the same gesture. In order to test this in a fair way, as the direction-based technique produces features of only eight varieties, the other two techniques also had their features decomposed according to mean direction. This direction was determined by using the start and end *x*-coordinates as well as the *y*-coordinate calculated through use of the regression equation. As both gesture engines use the average direction of a feature for analysis, this technique was felt to be appropriate. Three conditions, consisting of 20 samples, were analysed as a measure of robustness. These included a control condition, where gestures were drawn as consistently as possible, one where random variation was added to the drawing of each gesture, and a third, with gestures drawn smaller

	<i>Z</i>		<i>Circle</i>		<i>U</i>	
	Feature count	Number	Feature count	Number	Feature count	Number
Parabolic regression	3	50	3	17	3	1
			4	26	4	23
			5	7	5	20
					6	6
Circular regression	3	33	1	1	2	7
	4	17	2	1	3	28
			3	40	4	14
			4	8	5	1
Direction change	3	46	5	1	3	7
	4	4	6	19	4	20
			7	25	5	20
			8	4	6	3
			9	1		

Table 4.1: The distribution of results for each feature extraction technique and test case according to the number of features extracted and how often. Values in italics will be investigated further.

than normal. As in the sample generation for consistency, each test case was drawn more times than needed and the initial samples discarded.

4.5.3 Results

Consistency

The results for each gesture were analysed using Krippendorff's alpha, which is a measure of inter-rater reliability. This technique is used to determine the level of agreement between a set of coders on a number of variables. In this case, each drawn gesture was considered a rater, with the features within these gestures serving as the variables that each coder was assessing. The two regression techniques used the interval case of this metric, while the direction-based segmentation was calculated using the procedure for nominal data. As the test cases were not always recognised with the same number of features, this statistic was applied separately for each feature count, if there were more than 10 recognised gestures with this count. This minimum was selected to ensure there are enough samples to provide a reliable result as small sample sizes may produce spurious results. Table 4.1 summarises the number of features found with each count for the three test cases.

In order to apply Krippendorff's alpha, each feature had to be reduced to a single value. While this was a simple matter for the direction-based technique (as each feature is classified into one of eight directions), it proved more complex for the two regression methods. They were decomposed by calculating the mean end point for each feature and then using this value as the x -coordinate of the equation obtained through regression. Thus the produced y -coordinate was used in the inter-rater analysis. Descriptive statistics for the two regression techniques are provided in Table 4.2, although only the standard deviation has been included as the mean is only useful in the context of gesture recognition. The same has not been done for the direction-based technique as these values do not make sense when considering ordinal values.

The *Z* gesture was the most consistently recognised of the three, with the parabolic regression achieving the highest consistency ($\alpha = 0.965$). In addition, the number of features for all 50 test cases was consistent. While the direction-based method was less consistent, it still demonstrated a high degree of agreement ($\alpha = 0.797$). The circular regression, however, showed a much lower level of consistency (3 features: $\alpha = 0.251$; 4 features: $\alpha = 0.38$). This is also demonstrated when examining the standard deviations for this technique (for example for the first feature, 3 features: $SD = 2871.57$; 4 features: $SD = 3756.6$).

The inconsistency of the circular regression was not true across all test cases, however. In the case of the *Circle*, 80% of the samples were segmented into the same number of features. In addition, it had a fairly high Krippendorff's alpha value of 0.723. While the direction-based technique did not produce the same number of features as consistently, those in each sample were very highly correlated (6 features: $\alpha = 0.861$; 7 features: α

Feature number		1	2	3	4	5
		<i>SD</i>	<i>SD</i>	<i>SD</i>	<i>SD</i>	<i>SD</i>
<i>Z</i>	Parabolic	12.7	44.67	5.21		
	Circular	2871.57	730.14	2489.91		
<i>Circle</i>	Parabolic	3756.6	723.96	813.36	2515.17	
	Circular	26.44	23.16	10.54		
<i>U</i>	Parabolic	36.98	3770.67	322.03	35.8	
	Circular	34.4	47.11	87.3		
	Parabolic	2030.49	1873.02	66.043.82	116.48	
	Circular	149.93	669.67	152241.17	34.69	151.66
	Parabolic	100.18	91.99	119.98		
	Circular	39.42	84.38	57.6	63.69	

Table 4.2: Standard deviation of each feature for the two regression techniques.

= 0.921). The parabolic regression method produced interesting results. When a gesture was contained three features, the segmentation was highly consistent ($\alpha = 0.972$). However, when four features were constructed, there was no correlation across the samples ($\alpha = 0.002$).

Both regression techniques appear to have struggled with the *U* test case. The circular regression (3 features: $\alpha = 0.316$; 4 features: $\alpha = 0.448$) performed much more consistently than the parabolic regression (4 features: $\alpha = 0.018$; 5 features: $\alpha = 0$), perhaps due to the inclusion of straight-lines in the gesture (which the parabolic regression is unable to recognise). The direction-based technique was far more consistent, however (4 features: $\alpha = 0.824$; 5 features: $\alpha = 0.851$).

The mean α scores when first averaging within each test case and then across all gestures for each feature extraction technique show that the direction-based technique shows the greatest consistency ($M = 0.84$, $SD = 0.05$). The circular regression ($M = 0.51$) is moderately more consistent than the parabolic ($M = 0.49$). However, the parabolic regression ($SD = 0.48$) shows far more variance than does the circular regression ($SD = 0.19$).

Robustness

The results for each test case and extraction technique were analysed using a Two-way Main effects ANOVA. The two factors consisted of the sample a feature belonged to (*Control*, *Small* or *Random variation*) and the feature number within the gesture sequence. As the purpose of the analysis was to determine how similar the results within each sample were, the main effect for feature number and the interaction between this and the sample type was ignored (and, in fact, not calculated). In addition, it does not make sense to examine the means to determine where an effect was experienced more strongly. For this reason, only basic post-hoc testing was performed in order to determine where the differences lay, rather than the nature of these differences. Adjustments were made to handle missing data in the cases where the number of gestures per sample was not consistent.

An ANOVA was performed for each test case using all three feature extraction techniques, resulting in nine separate ANOVAs. A summary of the main effects can be found in Table 4.3. The full ANOVA tables have been included in Appendix A.

The majority of these tests showed significant results, indicating that there was significant variation in the feature extraction samples. In other words, the features were extracted significantly differently. All three test cases showed divergent results in the case of parabolic regression, while circular regression only showed significant differences for the *Circle* test case. The direction-based method reported results that were not statistically significant for the *Z* gesture only.

Post-hoc testing, using Tukey's HSD, was conducted for each significant result to determine whether this difference was present across all samples. The full results for each comparison can be found in Appendix A. Table 4.4 summarises these results by indicating where significant differences were found. The results show that in general the *Small* condition was divergent from both of the others. This is especially true of the *Circle* test case. The direction-based extraction for the *U* test case showed significant results across all groupings with the parabolic regression also showing a significant difference between the *Control* and *Random variation* condition.

		SS	df	MS	F	<i>p</i>
Parabolic regression	<i>Z</i>	19.206	2	9.603	10.398	0.000057*
	<i>Circle</i>	16.9	2	8.45	11.969	0.000013*
	<i>U</i>	46.66	2	23.330	5.5	0.004797*
Circular regression	<i>Z</i>	2.282	2	1.141	0.507	0.603396
	<i>Circle</i>	29.283	2	14.642	4.4184	0.013792*
	<i>U</i>	0.8405	2	0.8405	0.64117	0.424727
Direction-based	<i>Z</i>	2.059	2	1.029	1.689	0.187924
	<i>Circle</i>	249.403	2	124.701	114.053	< 0.000001*
	<i>U</i>	62.067	2	31.033	34.036	< 0.000001*

Table 4.3: Values for the Sample variable from the ANOVAs performed for each test case-feature extraction technique combination. A ‘*’ indicates a significance result where $\alpha = 0.05$.

	<i>Z</i>	<i>Circle</i>	<i>U</i>
Parabolic regression	Control - Random variation	Control - Small	Control - Small
	Small - Random variation	Small - Random variation	Small - Random variation
Circular regression		Control - Small	
		Small - Random variation	
Direction-based		Control - Small	Control - Small
		Small - Random variation	Small - Random variation
			Random variation - Control

Table 4.4: Summary of post-hoc testing using Tukey’s HSD. Each cell indicates where a significant difference was found and between which samples.

Further testing was conducted on the cases which showed no significant results. This was done by conducting a One-way ANOVA on the first feature of each non-significant test case. The first feature was chosen as it removes the need to handling missing data. These tests showed significant differences between all the gesture-extraction technique combinations tested (Circular: Z and $U - p < 0.000001$; Direction-based: $Z - p = 0.001434$). Post-hoc testing was also conducted on these results. This determined that the difference, in all three cases, lay between the *Small* sample and the other two samples. Full results are available in Appendix A.

4.5.4 Discussion and conclusion

The results from the two experiments allow us to determine which feature extraction technique is most appropriate based on its consistency and robustness.

The direction-based segmentation scheme is far more consistent than either of the other two techniques. While circular and parabolic regression did in some instances score higher on Krippendorff’s alpha than the direction-based method, they did not do so consistently. All alpha values for the direction-based technique, however, were above, or close to, 0.8. Thus, based on consistency alone, this method would appear to be the best technique since the other two showed far greater variation, especially in the case of the parabolic regression feature extraction.

However, robustness must also be considered. All the techniques failed the test of robustness to some extent. Much of this variation appears to be attributed to small gestures. Thus none of the techniques are appropriate for extracting gestures drawn below a certain size. This may be due to the sparseness of data, in the case of the regression methods and the lack of a large enough angle between features (due to the small size) in the direction-based case. The circular regression appears to have provided the most robust results overall, with the parabolic regression showing strong variation in all cases. However, when removing the *Small* sample from results, all three methods are fairly robust.

However, the limitations of this experiment must be considered. As the circular and parabolic regression results were converted to a form more similar to that of the direction-based technique, this may have skewed the results. In addition, as only one person was responsible for drawing the gestures used for analysis, this too may have affected the results. Finally, the fact that statistical techniques had to be used to handle missing data must also

be considered when interpreting these results.

In summary, the direction-based segmentation technique provided the most consistent results. All three techniques were fairly robust, except when considering small gestures, although the circular regression performed moderately better than the other two techniques in this respect. With this in mind, the direction-based method appears to be the best choice for integration with the two gesture engines. As all three methods provided roughly similar results on the tests of robustness, this choice is primarily based on the evaluation of consistency in extraction. The robustness results, however, suggest that attempts should be made to provide some form of training for users before they use the system. This, in turn, should decrease the divergence when drawing the same gesture. The high consistency of the direction-based technique can then be fully realised, resulting in features that are highly similar within the same gesture. This will greatly increase the probability of the gesture engines being able to accurately identify valid gestures.

Chapter 5

Experimentation

5.1 Introduction

In order to investigate the effects of interface type in touch-based gaming, a series of user experiments were conducted. This was necessary as the hypothesis under question examines a qualitative topic. That is, in order to determine which interface type is preferred, one needs to consider aspects related to enjoyment and gaming experience.

Two weeks of user testing were conducted, with each team member responsible for conducting fifteen sessions with their game. The design of these experiments is described in more detail below. In addition the results are presented in Section 5.3, with a discussion of their findings provided in Section 5.4.

5.2 Method

5.2.1 Independent and dependant variables

In order to investigate the effect of gestures on touch-based gaming, three independent variables were used. These include interface type, game genre and participant characteristics. Interface type consists of three levels, namely direct manipulation, gestural and mixed interface versions. Game genre is composed of Puzzle, Strategy and Duelling/Fighting conditions. Finally, the participant characteristics that were investigated are the effect of gaming experience and previous use of a touch-device. These will be contrasted with the results of a control group who have neither significant gaming, nor touch-device experience.

In all cases, the dependant variable was the level of flow experienced by each participant. While one could also assess presence or immersion in the context of gaming, it was felt that flow would be most appropriate because the level of sensory stimulation might not be sufficient for presence to be experienced and no well-validated measures of immersion exist. In addition, as immersion is a necessary by-product of flow, a measure of flow should encapsulate this. Finally, as the experiment aims to assess which interface version elicits the most enjoyment, and flow is said to describe a state that is “so gratifying that people are willing to do it for its own sake with little concern for what they will get out of it” [Csikzentmihalyi 1990], flow appears to be the most appropriate measure of gaming experience in this case.

5.2.2 Experimental design

The experiment consisted of a 3 (Interface type) $\times$ 3 (Genre) $\times$ 3 (Previous experience) repeated measures, two-way between subjects design. Interface type was assessed through repeated measures. That is, each participant played three versions of a game, each with a different interface type (direct manipulation, gestural and mixed). Participants were split into three mutually exclusive groups, where those in each group played a game of a different genre. Within each of these genre groups, participants were further subdivided according to various subject

characteristics. These included experience with touch-devices and gaming experience. A third group was formed as a control for these subject characteristics.

5.2.3 Instruments

Two primary instruments were used in the execution of this experiment. These include a pre-experiment questionnaire used to assign participants to experimental conditions as well as a measure of flow.

Pre-experiment questionnaire

The pre-experiment questionnaire was used to assess which game genre and participant characteristic group a subject should be placed in. In addition, it was used to determine the level of computer literacy possessed by participants. The questionnaire has been included as Appendix B. It consisted of a section gathering demographic information such as age, gender, the degree towards which the participant was studying and year of study. A 7-point Likert scale was used to assess computer literacy.

In addition to demographic information, data was collected on both the participant characteristics under study as well as gaming preferences. This was done using a number of 7-point Likert scales. In order to assess the participant characteristics, questions regarding how often subjects played games as well as the frequency with which they interacted with touch-devices and how comfortable they felt using them, were asked.

Finally, the frequency with which participants played games within each genre as well as how much they enjoyed each genre was assessed. The scores for each of these sections were used to assign subjects to experimental groups. The manner in which this was done is discussed in Section 5.2.5.

Flow assessment

The Flow State Scales, developed by Jackson et al. [2001], were used to assess flow. This questionnaire uses nine factors in order to assess an individual's flow state. It was chosen due to its relatively short administration time. In addition, the instrument has been shown to have high reliability ($\alpha = 0.83$) [Jackson et al. 2001] and internal consistency (Cronbach's $\alpha = 0.93$) [Tenenbaum et al. 1999]. Tests have also been done confirming the discriminant and construct validity of the measure [Marsh and Jackson 1999].

While multiple versions of the test exist, the choice was made to use the *Long General Flow State Scale*. The long scale was chosen as it provides a more thorough analysis of each factor. In addition, some research sheds doubt on the validity and reliability of the short scale [Tenenbaum et al. 1999]. As the questionnaire was initially developed to assess flow in physical settings, the *General* version was used as this is not specific to physical activity. Finally, the *Flow State* rather than *Dispositional* scale was used as the *Dispositional* scale requires the test to be administered during the execution of some flow-inducing activity. Due to the nature of the experiment, there were concerns over the time taken to enter a flow state. In addition, it was felt that answering a questionnaire during game-play would disrupt any flow state achieved. Thus the choice was made to administer the questionnaire after each game had been completed.

The questionnaire provided flow scores along nine dimensions, summarised in Table 5.1. In addition, these factors can be combined to create an overall flow state score. Thus the questionnaire provides nine dependant measures of flow with an additional higher-order factor. The questionnaire consisted of 36 questions that assessed the various aspects of flow using a 5-point Likert scale, ranging from *Strongly disagree* to *Strongly agree*. These questions asked participants to assess various aspects of their mental state during the task. A select portion of the questionnaire has been included as Appendix C.4.

5.2.4 Materials and setting

Each participant played three versions of the same game. Three games were developed for this purpose. These games consisted of a Puzzle, Strategy and Fighting/Duelling game. Within each game genre, three versions of the same game were developed in order to test the effects of different interaction techniques (direct manipulation, gestures and a combination of the two). Thus, nine game versions were used in total. However, subjects only played three of these games each, with all three belonging to the same game genre. Participants played the games on the

Flow dimension	Description
<i>Challenge-Skill Balance</i>	This describes an appropriate balance between the difficulty presented by a task and the ability or perceived ability of an individual to complete this task.
<i>Merging of Action and Awareness</i>	This refers to a state of deep immersion where an individual feels at one with their actions.
<i>Clear Goals</i>	A clarity of intention, aided by goals being set prior to beginning an activity.
<i>Unambiguous Feedback</i>	This refers to feedback that provides a clear awareness of how well one is progressing towards one's goals.
<i>Concentration on the Task at Hand</i>	A state where one has a single-minded focus on the task at hand, while ignoring any potential distractions.
<i>Sense of Control</i>	A feeling of total mastery over one's environment and actions.
<i>Loss of Self-Consciousness</i>	Concern for the self and other everyday issues disappears due to total concentration on the task at hand.
<i>Transformation of Time</i>	This describes a perceived distortion in the normal passage of time.
<i>Autotelic Experience</i>	This describes an experience that is intrinsically rewarding and not simply a means to an end.

Table 5.1: Summary of the nine flow dimensions measured by the Flow State Scales developed by Jackson et al. [2001].

Apple iPad. This was chosen due to its multi-touch capabilities as well as large screen size when compared to other commercially available multi-touch devices. While the Microsoft Surface has a larger screen, its prohibitive cost means that this device would not commonly be available to gamers which would reduce the ecological validity of our study to some degree.

Experiment setting

All participants completed the experiment in the same room in the Mezzanine level of the Computer Science building. The room contained only a table and the iPad, in order to minimise distractions. In addition, noise levels were kept consistently low during experimentation. Participants were placed in this room alone, with an experimenter available outside the room at all times should the participant have any questions regarding the task at hand.

5.2.5 Participants

Non-probabilistic, quota sampling was used to recruit 45 participants. Each participant was paid R20 for completing the experiment. Participants were also required to sign a consent form which explained their rights with regard to the study and also confirmed their understanding of what was required of them. This consent form is provided as Appendix C.1. All of the recruited participants were University of Cape Town students, the majority of whom were studying towards a degree in Computer Science, Engineering or a similar Science related field. Their average year of study was 2.4 ($SD = 1.48$). Of the participants, 35 were male and 10 female, with their age ranging from 18 to 29 ($M = 20.98$, $SD = 2.01$).

Of the 45 participants, 15 were assigned to each game genre. Of these 15, 5 were matched to each of the two participant characteristic conditions, with the remaining 5 serving as the control group. The number of participants was chosen based on recommendations that a ratio of at least five participants be observed for each independent variable [Bartlett et al. 2001]. The choice of how to assign participants to each group was determined through the use of the pre-experiment questionnaire. This questionnaire assessed participants' levels of gaming and touch-device experience. Table 5.2 summarises the scores for these questions within each participant characteristic group. In addition, it also provided a measure of what game genre participants enjoyed most of those being tested. This was done so that participants were assigned to a game genre they had expressed interest in, in order to prevent a simple dislike of the type of game being played from affecting results.

	<i>Touch-device experience</i>		<i>Gaming experience</i>	
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
Control group	3.2	1.26	3.8	1.24
Touch experience group	5.27	0.51	4.53	1.45
Gaming group	4	1.49	6.07	0.7

Table 5.2: Descriptive statistics of participant characteristics within each experience group across genre. All results were obtained using a 7-point Likert scale.

	<i>Touch-device experience</i>		<i>Gaming experience</i>	
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
Control group	2.4	1.14	3.2	1.1
Touch experience group	5.8	0.45	4	2.12
Gaming group	3.8	1.64	5.8	1.1

Table 5.3: Descriptive statistics of participant characteristics within the three groups that compose the Puzzle game condition. All results were obtained using a 7-point Likert scale.

An attempt was made to control for computer literacy amongst participants as differences in this variable could foreseeably influence the results achieved. This was done by recruiting participants from computer laboratories in the Computer Science building. In addition, a question was included on the pre-experiment questionnaire in order to assess this factor. Participants who did not score above the midpoint (4) on this measure were disregarded for the study. On the 7-point scale, the mean computer literacy score for all participants was 6.2 ($SD = 0.77$). No attempt was made to balance for factors such as race, age or gender as these were not felt to be particularly relevant to the topic at hand. In addition, controlling for these elements is beyond the scope of this study as it would have required a far greater number of participants.

Participant characteristics of Puzzle group

The Puzzle game group consisted of 15 participants. Of these, nine were male and six female, ranging in age from 18 to 24 ($M = 20.8$, $SD = 1.74$). Nine of these subjects were completing Computer Science related degrees with the remaining participants studying other Science disciplines or Accounting. The year of study ranged between 1 and 4 ($M = 2.4$, $SD = 0.99$). All individuals rated themselves as being highly computer literate on the 7-point scale ($M=6.2$, $SD = 0.77$). Table 5.3 summarises the scores on the two participant variables within the gaming experience, touch-device experience and control groups. Participants were assigned to this group due to their expressed interest in puzzle games. The mean score when subjects were asked how much they enjoyed puzzle games was 5.8 ($SD = 1.4$).

5.2.6 Control procedures

Repeated measures was used to assess interface type as it removes the need for matching between groups. However, using a repeated measures design introduces a number of caveats that must be handled. These can be classed as order effects and include fatigue, contrast and practice effects. The literature suggests using either counterbalancing or allowing enough time between each condition so that the order effects are reduced [Cozby 2005]. An attempt was made to use both techniques. That is, the order with which participants complete the experimental conditions was varied through randomisation. In addition, participants were required to complete the Flow State questionnaire between each condition. This increased the amount of time between each condition. A distractor task was not used as it would have increased the overall length of the experiment dramatically, perhaps introducing fatigue effects.

A tutorial session was included so that the effect of unfamiliarity with either gestures, or touch-devices in general, could be minimised. In this session, participants were instructed to draw the gestures that would be used in the game a certain number of times, with the tutorial only progressing once the gesture had been successfully recognised. This served to familiarise participants with both the system and this style of interaction.

While this session may have reduced between-group differences with regard to touch-device experience, it was felt to be necessary as difficulty in using or understanding how the gestural interface worked would have severely compromised the degree to which flow was elicited in this condition. In addition, the practice session should not have proved sufficient to provide participants with enough experience with touch-devices for this participant characteristic to have been strongly influenced.

The experimental system was designed so as to minimise experimenter bias. Instructions on how to complete the tasks were given to participants in written form and included in the games. This was done to decrease the degree of experimenter-participant interaction. More importantly, the randomisation and execution of each game was entirely automated. That is, a system was written to link all three games, the tutorial session and all instructions together so that no experimenter interaction was required throughout the experiment. As the experiment does not lend itself to single- or double-blind conditions, since participants are required to be aware of which condition they are completing in order to play the game, the automated system should remove the majority of experimenter bias. Attempts were also made to keep experimenter behaviour consistent before the experiment began. Finally, in order to reduce the number of demand characteristics, the purpose of the study was not explained to participants until they had completed it. This was done as any prior knowledge of the hypothesis under question may have influenced participants to attempt to either confirm or disconfirm it.

Finally, a number of situational effects were also considered. Participants all completed the study in the same setting with minimal distractions or external elements. In addition, in order to prevent the Hawthorne effect, participants were not observed during the experiment. That is, all participants played the game in a separate room. The experimenter was present outside this room should they have had any questions. The choice to not watch participants (rather than to watch all participants) was made as observation may decrease subject performance due to nervousness and other such factors.

5.2.7 Procedure

The following section describes the experimental procedure followed for each participant. Participants were seated and asked to sign a consent form and complete a demographic information sheet. They were then given written instruction on how to complete the experiment. These forms can be found in Appendix C.1, C.3 and C.2, respectively.

The experiment consisted of seven steps. Initially participants completed a tutorial session where they were asked to practice drawing the gestures that would be used in the game. After this they were instructed on how to play the first interface version of the game. Game-play lasted seven minutes. This time frame was controlled by a timer within the game. After this time had elapsed, game-play was interrupted and the participant was instructed to complete the first questionnaire. This process of instructions, seven minutes of game-play and then filling out the questionnaire was repeated for each of the other two game versions. Once all three games had been played, the participant was informed that the experiment was over and thanked for their participation. The entire process was automated and required no experimenter interaction. Once the participant had completed the experiment, they were paid R20 and given a sheet explaining the purpose of the experiment. This form is available in Appendix C.5. They were also able to ask any questions they may have had regarding the topic under study. The entire procedure took roughly 45 minutes to complete.

5.2.8 Pilot study

A pilot study was held in order to assess the overall experimental procedure. Five participants were recruited from the Computer Science Honours class. Each participant played through the combined system which linked together all three game versions with the tutorial screen and timing and logging code. The participants also completed the questionnaire between each game version. This was done so that an approximation of total experiment time could be made. Initially each game was played for ten minutes. However, upon reviewing total experiment time, the decision was made to reduce this to seven minutes per game, as the experiment was initially felt to be too lengthy. The pilot study also allowed the logging and randomisation code to be checked. Finally, participants were asked to provide comments on the thoroughness of the in-game instructions and tutorial text, as well as any general comments they may have had on the experimental procedure. Following this, small adjustments were made to

Effect	SS	df	MS	F	<i>p</i>
Intercept	136549.2	1	136549.2	2934.263	< 0.000001*
Genre	4.6	2	2.3	0.050	0.951663
Group	97.2	2	48.6	1.044	0.362406
Genre*Group	327.8	4	82.0	1.761	0.158147
Error	1675.3	36	46.5		
Interface	83.1	2	41.5	2.652	0.077416
Interface*Genre	218.8	4	54.7	3.493	0.011565*
Interface*Group	45.8	4	11.4	0.731	0.573833
Interface*Genre*Group	96.5	8	12.1	0.771	0.629683
Error	1127.7	72	15.7		

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

Table 5.4: Results of two-way (Genre-Participant characteristics) repeated measures (Interface) ANOVA conducted on the higher order flow factor results.

the tutorial screens to make them more informative. The overall design of the experiment remained unchanged, however.

5.3 Results

As described above, three independent variables (one of which was a repeated measures factor) and their interactions were analysed. These were assessed using nine dependant flow variables which can be aggregated into one higher-order factor. In order to analyse the results, a two-way repeated measures ANOVA was used for the higher-order factor, while a two-way repeated measures MANOVA was constructed with the results along the nine flow dimensions. The following sections describe the results of these analyses. Important findings are presented in bold.

5.3.1 Higher-order flow factor analysis

A two-way repeated measures ANOVA was conducted on the results of all three game genre groups. This ANOVA also considered the data from the three participant characteristics groups. The full results can be seen in Table 5.4. Mauchly's sphericity test was conducted to ensure normality within the data. The results of this test showed no deviations from the normality of variance assumptions required by ANOVA and so no correcting factors were used ($p = 0.360843$).

The only significant effect found was an interaction between *Interface type* and *Genre* ($p = 0.011565$, $\alpha = 0.05$). Tukey's HSD test was conducted on this interaction, but found no significant differences. A Fisher LSD test was then performed. The differences found are summarised in Table 5.5. However, these results should be interpreted cautiously as this test has been criticised for not sufficiently controlling for Type I errors [Stevens 1999]. Full results for both the Tukey HSD and Fisher LSD tests can be found in Appendix D.1. Figure 5.1 plots the cell means with error bars for this interaction. By comparing cell means along with the results of the post-hoc testing, one can see that within the Strategy game genre, the direct manipulation interface version is preferred to the mixed version ($M = 33.8$ and $M = 29.683$, respectively). Within the Fighting/Duelling condition, the gestural version elicits more flow than the direct manipulation version ($M = 34.31$ and $M = 30.3$, respectively). The results of the other significant interactions will not be discussed further as they are across both game genre and interface type. Thus with regard to the higher-order factor, significant differences in the degree of flow experienced according to interface type were only present in the Strategy and Fighting/Duelling conditions.

Factor 1		Factor 2		p
Genre	Interface type	Genre	Interface type	
Puzzle	Direct manipulation	Fighting/Duelling	Gestural	0.029672*
Strategy	Direct manipulation	Strategy	Mixed	0.005719*
Strategy	Mixed	Fighting/Duelling	Gestural	0.014762*
Fighting/Duelling	Direct manipulation	Fighting/Duelling	Gestural	0.006940*

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

Table 5.5: Summary of significant results of Fisher LSD conducted on the higher-order flow factor results.

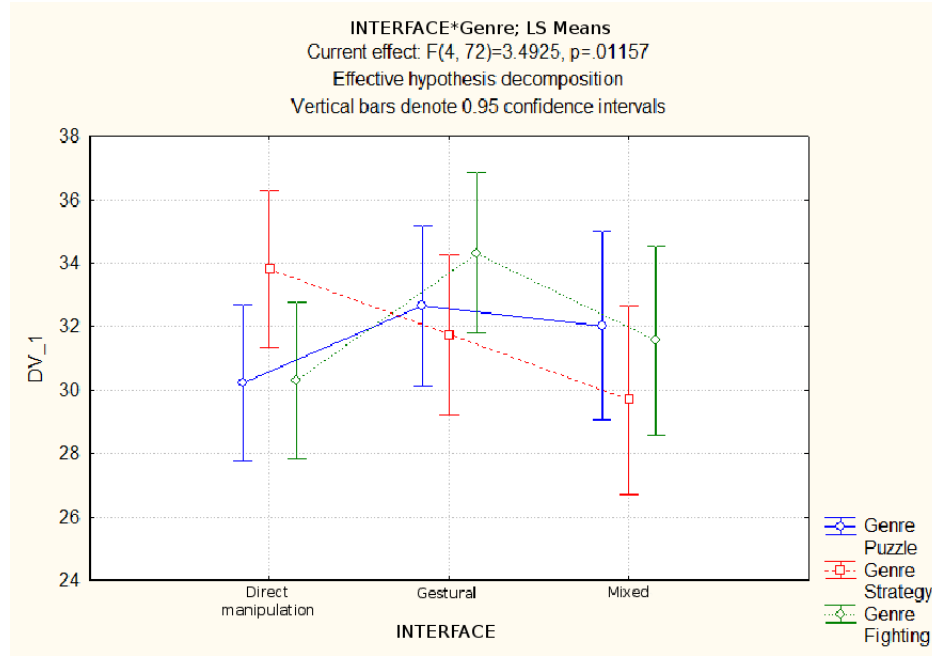


Figure 5.1: Cell means plot for *Genre*Interface* interaction effect of higher-order flow factor ANOVA.

5.3.2 Flow dimensions analysis

The results across all nine flow dimensions were analysed using a two-way (Genre-Participant characteristic group) repeated measures (Interface) MANOVA with nine dependent variables. Mauchly's sphericity test was conducted to ensure normality. Full results for the test can be found in Appendix D.2.1. The assumption of normality was found to hold true for all dependent variables apart from *Concentration on the task at hand* ($p = 0.035$) and *Transformation of time* ($p = 0.01$). The Greenhouse-Geisser estimate was used to correct for this in the case of these two variables.

The results of the MANOVA can be seen in Table 5.6. The reported p values refer to the results obtained using Wilks' lambda distribution, which is a multivariate generalisation of the F -distribution [Tredoux and Durheim 2002]. An α value of 0.05 was used throughout. All main effects were found to be significant (Genre: $p = 0.006511$; Group: $p = 0.043880$; Interface: $p = 0.000005$). In addition, the interaction between each pairwise combination of independent variables was also found to be significant (Genre*Group: $p = 0.006588$; Interface*Group: $p = 0.025903$; Interface*Genre: $p = 0.004767$). However, the overall interaction between all three independent variables was not found to be significant ($p = 0.644534$).

In order to better understand the nature of these interactions, post-hoc testing was conducted for each significant interaction along all dependent variables. A summary of these flow factors and their meaning can be

Effect	Value	F	Effect df	Error df	<i>p</i>
Intercept	0.009898	311.2176	9	28.0000	< 0.000001*
Genre	0.318534	2.4013	18	56.0000	0.006511*
Group	0.396239	1.8313	18	56.0000	0.043880*
Genre*Group	0.157573	1.8885	36	106.6663	0.006588*
Interface	0.099285	9.5760	18	19.0000	0.000005*
Interface*Genre	0.094389	2.3802	36	38.0000	0.004767*
Interface*Group	0.126765	1.9091	36	38.0000	0.025903*
Interface*Genre*Group	0.087785	0.9168	72	77.0695	0.644534

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

Table 5.6: Results of two-way (Genre-Participant characteristics) repeated measures (Interface) MANOVA conducted on the nine flow dimensions. Wilks' lambda distribution test is used.

found in Table 5.1. In addition, univariate repeated measures tests for each dependent variable were conducted. The full results of all these tests are contained in Appendix D.2.2 and Table 5.7 contains a summary of our findings. The following sections will describe the results of these tests according to these interactions and main effects.

*Interface*Genre*

Tukey's HSD test was conducted on this interaction for each dependent variable. In addition, univariate testing showed this interaction as significant for the *Unambiguous Feedback* and *Loss of Self-Consciousness* dimensions. The results discussed below refer to the findings of these tests. In the discussion only those results with some meaningful interpretation are mentioned. That is, significant differences between both different interface types and genres are not elaborated on.

A significant effect was shown between the direct manipulation version of the Puzzle game and all other game combinations except the Fighting/Duelling direct manipulation interface along the *Challenge-Skill Balance* dimension. In all cases, the mean for the direct manipulation Puzzle game was lower than the other combinations. This suggests that this condition elicited the lowest degree of flow along this dimension. In addition, the same was true of the direct manipulation Fighting/Duelling version when compared to the Fighting/Duelling gestural and mixed interfaces (Gestural: $p = 0.035257$; Mixed: $p = 0.023826$). No meaningful significant results were obtained for the *Merging of Action and Awareness* dimension.

With regard to the results of post-hoc testing along the *Clear Goals* dimension, the only significant result of note is the lower score of direct manipulation in the Fighting/Duelling condition when compared to this genre's other two interface types (Gestural: $p = 0.001737$; Mixed: $p = 0.002781$). *Unambiguous Feedback* showed a significant interaction between *Interface* and *Genre* in the univariate testing. Post-hoc testing was also conducted for this variable. The only meaningful significant results was a difference between the Fighting/Duelling direct manipulation and gestural versions ($p = 0.000921$). A cell mean plot indicated that the gestural version elicited more flow along this dimension than the direct manipulation version.

Post-hoc testing did not reveal any significant results for either *Concentration on the Task at Hand* or *Sense of Control* for this interaction. *Loss of Self-Consciousness* showed significant results in both Tukey's HSD and the univariate testing. The mixed interface version of the Strategy game was shown to produce less *Loss of Self-Consciousness* than either of the other two interface versions (Gestural: $p = 0.016831$; Direct manipulation: $p = 0.000233$). The *Transformation of Time* and *Autotelic Experience* dimensions showed a significant difference between the Strategy direct manipulation and mixed versions. For both variables, the direct manipulation interface elicited more flow than the mixed one.

When examining the cell means plots of the interaction across the nine dependent variables, in general, all interactions are ordinal. That is, they have simple effects that lie in the same direction [Tredoux and Durrheim 2002]. This ordinal interaction holds true more so for the Strategy and Fighting/Duelling games than for the Puzzle and either of the other two. Overall, when considering these graphs, the majority of disagreement between genres seems to lie with the mixed interface. When disregarding this condition, the disordinal effects are only introduced by the Strategy game. Thus it appears that there is more overall agreement on interface preference

Factor	Condition 1					Condition 2				<i>p</i>
	Interface	Genre	Group	<i>M</i>		Interface	Genre	Group	<i>M</i>	
Challenge-Skill Balance	DM			11.622	<	G			15.289	0.000111*
	DM			11.622	<	M			14.933	0.000111*
	DM	P		9.333	<	G	P		14.067	0.000293*
	DM	P		9.333	<	M	P		13.200	0.004324*
	DM	F		12.400	<	G	F		15.600	0.000152*
	DM	F		12.400	<	M	F		15.733	0.000145*
	DM		G	11.467	<	G		G	15.667	0.001410*
	DM		T	12.067	<	G		T	16.400	0.000908*
	DM		T	12.067	<	M		T	16.667	0.000405*
Merging of Action and Awareness	DM			13.667	<	G			15.622	0.001201*
	DM			13.667	<	M			15.511	0.002278*
Clear Goals	DM			12.578	<	G			15.356	0.000114*
	DM			12.578	<	M			15.200	0.000121*
	DM	F		12.467	<	G	F		16.400	0.001737*
	DM	F		12.467	<	M	F		16.267	0.002781*
	DM		G	12.000	<	G		G	15.667	0.000855*
	DM		G	11.533	<	M		G	15.200	0.004454*
Unambiguous Feedback	DM	F		10.867	<	G	F		15.267	0.000921*
Concentration on the Task at Hand	DM			13.267	<	G			15.511	0.005088*
Sense of Control	DM			12.467	<	G			14.889	0.000209*
	M			13.356	<	G			14.889	0.018505*
	DM		G	11.533	<	G		G	15.133	0.039190*
	DM		T	11.800	<	G		T	15.600	0.004790*
Loss of Self-Consciousness	M	S		10.333	<	G	S		14.600	0.016831*
	M	S		10.333	<	DM	S		16.333	0.000233*
Transformation of Time	M			13.156	<	DM			15.556	0.001579*
	M	S		11.133	<	DM	S		15.067	0.025650*
Autotelic Experience	G			12.444	<	DM			14.778	0.003083*
	M			12.156	<	DM			14.778	0.000863*
	M	S		9.333	<	DM	S		13.933	0.006637*
	G		G	10.667	<	DM		G	14.667	0.030455*

Key:

Interface: DM - Direct manipulation, G - Gestural, M - Mixed

Genre: P - Puzzle, S - Strategy, F - Fighting/Duelling

Group: G - Gaming experience, T - Touch-device experience, C - Control

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

Table 5.7: Summary of all Tukey HSD post-hoc testing and univariate results decomposed according to *Interface*, *Genre* and *Group*. Only meaningful significant results are included above. Condition 1 and 2 refer to the two combinations of *Interface*, *Genre* or *Group* that produced a significant difference. Blank column entries for a row mean this factor is not included in the combination. The inequality indicates the nature of the significant result, as can be seen by comparing their means.

between the Strategy and Fighting/Duelling games, but that disordinal interactions are present between Strategy and the other two genres when considering only the direct manipulation and gestural versions. Most importantly, no significant differences were found between the same interface type and two or more different *Genres* across any of the variables. **In addition, the nearly parallel nature of the mean plots for each genre across a number of dimensions, suggests that while some interface version may be significantly preferred within a single genre, there are no significant overall inter-genre difference with regard to interface preference.**

Interface*Group

No significant univariate results were found for this interaction. Tukey's HSD test, however, revealed a number of significant effects between conditions of this interaction. In general, no significant or meaningful results were found for *Merging of Action and Awareness*, *Unambiguous Feedback*, *Concentration on the Task at Hand*, *Loss of Self-Consciousness* or *Transformation of Time*. Once again, only the meaningful differences are discussed below.

Challenge-Skill Balance showed a significant difference between the Gaming experience direct manipulation and gestural versions ($p = 0.001410$). Significant differences were also found within the Touch-device experience group between the direct manipulation interface and both of the other versions (Gestural: $p = 0.000908$; Mixed: 0.000405). In all three cases, the direct manipulation interface elicited less flow along this dimension than the other interfaces. Post-hoc testing on the *Clear Goals* variable showed the direct manipulation interface produced significantly less flow than either of the other two versions (Gestural: $p = 0.000855$; Mixed: $p = 0.004454$). The direct manipulation version was also shown to elicit significantly less flow along the *Sense of Control* dimension than the gestural version for both the Gaming and Touch-device experience groups ($p = 0.039190$ and $p = 0.004790$, respectively). Finally, in the *Autotelic Experience* condition, the gestural interface produced less flow than the direct manipulation version ($p = 0.030455$).

Ordinal interactions between the Control and Touch-device experience groups were present along all dimensions, except *Sense of Control*. In addition, in many of the cell mean plots, the interaction between the Control and Touch-device experience groups is almost parallel between some or all of the interface versions. This, along with the results of the post-hoc testing suggest that there may be very little difference between these groups. The same was not true between Gaming experience and either of the other two groups, where a disordinal interaction between one or both of the other groups was present along every dimension. This suggests some difference between how flow is experienced within the Gaming experience group and the other two groups. The post-hoc testing supports this to some degree, showing significant differences between the Gaming experience conditions and those of the Touch-device experience and Control groups. However, none of these differences lie across the same interface version. Multivariate contrast analysis was used to assess this further. This showed a significant difference between the Gaming experience and Control groups ($p = 0.014123$). When comparing means for these two groups both on a variable by variable basis, as well as overall, the Gaming experience group appears to experience more flow than the Control group. While no significant difference was found between Gaming and Touch-device experience, a significant difference was present between the Control and Touch-device experience groups ($p = 0.035751$). When comparing per-variable means, the Touch-device experience group seems to experience more flow along all dimensions except *Concentration on the Task at Hand* and *Sense of Control*. The overall means also suggest that more flow is elicited in those with touch-device experience than without. **This suggests that both gaming and previous touch-device experience increase the degree to which a flow state is experienced.**

Interface main effect

Univariate testing revealed a significant *Interface* main effect for all dependent variables. In addition, a significant interaction between *Interface type* and *Genre* was found for *Unambiguous Feedback* and *Loss of Self-Consciousness*. For this reason, the main effect in these two cases will not be investigated further. Tukey's HSD was performed on the *Interface* main effect for all other dependent measures.

Significant differences were found between the direct manipulation interface and both the gestural and mixed interfaces for *Challenge-Skill Balance* ($p = 0.000111$ in both cases), *Merging of Action and Awareness* (Gestural: $p = 0.001201$; Mixed: $p = 0.002278$), *Clear Goals* (Gestural: $p = 0.000114$; Mixed: $p = 0.000121$) and *Autotelic Experience* (Gestural: $p = 0.003083$; Mixed: $p = 0.000863$). In the case of the first three variables, the cell means plots indicate that the direct manipulation version elicited less flow along these dimensions than either the gestural or mixed interface versions. The direct manipulation interface elicited more flow along the *Autotelic Experience* dimension than the other two interface types, however.

The *Concentration on the Task at Hand* dimension was shown to produce a higher flow effect in the gestural interface when compared to the direct manipulation interface ($p = 0.005088$). In the case of *Sense of Control*, the gestural interface was shown to elicit this dimension more highly than either of the other two interfaces (Direct manipulation: $p = 0.000209$; Mixed: $p = 0.018505$). Finally, the direct manipulation interface elicited more flow along the *Transformation of Time* dimension than the mixed interface ($p = 0.001579$).

Multivariate contrast analysis was used to attempt to collapse these results across all dependent variables. This provided significant results for all possible interface pairwise combinations (Direct manipulation - Gestural: $p = 0.000002$; Direct manipulation - Mixed: $p = 0.000012$; Gestural - Mixed: $p = 0.033951$). **Overall means for these three conditions show that more flow is elicited in the gestural condition than in either of the other two and that the mixed interface produces more flow than the direct manipulation interface.**

5.3.3 Puzzle game results

Multivariate contrast analysis was used to determine the effects of *Interface* and *Group* within the Puzzle game genre. This was chosen instead of a separate MANOVA as there would then be more dependent variables than subjects in each of the subject participant groups which would decrease both the power and validity of the analysis.

The results of the contrast analysis for *Interface* show significant differences between the gestural version and both the direct manipulation and mixed interfaces (Direct manipulation: $p = 0.000031$; Mixed: $p = 0.014434$). When examining the overall means for these conditions, they show that the gestural version elicits more flow than either the direct manipulation or mixed versions. The interaction between *Interface* and *Group* was also subjected to Multivariate contrast analysis. The only significant effect was found between the Gaming and Control conditions ($p = 0.008096$). Cell means for these two conditions suggest that those with significant gaming experience more readily experience flow than those with limited gaming experience.

Both of these contrast analyses provide findings that correspond with those found across all three genres. However, the lack of any difference between the mixed and direct manipulation interfaces, as well as between Touch-experience and Control groups, may suggest that these differences are introduced by the other genres.

5.4 Discussion

The results provided through the experimentation described in the previous section allow us to determine the effect of interaction style on the overall level of enjoyment, or flow, experienced while playing a game. In addition, the investigation of genre and the subject characteristics of gaming and touch-device experience allow for a greater understanding of the contributing factors related to this experience. This section will discuss and contrast the results obtained with regard to determining whether there is a difference in the level of flow experienced when using a gestural, direct manipulation or mixed interface. In addition, the effect of game genre and subject characteristics on this factor will be reviewed. Finally, a discussion of the results specific to the Puzzle game genre will be included.

5.4.1 Effects of game genre on interface choice

The significant interaction between interface and game genre suggests that the genre does effect the degree to which flow is elicited by each interaction style. However, the nature of this effect is less clear. All of the data showing a significant interaction between interface type and genre either referred to some effect between a genre-interface combination and another interface type within the same genre or a combination that included a different interface and genre type. In other words, no significant interactions were found between genre across the same interface type. In addition, when a significant difference between two interface types within the same genre was found, either other genres showed the same differences or no other genre showed a significant difference between any of their interface types. This is illustrated in the *Challenge-Skill Balance* dimension, where within both the Fighting and Puzzle games, significant differences were found between direct manipulation and the other two interfaces. These differences were in the same direction, however. All other flow factors which show significant differences between interface types, exhibit these differences within one genre only, while the other genres show no significant between-interface differences. Furthermore, when a significant difference was found between two interfaces within a genre, both the interfaces identified and the direction of the interaction were corroborated by other significant findings within the *Interface* main effect as well as the *Interface*Group* interaction.

However, a noticeable discrepancy is present when considering the results of this interaction for the higher-order factor. That is, the direct manipulation interface appears to elicit more flow than either of the other two versions for the Strategy game. This contradicts the findings of a number of individual flow factors. While the difference between the direct manipulation and gestural interface is not significant, direct manipulation appears to elicit significantly more flow than the mixed interface along this higher-order factor. As this factor is an aggregate of the nine individual flow factors, this may be due to the significant differences found between these interface version within the *Loss of Self-Consciousness*, *Transformation of Time* and *Autotelic Experience* dimensions. These significant differences appear to have greatly affected the results of the Strategy game with regard to the higher-order factor. However, as this factor is a combination of the results of the individual flow factors, it may

be more appropriate to examine their results, rather than their aggregate, in order to obtain a more fine-grained understanding of how genre affects interface preference.

The ordinal interaction present across most flow dimensions, also suggests that the effect of genre is consistently in the same direction. In other words, the interface versions elicit similar levels of flow across game genre. In addition, the almost parallel nature of these lines would suggest that this interaction is fairly weak. This ordinal interaction is especially true when considering the Strategy and Fighting/Duelling games. The nature of the Puzzle game may explain why it does not as closely match the means of the previous two genres. That is, it may be more difficult to elicit flow, or the levels along which flow was elicited may have varied, with such a game, when compared to Strategy or Fighting/Duelling games. One of the characteristics present in the other two genres but not in the Puzzle game, were real-time constraints placed on the user during game-play. In the Puzzle game, unlike in the other two game genres, where issues of timing had to be considered, the user was able to spend as long as they liked on any particular element. This may have led to the participant either not experiencing flow as strongly or simply eliciting flow along different dimensions to those of the other two games. Similar reasoning may be used to explain the disordinal interaction present for the mixed condition when considering the Strategy game. Furthermore, when removing the mixed condition, there is agreement in the distribution of means between all three genres. That is, the interactions between direct manipulation and the gestural interface are ordinal across the three genres for all flow factors when considering the results of the MANOVA. The discrepancies introduced by the mixed interface version may be due to the differences in the ways in which direct manipulation and gestural commands were combined within each genre. That is, while in the other two interface versions all commands were either gestural or direct manipulation, the manner in which these two interaction techniques were combined in the mixed version may have resulted in additional variance related to their combination rather than the actual genre.

It should also be noted that any discrepancies may be specific to the actual games, rather than the genre as a whole. As there is considerable variation in game design, even within a game genre, additional testing would be required to definitively say that these differences are due to game genre as a whole, rather than just the specific attributes of these individual games. Thus, it appears that, while genre does affect the degree to which flow is experienced, this effect is not easily explained. Genre appears to accentuate or dampen the degree to which flow is experienced within interface types along specific flow dimensions. However, these variations are not large or consistent enough to suggest that genre plays a definitive role in which interface type is preferred. Rather, the presence of a significant interaction seems to suggest that genre effects the degree to which flow is experienced within the larger overall direction shared by all genres, rather than serving as a primary contributing factor in the preference of interaction technique.

5.4.2 Interaction between participant characteristics and interaction style

The effect of previous gaming and touch-device experience was explored in relation to interaction style preferences. A significant interaction was found between these two variables. However, as with the *Interface*Genre* interaction, understanding the nature of this interaction is made more complex by the number of dependent variables that were assessed. As with the *Interface*Genre* interaction, no significant differences were found within an interface type across the participant characteristic groups. However, differences were found within specific groups between different interface versions, but this was only present in the Gaming and Touch-device experience conditions. In other words, the Control group showed no significant differences between interface types.

Where significant differences were found, both the nature and direction of these differences were consistent with the other significant interactions found for that flow dimension. In addition, when differences were present between interface types within both the Gaming and Touch-device experience groups for a specific flow factor, the direction and interface types affected were shared. However, significant results for the Touch-device experience group were only obtained for flow dimensions where the same was found for the Gaming experience condition. The converse was not always the case: the gaming experience group showed far more significant between-interface differences than either of the two other groups. As a significant result implies that the means of the two conditions were sufficiently far apart, this may mean that those in the Gaming experience condition experienced or did not experience flow more strongly than those in the other two experimental groups. That is to say, when a flow dimension elicited large amounts of flow, those in the Gaming experience group experienced this to a greater degree than those in the other two groups, while the same is true of the converse. That is, when flow was only

weakly experienced, those in the Gaming experience group experienced less flow than individuals with only limited gaming experience. Essentially, gamers seem to experience an amplification of flow both positively and negatively. This may be due to a number of factors. Firstly, due to previous gaming experience, these individuals may be able to more readily enter a flow state than those in the other two conditions. In addition, they may thus be more sensitive to elements that disrupt this flow state. A second explanation relates to personality factors that predispose certain individuals to being more amenable to entering a flow state [Csikzentmihalyi 1990]. That is, those with gaming experience may have a personality type that predisposes them to entering a flow state more readily, and because of this, enjoy playing games more so than those who lack this trait. This in turn may explain why they play games more often than other individuals and also why they are more sensitive to variations in flow when compared to those in the Touch-device experience or Control groups. However, further testing would be required to definitively accept either hypothesis as the cause of this variation.

The Multivariate contrast analysis showed significant differences between the Control and other groups. The mean comparison for this test seems to suggest that possessing either Gaming or Touch-device experience does increase the degree of flow experienced overall. That is, individuals who possess these participant characteristics more readily enter a flow state than those who have both limited gaming and touch-device experience, regardless of interface type. However, as those in the Touch-device experience group had more gaming experience (as assessed by the pre-experiment questionnaire) than those in the Control group, gaming experience may still be the active factor in this case. Without further experimentation, however, it is not possible to further separate these variables. With this said, the results are not surprising overall, as an unfamiliarity with either an interaction technique or an application-type may cause frustration as well as feelings of confusion, which in turn, would decrease the amount of flow experienced, regardless of interface type.

Thus, while interface preference appears to have been fairly consistent both across groups, as well as when compared to the significant results for the *Interface* main effect and *Interface*Genre* interaction, participant characteristics do appear to have influenced the flow results. It seems that those with touch-device or gaming experience are more likely to enter a flow state of a higher degree across most flow dimensions, regardless of interface type. In addition, individuals who regularly play games appear to be more sensitive to overall variations present for each interface type within the flow dimensions. Thus, when considering interface type, participant characteristics do appear to influence the degree of the flow state experienced, rather than its specific nature.

5.4.3 Flow and the effect of interaction technique

Due to the nature of the data, it is fairly difficult to make clear statements with regard to which interaction technique elicits more flow. Despite this, a number of patterns seem to have emerged. As the nature of the interactions between *Interface* and both *Genre* and *Group* have been reviewed above, only the results specific to differences in flow between the various interfaces (including when these differences contain interactions with other variables) will be discussed in this section.

A number of dimensions indicated a clear preference for the gestural and mixed interface over the direct manipulation version. These include *Challenge-Skill Balance*, *Merging of Action and Awareness* and *Clear Goals*. The preference in the *Challenge-Skill Balance* dimension may indicate that the addition of gestural controls affects the level of difficulty present in playing the games. However, without further testing it is not possible to specify whether this increases or decreases the difficulty of the game. The results for *Merging of Action and Awareness* seem to suggest that interfaces containing gestural elements increase the degree to which immersion is experienced. This is to be expected, in a sense, as the combination of a touch-based interaction technique with more fluid gestural controls may remove some of the perceived distance present between the user, device and application. The explanation for the preference of the gestural and mixed interfaces in the case of *Clear goals* is less apparent, however. As all interface versions shared the same goals within each game genre, this difference seems to suggest that these goals were more apparent when using gestural or a combination of gestures and direct manipulation. Finally, the mixed version does not elicit a significantly greater flow state along any other flow dimensions. The presence of differences only in cases where the same was found between the gestural and direct manipulation interfaces suggests that the source of the higher flow state in the mixed version may be the inclusion of gestural commands.

The gestural interface was shown to elicit more flow than the direct manipulation interface along the *Un-*

ambiguous Feedback, *Concentration on the Task at Hand* and *Sense of Control* dimensions. This result, in the case of *Unambiguous Feedback* and *Sense of Control*, is unexpected. Due to the general unfamiliarity of users with regard to gestures, one might expect that the direct manipulation interface would provide users with more of a feeling of control. The mixed interface was also shown to elicit significantly less flow than the gestural version along this dimension. In this case, one interpretation is that the use of direct manipulation controls within this interface may have negated any positive effects associated with the included gestural controls. A second explanation is that the inclusion of direct manipulation controls as well as gestural ones may have confused users. With regard to *Unambiguous Feedback*, as the feedback given when a command was executed was the same across interface type, this difference suggests that the use of gestures enhanced the degree to which users were able to interpret this information. However, the mechanism behind this is unclear and would require further testing. Finally, as gestures have been shown to increase the cognitive load placed upon users [Nielsen et al. 2004], they may have had to concentrate more in the gestural version than in the other two in order to successfully interact with the game. This would potentially explain the higher flow state experienced in the gestural version when considering *Concentration on the Task at Hand*.

The *Loss of Self-Consciousness* dimension showed a significantly lower flow score for the mixed interface than either of the other two versions. No significant interaction was found between the direct manipulation and gestural interfaces, however. This suggests that the combination of interaction techniques plays some role in decreasing the degree to which users are able to disassociate from themselves and immerse themselves in the game-play. Once again, the reasons for this are unclear and a deeper understanding of this phenomenon would require further investigation. A similar effect was found for *Transformation of Time*, except in this case only the direct manipulation interface was shown to elicit significantly more flow than the mixed version. Again, the potential confusion and possible additional cognitive load present when combining multiple interaction techniques should perhaps be considered.

Finally, with regard to *Autotelic Experience*, direct manipulation elicited significantly more flow than either the gestural or mixed interfaces. This is surprising, as one might expect that the use of gestures would provide more of an intrinsically rewarding activity due to the process involved in executing each command. However, the increased cognitive load as well as learning requirements present in the gestural case may have influenced this. That is, in the direct manipulation version, the user is able to play the game with only minimal game-play related learning required. In the gestural interface, the amount of learning required is compounded by the use of gestures. This, in turn, may detract from the intrinsic enjoyment value of the game. Increased familiarity with the gestural commands might lessen this difference if it is indeed due to the cognitive demands placed on the user by the gestural interface.

Overall, further experimentation is required when attempting to explain the differences between the various interfaces on a dimension by dimension basis. While possible explanations for these discrepancies have been presented above, they are not conclusive. In addition, these factor results do not provide an overall view of the appropriateness of interaction techniques within gaming. They do, however, present an interesting pattern.

That is, it appears that the gestural interface elicits flow more highly and consistently than either of the other two interfaces. This is further confirmed by the Multivariate contrast analysis. This analysis also showed that the mixed interface version was preferred to the direct manipulation one. In addition, the mixed interface does not elicit significantly more flow than the direct manipulation interface in any conditions except where the same is true of the gestural interface. These two factors may suggest this effect is due to the inclusion of gestures in the mixed interface. The fact that direct manipulation was felt to be more intrinsically rewarding seems to contradict the other results, however. This suggests the need for further testing to determine whether this is due to an unfamiliarity with gestural techniques, and an increase in the cognitive load placed on the user. Overall, however, when considering gaming experience, it appears that gestures are the most successful at eliciting a higher flow state.

The consistency of results regardless of genre or participant characteristics suggests that it was indeed interface type which determined the way in which flow was experienced in these games. Within this, the inclusion of gestures in an interface appears to increase the level of flow experienced by participants. Due to the dimensionality present in the results, these findings should be interpreted cautiously. However, the consistency with which the gestural interface elicits more flow, as well as the results of the contrast analysis, suggest that gestures do indeed provide the most enjoyable gaming experience in this case.

5.4.4 Results specific to the Puzzle game

The Multivariate contrast analysis conducted on the results of the Puzzle game provided information that corresponds with the data described above. This is unsurprising given the lack of any large inter-genre differences when examining interface preferences.

Within this genre, the gestural interface was preferred over either of the other two interaction techniques, although no significant difference was found between the direct manipulation and mixed interfaces. Given that the game revolved around a three-dimensional shape, the preference for gestural commands may have been due to the fact that they allowed the user to feel as if they were interacting with the shape on a more physical level. This feeling would, in turn, have increased the degree of immersion experienced. As a level of immersion is required for flow to be experienced [Sweetser and Wyeth 2005], this would have increased the overall flow state experienced.

The contrast analysis also showed a significant difference between the Gaming and Control groups. As discussed above, this may be due to either the personality characteristics of those who play games frequently, or due to their more extensive experience in entering a flow state.

Perhaps the most important feature of these results, however, is that they correlate with the results obtained overall. Given that the Puzzle game exhibited a number of distinct design differences when compared to the other two genres, this reaffirms the hypothesis that genre does not overtly influence interface preference. These differences include the exclusion of elements such as real-time constraints as well as a difference in the actual aim of the game when compared to the other two genres. That is, within the Puzzle game, the player attempted to achieve as high a score as possible, while the other two genres required the player to defeat some opponent.

The fact that the results of the Puzzle game corroborate those achieved for all three games suggests that genre, and more specifically, differences in game-play and game aims, do not play an important role in the degree to which flow is experienced with regard to interaction style. Moreover, the lack of significant differences along this dimension further reaffirms the idea that the flow differences present are indeed due to the choice of interaction technique.

Chapter 6

Conclusion

Multi-touch gesture-based gaming is a relatively unexplored topic. This means that there is a considerable lack of existing information on the way in which interaction techniques influence the gaming experience on such devices. Given the recent surge in commercially available touch-devices, in addition to the fact that this technology may provide a more immersive gaming environment, it appears that research in this area may produce valuable information related to improving gaming experience. This project aimed to address this issue by investigating the effect of direct manipulation and gestures on the level of flow elicited during game-play. In addition, an interface using a mix of these interaction techniques was constructed in order to assess their combination. Additional elements, such as game genre, as well as previous gaming or touch-device experience and their effects on game enjoyment in relation to interaction style, were also investigated.

In order to conduct this study, a number of components were developed. These included a feature extraction module, two separate gesture recognition engines as well as three games, each of a different genre. These components were integrated for the purpose of user experimentation.

Experimentation revealed a significant difference between the interaction styles. That is, the gestural interface was found to elicit a higher degree of flow than either of the other two interaction techniques. Furthermore, as the mixed interaction style game was more successful in creating a flow state than the direct manipulation version, but only in conditions where the same was true of the gestural interface, one might conclude that this was due to the inclusion of gestural commands within the mixed interface. The direct manipulation interface was found to be more intrinsically rewarding, however. This may be due to the increased cognitive load placed on the user during game-play, as memorisation and recall of gestures is required. Overall, however, the gestural interface appears to elicit the highest degree of flow, when compared to either of the other two interfaces. This was found to be consistent across game genre and previous gaming and touch-device experience.

While game genre and specific participant characteristics do not appear to influence which interface version is preferred, both variables show variation when considering the degree to which each interface elicits flow. Variations were found between the level of flow experienced when playing games of different genres. The Strategy and Fighting/Duelling games both appear to have produced more flow than the Puzzle game. This variation, however, was relatively consistent across interface type, suggesting that some genres may, in general, elicit flow to a greater or lesser extent. Similar results were present when considering the effect of previous gaming and touch-device experience. Individuals who lacked either significant gaming or touch-device experience appear to have experienced less of a flow state than individuals who possess experience in these areas, regardless of interface type. Gaming experience appears to be a stronger predictor for this effect than touch-device experience. In addition, gaming experience seems to suggest a greater sensitivity to overall flow trends, with individuals experiencing more extreme flow states both positively and negatively. This may be due to the fact that previous gaming experience allows individuals to more easily enter a flow state. The difference between those with gaming or touch-device experience and those without may also be due to a lack of familiarity with either touch-based interaction, or the style of application in general. Overall, however, genre and previous gaming and touch-device experience should be viewed as predictors for the degree to which flow can be experienced, rather than indicators of which interface version is most enjoyable.

A number of limitations within the present study should be noted, however. The first of these is the small sample size when considering the number of variables. That is, for the genre and participant characteristic variables, only 15 participants per condition were used. This is further compounded by the fact that the measure used assessed flow along nine separate dimensions. Thus, one should be cautious when attempting to generalise the findings of this study to a larger population. In addition, as computer literacy was controlled for, the generalisability of the study is further limited. Furthermore, as only one game from each genre was tested, the results related to genre may be due to the characteristics of the individual games rather than being indicative of that genre in general. The specific nature of the games tested can be expanded on further in this regard. That is, as the games were designed with gestural interaction in mind, this may have influenced the results.

Thus, although gestures were found to elicit more flow, they should not be seen as a panacea. Rather, the decision to use gestures rather than some other interaction technique for multi-touch gaming should be consciously examined within the context of the specific game being developed. While gestures may prove ideal for some games, their use and associated cognitive load, may detract from the overall flow state achieved in other games. With this said, the results seem to indicate that for games where their use is appropriate, the inclusion of gestural controls when compared to other interaction techniques, may increase the degree of flow, and thus overall game enjoyment, experienced by players.

6.1 Future work

A number of elements within this project could be expanded on further in future work. The first of these relates to the sample size used for experimentation. A similar experiment with a larger sample size would provide more conclusive results. In addition, it would allow for the investigation of additional subject characteristics, such as computer literacy, with regard to their effect on the degree of flow experienced when using different interaction techniques. Furthermore, the addition of a pre-experiment questionnaire used to assess an individual's propensity to experience flow could also provide valuable information with regard to subject characteristics. This would also help to differentiate between gaming experience and flow propensity.

An investigation into the effects of game genre on interaction technique preference would also benefit from a larger sample size. This is due to the fact that the existing sample size limited the number of games it was possible to test within each genre. The inclusion of multiple games per genre, as well as additional genres, could provide more generalisable results in this regard. Further research could also examine the reasons for the differences in the amount of flow elicited by each interaction technique across the various flow dimensions. In particular, further study into why the direct manipulation version elicits more flow along the *Autotelic Experience* dimension than the other interfaces may prove valuable. This could be expanded further to include an investigation of whether there are specific gestural or direct manipulation components that can be shown to increase the possibility of experiencing a flow state. In addition, means of reducing the cognitive load placed on users by gestures could be investigated within the context of computer games.

Overall, however, similar studies with a wider range of games and a larger sample size could greatly improve the generalisability of this work. In addition, the investigation of further variables related to this concept would help to determine more conclusively when the use of gestures is most appropriate in multi-touch gaming.

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Appendix A

Feature extraction evaluation results

A.1 Parabolic regression

A.1.1 Z gesture

Univariate Analysis of Variance

Effect	SS	df	MS	F	<i>p</i>
Intercept	4657.729	1	4657.729	5043.171	< 0.000001*
Sample	19.206	2	9.603	10.398	0.000057*
Feature number	283.873	2	141.937	153.682	< 0.000001*
Error	147.771	160	0.924		

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

Tukey HSD test

Group	Small	Control	Random divergence
Small		0.998108	0.011347*
Control	0.998108		0.006701*
Random divergence	0.011347*	0.006701*	
Group means	5.3556	5.3667	5.9
Error: Between MS = 0.92357, df = 160.00			

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

A.1.2 Circle gesture

Univariate Analysis of Variance

Effect	SS	df	MS	F	<i>p</i>
Intercept	4234.05	1	4234.050	5997.238	< 0.000001*
Sample	16.900	2	8.450	11.969	0.000013*
Feature number	1082.500	2	541.250	766.643	< 0.000001*
Error	123.550	175	0.706		

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

Tukey HSD test

Group	Small	Control	Random divergence
Small		0.000086*	0.000086*
Control	0.000086*		1.000000
Random divergence	0.000086*	1.000000	
Group means	4.4167	5.0667	5.0667
Error: Between MS = 0.706, df = 175.00			

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

A.1.3 U gesture**Univariate Analysis of Variance**

Effect	SS	df	MS	F	<i>p</i>
Intercept	4606.645	1	4606.645	1086.091	< 0.000001*
Sample	46.660	2	23.330	5.500	0.004797*
Feature number	197.660	3	65.887	15.534	< 0.000001*
Error	767.710	181	4.241		

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

Tukey HSD test

Group	Small	Control	Random divergence
Small		0.000537*	0.000023*
Control	0.000537*		0.248016
Random divergence	0.000023*	0.248016	
Group means	7.0476	5.5600	5.0143
Error: Between MS = 4.2415, df = 181.00			

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

A.2 Circular regression**A.2.1 Z gesture****Univariate Analysis of Variance**

Effect	SS	df	MS	F	<i>p</i>
Intercept	4866.811	1	4866.811	2161.471	< 0.000001*
Sample	2.282	2	1.141	0.507	0.603396
Feature number	133.666	2	66.833	29.682	< 0.000001*
Error	349.001	155	2.252		

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

Additional testing: First feature only

Univariate Analysis of Variance

Effect	SS	df	MS	F	<i>p</i>
Intercept	1664.267	1	1664.267	1287.153	< 0.000001*
Sample	80.033	2	40.017	30.949	< 0.000001*
Error	73.700	57	1.293		

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

Tukey HSD test

Group	Small	Control	Random divergence
Small		0.000119*	0.000119*
Control	0.000119*		1.000000
Random divergence	0.000119*	1.000000	
Group means	6.9000	4.4500	4.4500
Error: Between MS = 1.2930, df = 57.000			

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

A.2.2 Circle gesture

Univariate Analysis of Variance

Effect	SS	df	MS	F	<i>p</i>
Intercept	1175.335	1	1175.335	354.6819	< 0.000001*
Sample	29.283	2	14.642	4.4184	0.013792*
Feature number	437.703	3	145.901	44.0287	< 0.000001*
Error	460.614	139	3.314		

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

Tukey HSD test

Group	Small	Control	Random divergence
Small		0.000022*	0.000022*
Control	0.000022*		0.333543
Random divergence	0.000022*	0.333543	
Group means	7.6364	4.3273	4.7941
Error: Between MS = 3.3138, df = 139.00			

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

A.2.3 U gesture

Univariate Analysis of Variance

Effect	SS	df	MS	F	<i>p</i>
Intercept	264.56	1	264.56	372.59	< 0.000001*
Sample	0.8405	2	0.84050	0.64117	0.424727
Feature number	31.2050	3	31.20500	23.80444	0.000003*
Error	173.0375	132	1.31089		

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

Additional testing: First feature only

Univariate Analysis of Variance

Effect	SS	df	MS	F	<i>p</i>
Intercept	2343.750	1	2343.750	5031.780	< 0.000001*
Sample	24.700	2	12.350	26.514	< 0.000001*
Error	26.550	57	0.466		

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

Tukey HSD test

Group	Small	Control	Random divergence
Small		0.000119*	0.000119*
Control	0.000119*		0.625879
Random divergence	0.000119*	0.625879	
Group means	5.3500	6.8000	6.6000
Error: Between MS = 0.46579, df = 57.000			

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

A.3 Direction-based segmentation

A.3.1 Z gesture

Univariate Analysis of Variance

Effect	SS	df	MS	F	<i>p</i>
Intercept	5117.257	1	5117.257	8398.266	< 0.000001*
Sample	2.059	2	1.029	1.689	0.187924
Feature number	310.837	2	155.419	255.068	< 0.000001*
Error	96.882	159	0.609		

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

Additional testing: First feature only

Univariate Analysis of Variance

Effect	SS	df	MS	F	<i>p</i>
Intercept	1480.067	1	1480.067	3800.171	< 0.000001*
Sample	5.733	2	2.867	7.360	0.001434*
Error	22.200	57	0.389		

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

Tukey HSD test

Group	Small	Control	Random divergence
Small		0.009938*	0.002336*
Control	0.009938*		0.868449
Random divergence	0.002336*	0.868449	
Group means	5.4000	4.8000	4.7000
Error: Between MS = 0.38947, df = 57.000			

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

A.3.2 Circle gesture**Univariate Analysis of Variance**

Effect	SS	df	MS	F	<i>p</i>
Intercept	4363.194	1	4363.194	3990.604	< 0.000001*
Sample	249.403	2	124.701	114.053	< 0.000001*
Feature number	178.189	3	178.189	162.973	< 0.000001*
Error	126.831	116	1.093		

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

Tukey HSD test

Group	Small	Control	Random divergence
Small		0.000113*	0.000113*
Control	0.000113*		0.675982
Random divergence	0.000113*	0.675982	
Group means	4.0000	6.9487	7.1463
Error: Between MS = 1.0934, df = 116.00			

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

A.3.3 U gesture**Univariate Analysis of Variance**

Effect	SS	df	MS	F	<i>p</i>
Intercept	3286.533	1	3286.533	3604.518	< 0.000001*
Sample	62.067	2	31.033	34.036	< 0.000001*
Feature number	93.633	3	93.633	102.693	< 0.000001*
Error	105.767	116	0.912		

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

Tukey HSD test

Group	Small	Control	Random divergence
Small		0.000120*	0.000113*
Control	0.000120*		0.004006*
Random divergence	0.000113*	0.004006*	
Group means	4.3000	5.3500	6.0500
Error: Between MS = 0.91178, df = 116.00			

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

Appendix B

Pre-experiment questionnaire

Name: _____
Age: _____
Qualification towards which you are studying: _____
Year of study: _____
Contact number: _____
Email address: _____

Preferred day:

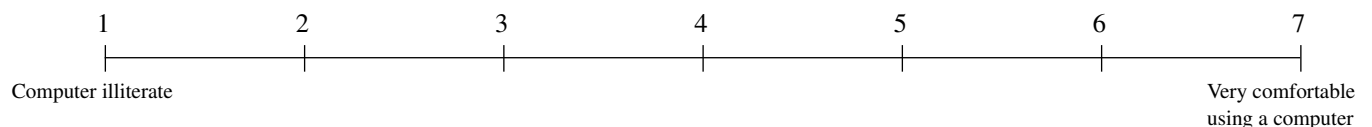
Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
27/09/10	28/09/10	29/09/10	30/09/10	01/10/10	02/10/10	03/10/10
04/10/10	05/10/10	06/10/10	07/10/10	08/10/10	09/10/10	10/10/10

Preferred time:

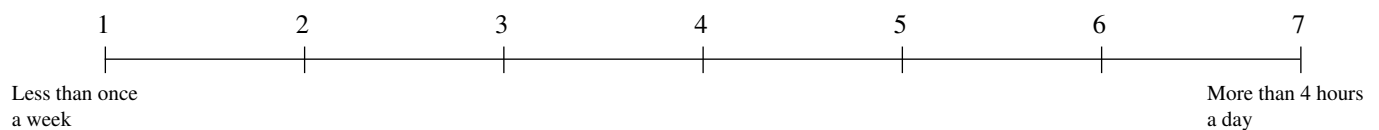
08:00	09:00	10:00	11:00	12:00	13:00	14:00	15:00	16:00	17:00
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The following questions use a 7-point rating scale. Select the option which represents the degree that you feel best represents yourself.

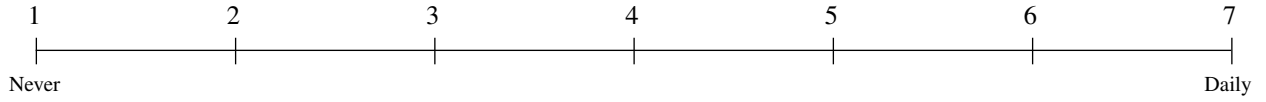
Computer literacy



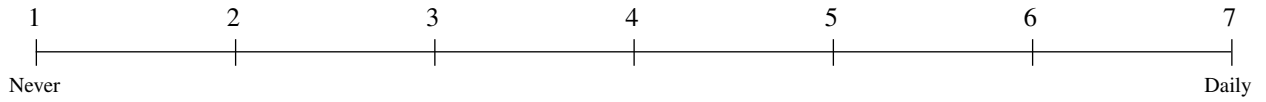
How much time do you spend using a computer on average?



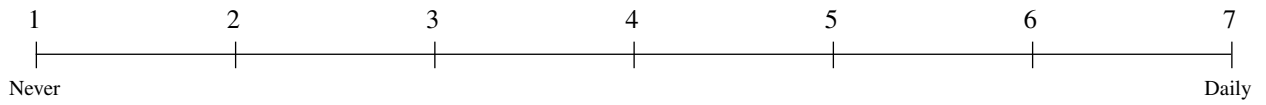
How often do you play games



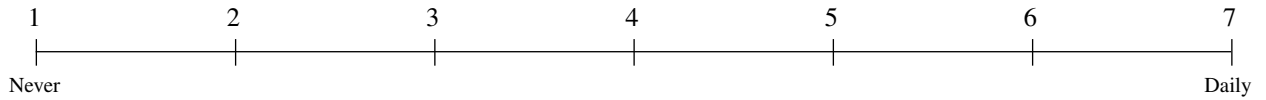
How often do you play strategy games



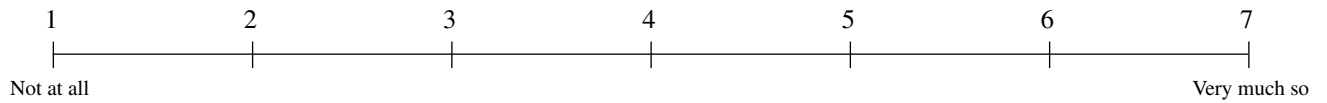
How often do you play puzzle games



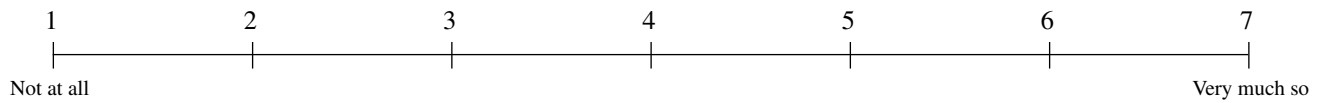
How often do you play fighting/duelling games



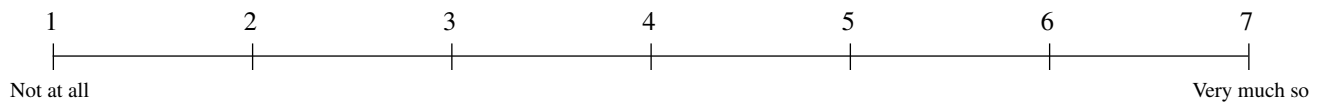
How much do you enjoy strategy games



How much do you enjoy puzzle games



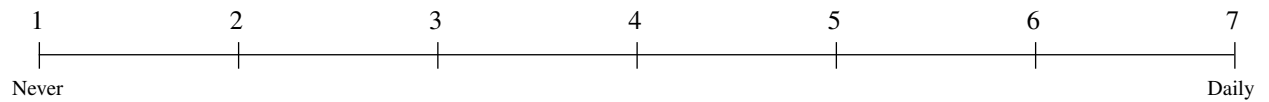
How much do you enjoy fighting/duelling games



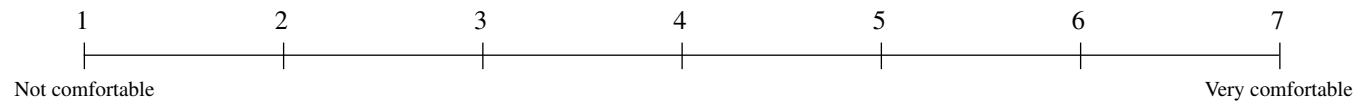
Do you own a touch-device (e.g. iPhone, iPod Touch, Android phone, etc.)?

Yes	No
-----	----

How often do you use touch-devices on average?



How comfortable do you feel using a touch-device?



Appendix C

User experimentation forms

C.1 Consent form

University of Cape Town

Department of Computer Science

Consent form

Researcher: _____

- I agree to participate in this experiment.
- I agree to my responses being used for education and research.
- I understand that my personal details will be used in aggregate form only, so that I will not be personally identifiable.
- I understand that I am under no obligation to take part in this project.
- I understand I have the right to withdraw from this experiment at any stage.
- I have read this consent form and the information it contains and had the opportunity to ask questions about them.

Name of Participant: _____

Signature of Participant: _____

Date: _____

C.2 Introductory instruction form

Gaming on the iPad introduction

Thank you for agreeing to participate in this study. Please read this information carefully as it explains the experimental procedure.

The experiment will take approximately 45 minutes and will begin with a practice session where you will be asked to draw a number of gestures. Once this has been completed, you will play a game three times. Between each game session you will be asked to describe your game-play experience by completing a questionnaire. Please ensure that you fill out the questionnaire marked with the same number as instructed within the game. Further instructions on how to play the game will be given within the game. Please play each game until told to stop.

A demographic information form has been included with the questionnaires. Please fill this out before you begin the practice session.

If you have any further questions, please ask the available experimenter.

C.3 Demographic information form

Demographic information

Age: _____
Gender: _____
Qualification towards which you are studying: _____
Year of study: _____

Please rate your level of computer literacy:

1 2 3 4 5 6 7
|-----|
Illiterate Highly literate

Please rate how often you play games:

1 2 3 4 5 6 7
|-----|
Never Daily

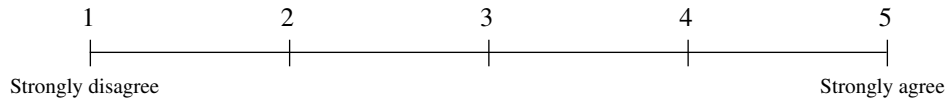
Please rate your experience with touch-devices:

1 2 3 4 5 6 7
|-----|
No experience Very experienced

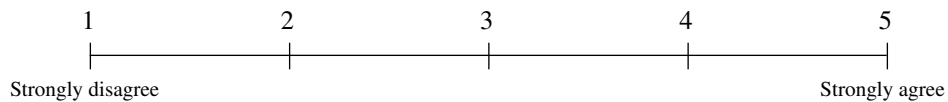
C.4 Flow State Scale

Due to copyright limitations, only a select portion of the questionnaire is able to be reproduced. Sample questions are included below.

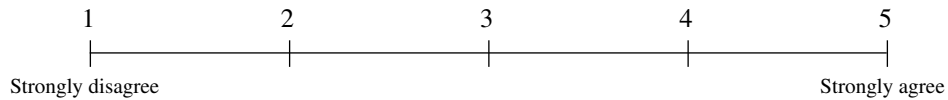
I was challenged, but I believed my skills would allow me to meet the challenge



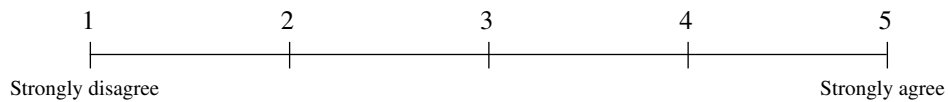
It was no effort to keep my mind on what was happening



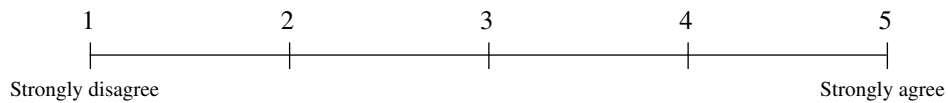
I loved the feeling of what I was doing, and want to capture this feeling again



I had a feeling of total control over what I was doing



It felt like time went by quickly



C.5 Post-experiment form

Post-experiment explanation

Thank you for taking part in this experiment.

The research question being investigated was whether gestures make touch-based games more enjoyable and whether there are genre dependent differences in this.

In order to investigate this, three game version were designed. These included a gestural interface, direct manipulation interface and a version that contained a mix of the previous two. The questionnaire you answered assesses the level of flow present in each condition. Flow can be understood as the intrinsic enjoyment, or fun, one gets out of performing some activity. In this case, we're using flow to determine which game version was the most enjoyable to play.

In order to determine whether there are genre dependent differences in the appropriateness of gestures in gaming, three different games were developed. These included a strategy, puzzle and duelling/fighting game. Once we have all the results, comparisons will be done both across game version and genre.

Once again, thank you for participating in our experiment. If you have any questions, feel free to ask.

Appendix D

Full statistical results

D.1 Higher-order flow factor: Post-hoc testing

Tukey HSD test

Genre	Interface	Puzzle			Strategy			Duelling		
		DM	G	M	DM	G	M	DM	G	M
Puzzle	DM		0.754183	0.940225	0.591897	0.995853	0.999999	1.000000	0.407407	0.998332
	G	0.754183		0.999968	0.999442	0.999919	0.809728	0.941348	0.992161	0.999676
	M	0.940225	0.999968		0.988522	1.000000	0.941348	0.991065	0.945889	1.000000
Strategy	DM	0.591897	0.999442	0.988522		0.882400	0.119458	0.628191	0.999999	0.952232
	G	0.995853	0.999919	1.000000	0.882400		0.887037	0.997415	0.899003	1.000000
	M	0.999999	0.809728	0.941348	0.119458	0.887037		0.999996	0.251894	0.984608
Duelling	DM	1.000000	0.941348	0.991065	0.628191	0.997415	0.999996		0.139794	0.994157
	G	0.407407	0.992161	0.945889	0.999999	0.899003	0.251894	0.139794		0.606237
	M	0.998332	0.999676	1.000000	0.952232	1.000000	0.984608	0.994157	0.606237	

Error: Between; Within; Pooled MS = 25.953, df = 82.162

DM - Direct manipulation, G - Gestural, M - Mixed

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

Fisher LSD test

Genre	Interface	Puzzle			Strategy			Duelling		
		DM	G	M	DM	G	M	DM	G	M
Puzzle	DM		0.09653	0.21276	0.05640	0.41217	0.78190	0.95725	0.02967*	0.47007
	G	0.09653		0.67084	0.53229	0.62980	0.11662	0.21328	0.36816	0.56191
	M	0.21276	0.67084		0.34054	0.87931	0.21328	0.35880	0.21983	0.80254
Strategy	DM	0.05640	0.53229	0.34054		0.15700	0.00571*	0.06344	0.78190	0.22992
	G	0.41217	0.62980	0.87931	0.15700		0.16033	0.44320	0.16867	0.92173
	M	0.78190	0.11662	0.21328	0.00571*	0.16033		0.74111	0.01476*	0.31858
Duelling	DM	0.95725	0.21328	0.35880	0.06344	0.44320	0.74111		0.00694*	0.38991
	G	0.02967*	0.36816	0.21983	0.78190	0.16867	0.01476*	0.00694*		0.05952
	M	0.47007	0.56191	0.80254	0.22992	0.92173	0.31858	0.38991	0.05952	

Error: Between; Within; Pooled MS = 25.953, df = 82.162

DM - Direct manipulation, G - Gestural, M - Mixed

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

D.2 MANOVA: Full results

D.2.1 Mauchly's sphericity test

Dependant variable	W	Chi-Sqr.	df	<i>p</i>
Challenge-Skill Balance	0.997321	0.093878	2	0.954146
Merging of Action and Awareness	0.951121	1.753994	2	0.416030
Clear Goals	0.992831	0.251808	2	0.881699
Unambiguous Feedback	0.919297	2.945099	2	0.229340
Concentration on the Task at Hand	0.825177	6.725492	2	0.034640*
Sense of Control	0.980870	0.676050	2	0.713178
Loss of Self-Consciousness	0.885833	4.242939	2	0.119855
Transformation of Time	0.768128	9.232958	2	0.009888*
Autotelic Experience	0.931844	2.470641	2	0.290742

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

D.2.2 Univariate tests

Challenge-Skill Balance

Effect	Value	F	Effect df	Error df	<i>p</i>
Interface	368.0148	2	184.0074	26.52536	<0.000001*
Interface*Genre	14.2963	4	3.5741	0.51522	0.724746
Interface*Group	30.5185	4	7.6296	1.09984	0.363333
Interface*Genre*Group	30.3704	8	3.7963	0.54725	0.816957
Error	499.4667	72	6.9370		

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

Merging of Action and Awareness

Effect	Value	F	Effect df	Error df	<i>p</i>
Interface	108.5778	2	54.28889	8.758889	0.000394*
Interface*Genre	34.7111	4	8.67778	1.400060	0.242642
Interface*Group	21.6889	4	5.42222	0.874813	0.483401
Interface*Genre*Group	42.7556	8	5.34444	0.862265	0.552170
Error	446.2667	72	6.19815		

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

Clear Goals

Effect	Value	F	Effect df	Error df	<i>p</i>
Interface	219.2444	2	109.6222	17.49291	0.000001*
Interface*Genre	50.8444	4	12.7111	2.02837	0.099458
Interface*Group	30.0889	4	7.5222	1.20035	0.318159
Interface*Genre*Group	68.6222	8	8.5778	1.36879	0.224885
Error	451.2000	72	6.2667		

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

Unambiguous Feedback

Effect	Value	F	Effect df	Error df	<i>p</i>
Interface	131.3926	2	65.69630	9.166925	0.000284*
Interface*Genre	89.4519	4	22.36296	3.120413	0.020006*
Interface*Group	18.7852	4	4.69630	0.655297	0.625051
Interface*Genre*Group	12.3704	8	1.54630	0.215762	0.987129
Error	516.0000	72	7.16667		

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

Concentration on the Task at Hand

Effect	Value	F	Effect df	Error df	<i>p</i>
Interface	117.3778	2	58.68889	5.456612	0.006216*
Interface*Genre	54.5778	4	13.64444	1.268595	0.290307
Interface*Group	24.7111	4	6.17778	0.574380	0.682091
Interface*Genre*Group	61.6000	8	7.70000	0.715909	0.676722
Error	774.4000	72	10.75556		

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

Sense of Control

Effect	Value	F	Effect df	Error df	<i>p</i>
Interface	135.1259	2	67.56296	9.924918	0.000156*
Interface*Genre	14.3407	4	3.58519	0.526659	0.716458
Interface*Group	63.4074	4	15.85185	2.328618	0.064197
Interface*Genre*Group	36.3259	8	4.54074	0.667029	0.718653
Error	490.1333	72	6.80741		

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

Loss of Self-Consciousness

Effect	Value	F	Effect df	Error df	<i>p</i>
Interface	164.0444	2	82.02222	7.719066	0.000918*
Interface*Genre	182.0889	4	45.52222	4.284071	0.003637*
Interface*Group	32.3556	4	8.08889	0.761241	0.553929
Interface*Genre*Group	85.1111	8	10.63889	1.001220	0.442816
Error	765.0667	72	10.62593		

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

Transformation of Time

Effect	Value	F	Effect df	Error df	<i>p</i>
Interface	129.7333	2	64.86667	6.616547	0.002302*
Interface*Genre	44.8889	4	11.22222	1.144692	0.342549
Interface*Group	55.2889	4	13.82222	1.409898	0.239375
Interface*Genre*Group	74.8889	8	9.36111	0.954855	0.477916
Error	705.8667	72	9.80370		

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

Autotelic Experience

Effect	Value	F	Effect df	Error df	<i>p</i>
Interface	186.0593	2	93.02963	8.855279	0.000364*
Interface*Genre	75.2296	4	18.80741	1.790234	0.140164
Interface*Group	42.6519	4	10.66296	1.014983	0.405488
Interface*Genre*Group	116.3259	8	14.54074	1.384100	0.218138
Error	756.4000	72	10.50556		

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

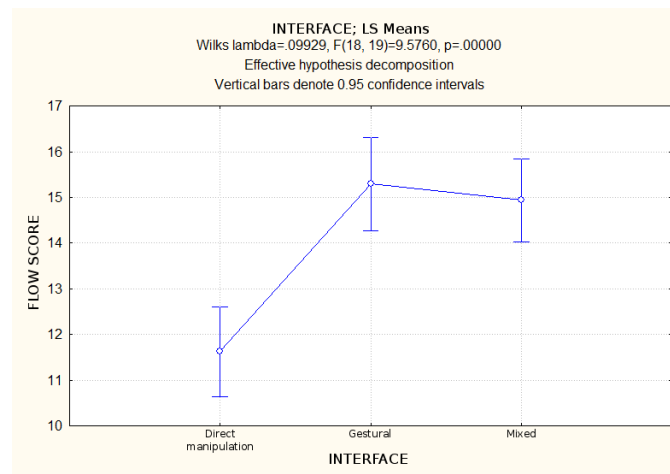
D.2.3 Tukey's HSD test

Main effect: *Interface*

Challenge-Skill Balance

	Direct manipulation	Gestural	Mixed
Direct manipulation		0.000111*	0.000111*
Gestural	0.000111*		0.798469
Mixed	0.000111*	0.798469	
Error: Within MS = 6.9370, df = 72.000			

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

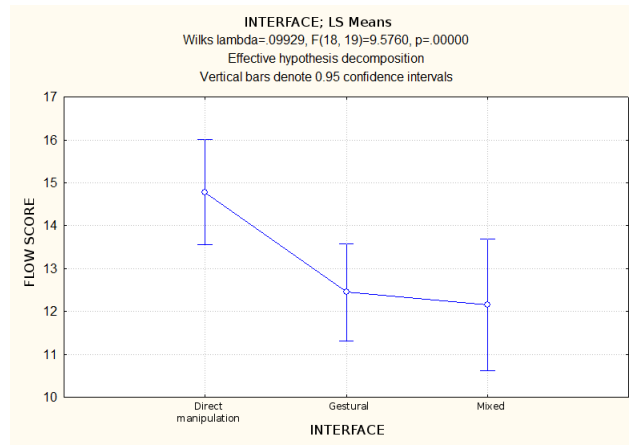


Challenge-Skill Balance: Cell mean plot

Merging of Action and Awareness

	Direct manipulation	Gestural	Mixed
Direct manipulation		0.001201*	0.002278*
Gestural	0.001201*		0.975679
Mixed	0.002278*	0.975679	
Error: Within MS = 6.1981, df = 72.000			

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

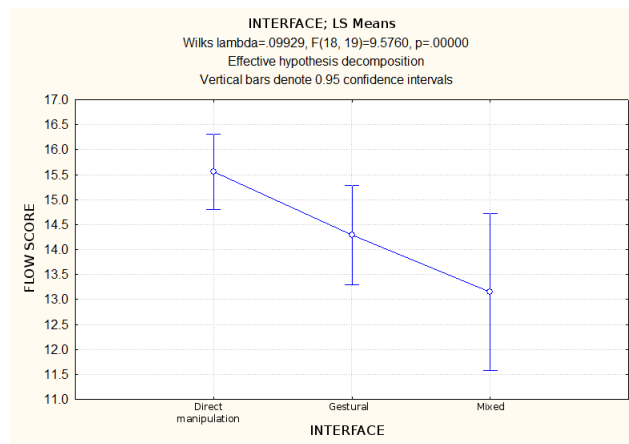


Merging of Action and Awareness: Cell mean plot

Clear Goals

	Direct manipulation	Gestural	Mixed
Direct manipulation		0.000114*	0.000121*
Gestural	0.000114*		0.953362
Mixed	0.000121*	0.953362	
Error: Within MS = 6.2667, df = 72.000			

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

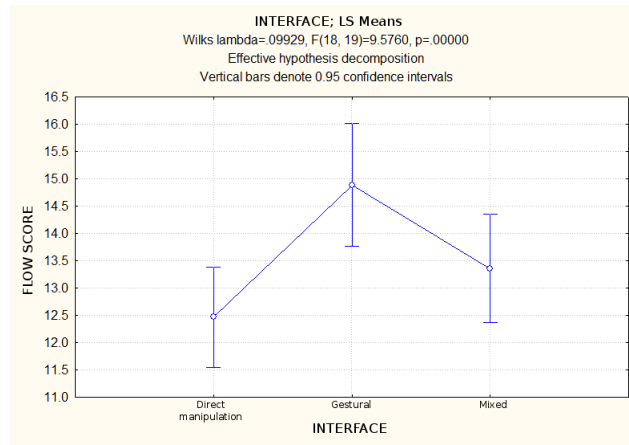


Clear Goals: Cell mean plot

Concentration on the Task at Hand

	Direct manipulation	Gestural	Mixed
Direct manipulation		0.005088*	0.521689
Gestural	0.005088*		0.086483
Mixed	0.521689	0.086483	
Error: Within MS = 10.756, df = 72.000			

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

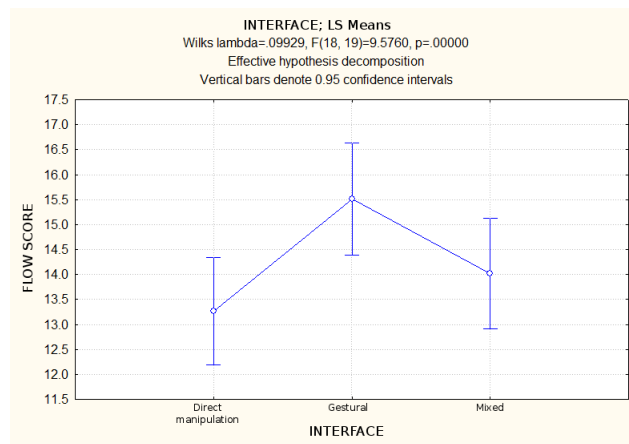


Concentration on the Task at Hand: Cell mean plot

Sense of Control

	Direct manipulation	Gestural	Mixed
Direct manipulation		0.000209*	0.245576
Gestural	0.000209*		0.018505*
Mixed	0.245576	0.018505*	
Error: Within MS = 6.8074, df = 72.000			

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

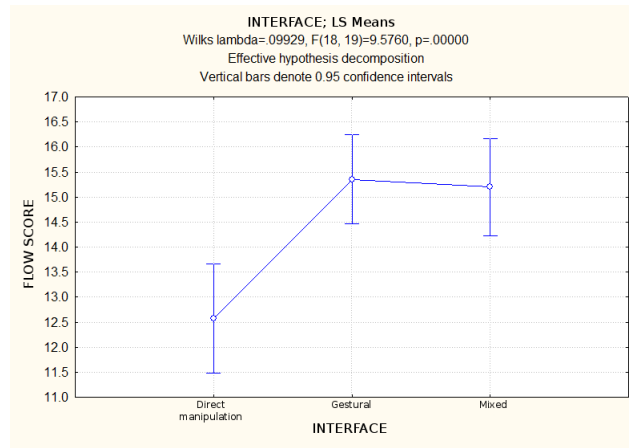


Sense of Control: Cell mean plot

Transformation of Time

	Direct manipulation	Gestural	Mixed
Direct manipulation		0.140792	0.001579*
Gestural	0.140792		0.205955
Mixed	0.001579*	0.205955	
Error: Within MS = 9.8037, df = 72.000			

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).

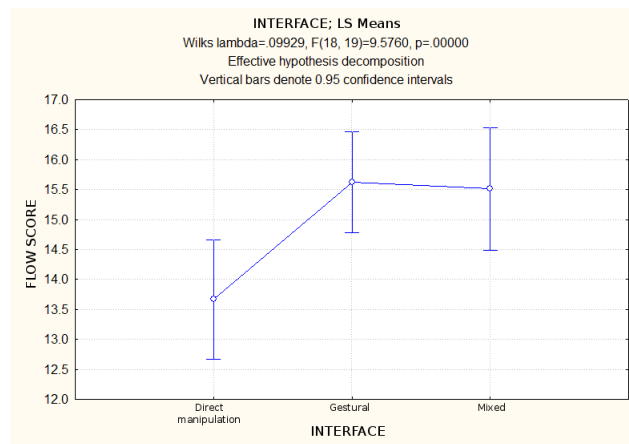


Transformation of Time: Cell mean plot

Autotelic Experience

	Direct manipulation	Gestural	Mixed
Direct manipulation		0.003083*	0.000863*
Gestural	0.003083*		0.906370
Mixed	0.000863*	0.906370	
Error: Within MS = 10.506, df = 72.000			

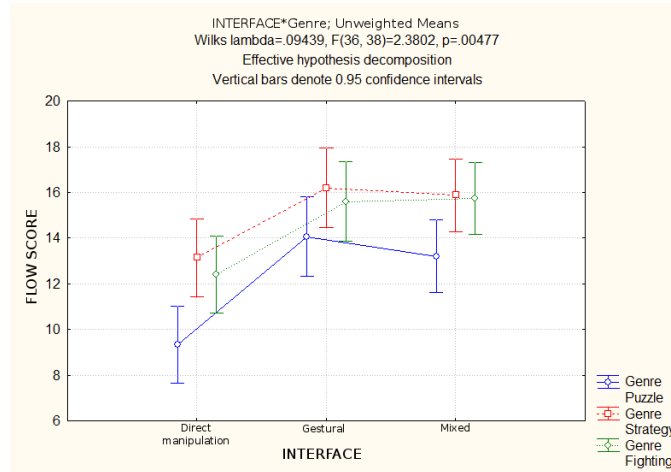
* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).



Autotelic Experience: Cell mean plot

Interaction effect: *Interface*Genre*

Challenge-Skill Balance										
Genre		Puzzle			Strategy			Duelling		
		DM	G	M	DM	G	M	DM	G	M
Puzzle	DM		0.000293*	0.004324*	0.041253*	0.000136*	0.000140*	0.192342	0.000152*	0.00014*5
	G	0.000293*		0.992289	0.996715	0.666131	0.833858	0.885127	0.925487	0.885127
	M	0.004324*	0.992289		1.000000	0.216286	0.365108	0.998927	0.511983	0.436359
Strategy	DM	0.041253*	0.996715	1.000000		0.051325	0.121302	0.999449	0.473768	0.400057
	G	0.000136*	0.666131	0.216286	0.051325		0.999994	0.041253*	0.999876	0.999982
	M	0.000140*	0.833858	0.365108	0.121302	0.999994		0.087614	1.000000	1.000000
Duelling	DM	0.192342	0.885127	0.998927	0.999449	0.041253*	0.087614		0.035257*	0.023826*
	G	0.000152*	0.925487	0.511983	0.473768	0.999876	1.000000	0.035257*		1.000000
	M	0.000145*	0.885127	0.436359	0.400057	0.999982	1.000000	0.023826*	1.000000	
Error: Between; Within; Pooled MSE = 10.256, df = 89.300										
DM - Direct manipulation, G - Gestural, M - Mixed										
* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).										

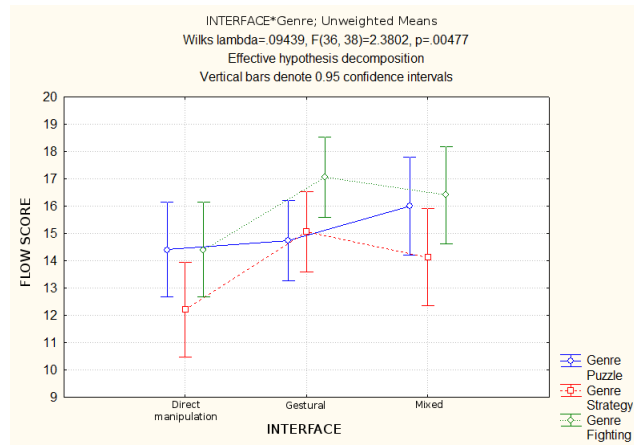


Challenge-Skill Balance: Cell mean plot

Merging of Action and Awareness

Genre		Puzzle			Strategy			Duelling		
		DM	G	M	DM	G	M	DM	G	M
Puzzle	DM		0.999990	0.707550	0.619478	0.999710	1.000000	1.000000	0.356435	0.731314
	G	0.999990		0.896785	0.427308	0.999999	0.999868	0.999999	0.541593	0.880870
	M	0.707550	0.896785		0.039442*	0.996519	0.798170	0.903059	0.991350	0.999994
Strategy	DM	0.619478	0.427308	0.039442*		0.056324	0.464022	0.619478	0.002246*	0.014382*
	G	0.999710	0.999999	0.996519	0.056324		0.982032	0.999710	0.731314	0.964744
	M	1.000000	0.999868	0.798170	0.464022	0.982032		1.000000	0.235152	0.580610
Duelling	DM	1.000000	0.999999	0.903059	0.619478	0.999710	1.000000		0.097875	0.416919
	G	0.356435	0.541593	0.991350	0.002246*	0.731314	0.235152	0.097875		0.998175
	M	0.731314	0.880870	0.999994	0.014382*	0.964744	0.580610	0.416919	0.998175	

Error: Between; Within; Pooled MSE = 10.115, df = 83.085
DM - Direct manipulation, G - Gestural, M - Mixed
* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).



Merging of Action and Awareness: Cell mean plot

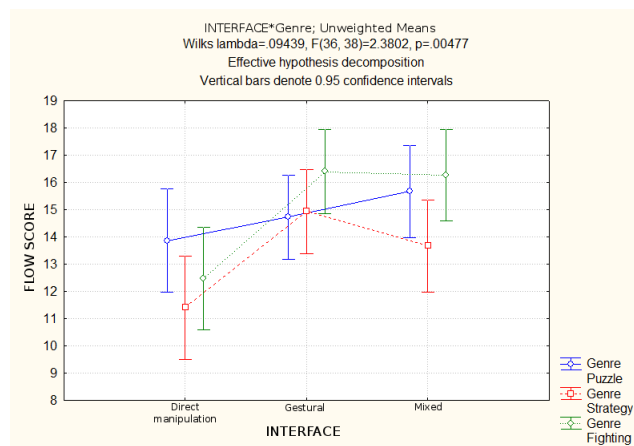
Clear Goals

Genre		Puzzle			Strategy			Duelling		
		DM	G	M	DM	G	M	DM	G	M
Puzzle	DM		0.989211	0.569355	0.501836	0.992740	1.000000	0.959756	0.464693	0.539541
	G	0.989211		0.982642	0.133101	1.000000	0.992740	0.615328	0.895625	0.932726
	M	0.569355	0.982642		0.016400*	0.999515	0.758465	0.170510	0.999515	0.999891
Strategy	DM	0.501836	0.133101	0.016400*		0.007097*	0.259318	0.992740	0.002280*	0.003296*
	G	0.992740	1.000000	0.999515	0.007097*		0.899634	0.501836	0.947441	0.969875
	M	1.000000	0.992740	0.758465	0.259318	0.899634		0.984313	0.359334	0.428405
Duelling	DM	0.959756	0.615328	0.170510	0.992740	0.501836	0.984313		0.001737*	0.002781*
	G	0.464693	0.895625	0.999515	0.002280*	0.947441	0.359334	0.001737*		1.000000
	M	0.539541	0.932726	0.999891	0.003296*	0.969875	0.428405	0.002781*	1.000000	

Error: Between; Within; Pooled MSE = 10.667, df = 80.578

DM - Direct manipulation, G - Gestural, M - Mixed

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).



Clear Goals: Cell mean plot

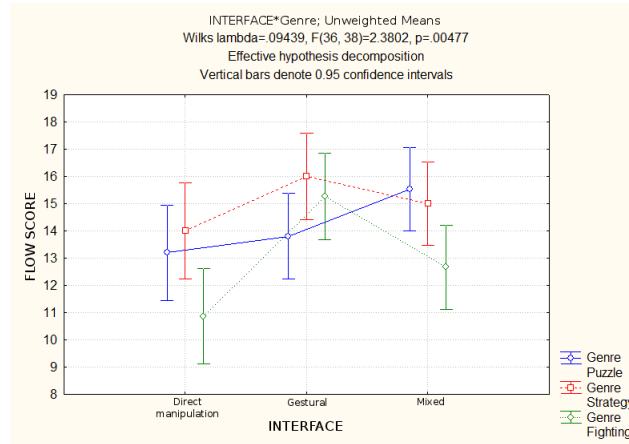
Unambiguous Feedback

Genre		Puzzle			Strategy			Duelling		
		DM	G	M	DM	G	M	DM	G	M
Puzzle	DM		0.999514	0.306979	0.998656	0.259906	0.809040	0.507560	0.666895	0.999936
	G	0.999514		0.699209	1.000000	0.587720	0.978651	0.205774	0.930857	0.985117
	M	0.306979	0.699209		0.912301	0.999977	0.999936	0.002635*	1.000000	0.231775
Strategy	DM	0.998656	1.000000	0.912301		0.517559	0.982434	0.140213	0.970262	0.959651
	G	0.259906	0.587720	0.999977	0.517559		0.982434	0.000663*	0.999289	0.092105
	M	0.809040	0.978651	0.999936	0.982434	0.982434		0.012429*	1.000000	0.507560
Duelling	DM	0.507560	0.205774	0.002635*	0.140213	0.000663*	0.012429*		0.000921*	0.655002
	G	0.666895	0.930857	1.000000	0.970262	0.999289	1.000000	0.000921*		0.181217
	M	0.999936	0.985117	0.231775	0.959651	0.092105	0.507560	0.655002	0.181217	

Error: Between; Within; Pooled MSE = 9.6370, df = 95.455

DM - Direct manipulation, G - Gestural, M - Mixed

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).



Unambiguous Feedback: Cell mean plot

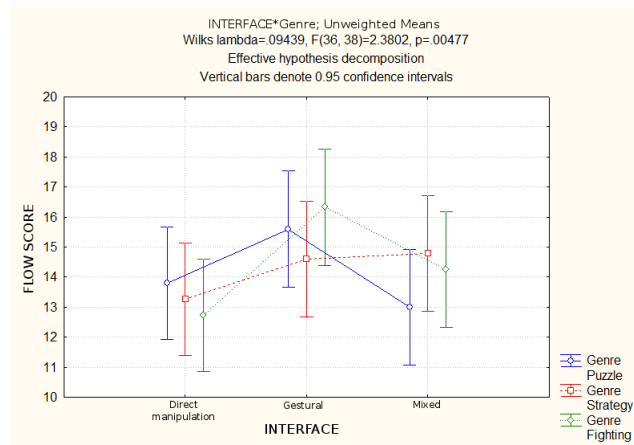
Concentration on the Task at Hand

Genre		Puzzle			Strategy			Duelling		
		DM	G	M	DM	G	M	DM	G	M
Puzzle	DM		0.850521	0.999071	0.999982	0.999604	0.997884	0.996656	0.613771	0.999993
	G	0.850521		0.435262	0.712829	0.997884	0.999604	0.444399	0.999793	0.985030
	M	0.999071	0.435262		1.000000	0.954467	0.912784	1.000000	0.243183	0.989254
Strategy	DM	0.999982	0.712829	1.000000		0.970396	0.933881	0.999982	0.350459	0.997884
	G	0.999604	0.997884	0.954467	0.970396		1.000000	0.894718	0.928710	1.000000
	M	0.997884	0.999604	0.912784	0.933881	1.000000		0.827872	0.964498	0.999982
Duelling	DM	0.996656	0.444399	1.000000	0.999982	0.894718	0.827872		0.081952	0.933881
	G	0.613771	0.999793	0.243183	0.350459	0.928710	0.964498	0.081952		0.728875
	M	0.999993	0.985030	0.989254	0.997884	1.000000	0.999982	0.933881	0.728875	

Error: Between; Within; Pooled MSE = 13.300, df = 100.63

DM - Direct manipulation, G - Gestural, M - Mixed

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).



Concentration on the Task at Hand: Cell mean plot

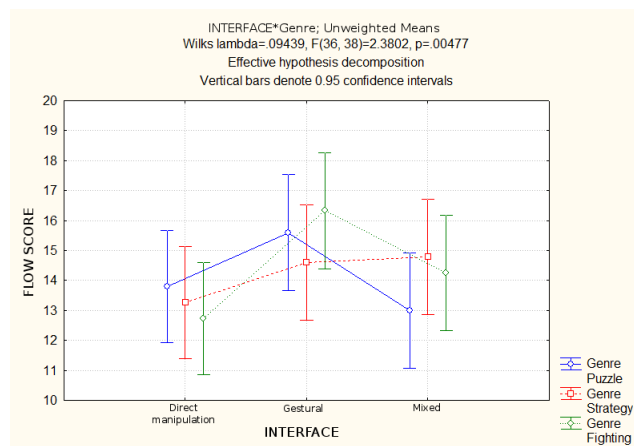
Sense of Control

Genre		Puzzle			Strategy			Duelling		
		DM	G	M	DM	G	M	DM	G	M
Puzzle	DM		0.156260	1.000000	0.999807	0.986972	0.999996	0.991032	0.910076	0.999999
	G	0.156260		0.209600	0.178407	0.966073	0.687022	0.073159	0.997644	0.651448
	M	1.000000	0.209600		0.999236	0.994032	1.000000	0.981625	0.942670	1.000000
Strategy	DM	0.999807	0.178407	0.999236		0.576016	0.969392	0.999987	0.615103	0.996172
	G	0.986972	0.966073	0.994032	0.576016		0.995220	0.615103	0.999987	0.998629
	M	0.999996	0.687022	1.000000	0.969392	0.995220		0.942670	0.981625	1.000000
Duelling	DM	0.991032	0.073159	0.981625	0.999987	0.615103	0.942670		0.096567	0.833036
	G	0.910076	0.997644	0.942670	0.615103	0.999987	0.981625	0.096567		0.894454
	M	0.999999	0.651448	1.000000	0.996172	0.998629	1.000000	0.833036	0.894454	

Error: Between; Within; Pooled MSE = 11.300, df = 82.059

DM - Direct manipulation, G - Gestural, M - Mixed

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).



Sense of Control: Cell mean plot

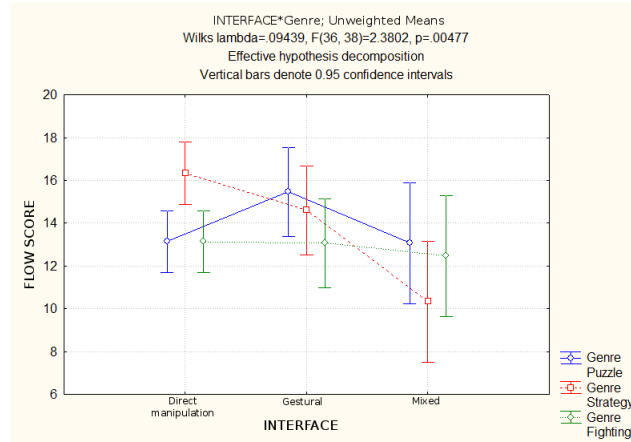
Loss of Self-Consciousness

Genre		Puzzle			Strategy			Duelling		
		DM	G	M	DM	G	M	DM	G	M
Puzzle	DM		0.575353	1.000000	0.480828	0.988243	0.657736	1.000000	1.000000	0.999963
	G	0.575353		0.537515	0.999729	0.999729	0.030127*	0.837733	0.815617	0.569390
	M	1.000000	0.537515		0.452000	0.984363	0.686339	1.000000	1.000000	0.999983
Strategy	DM	0.480828	0.999729	0.452000		0.871502	0.000233*	0.480828	0.452000	0.230124
	G	0.988243	0.999729	0.984363	0.871502		0.016831*	0.988243	0.984363	0.894897
	M	0.657736	0.030127*	0.686339	0.000233*	0.016831*		0.657736	0.686339	0.894897
Duelling	DM	1.000000	0.837733	1.000000	0.480828	0.988243	0.657736		1.000000	0.999753
	G	1.000000	0.815617	1.000000	0.452000	0.984363	0.686339	1.000000		0.999888
	M	0.999963	0.569390	0.999983	0.230124	0.894897	0.894897	0.999753	0.999888	

Error: Between; Within; Pooled MSE = 17.422, df = 82.800

DM - Direct manipulation, G - Gestural, M - Mixed

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).



Loss of Self-Consciousness: Cell mean plot

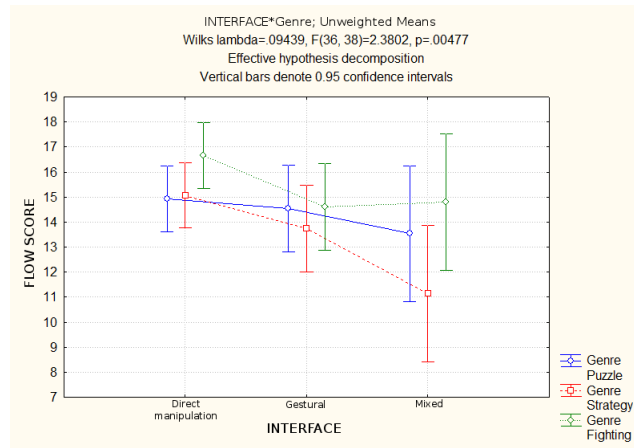
Transformation of Time

Genre		Puzzle			Strategy			Duelling		
		DM	G	M	DM	G	M	DM	G	M
Puzzle	DM		0.999993	0.948404	1.000000	0.994639	0.158720	0.945887	1.000000	1.000000
	G	0.999993		0.993688	0.999987	0.999723	0.282980	0.841962	1.000000	1.000000
	M	0.948404	0.993688		0.973679	1.000000	0.737052	0.392091	0.997641	0.992255
Strategy	DM	1.000000	0.999987	0.973679		0.961020	0.025650*	0.965962	0.999996	1.000000
	G	0.994639	0.999723	1.000000	0.961020		0.371420	0.484334	0.999501	0.997641
	M	0.158720	0.282980	0.737052	0.025650*	0.371420		0.004822*	0.258860	0.194716
Duelling	DM	0.945887	0.841962	0.392091	0.965962	0.484334	0.004822*		0.677086	0.783729
	G	1.000000	1.000000	0.997641	0.999996	0.999501	0.258860	0.677086		1.000000
	M	1.000000	1.000000	0.992255	1.000000	0.997641	0.194716	0.783729	1.000000	

Error: Between; Within; Pooled MSE = 14.722, df = 88.291

DM - Direct manipulation, G - Gestural, M - Mixed

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).



Transformation of Time: Cell mean plot

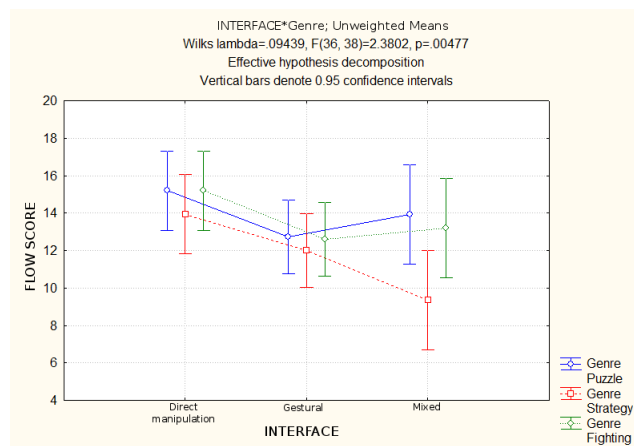
Autotelic Experience

Genre		Puzzle			Strategy			Duelling		
		DM	G	M	DM	G	M	DM	G	M
Puzzle	DM		0.492065	0.976725	0.996510	0.528253	0.010712*	1.000000	0.774791	0.937475
	G	0.492065		0.983392	0.997617	0.999940	0.444092	0.820797	1.000000	0.999998
	M	0.976725	0.983392		1.000000	0.948262	0.100395	0.996510	0.995031	0.999940
Strategy	DM	0.996510	0.997617	1.000000		0.783253	0.006637*	0.996510	0.995031	0.999940
	G	0.528253	0.999940	0.948262	0.783253		0.384045	0.528253	0.999987	0.997617
	M	0.010712*	0.444092	0.100395	0.006637*	0.384045		0.010712*	0.499824	0.271660
Duelling	DM	1.000000	0.820797	0.996510	0.996510	0.528253	0.010712*		0.418945	0.750615
	G	0.774791	1.000000	0.995031	0.995031	0.999987	0.499824	0.418945		0.999883
	M	0.937475	0.999998	0.999940	0.999940	0.997617	0.271660	0.750615	0.999883	

Error: Between; Within; Pooled MSE = 18.648, df = 78.186

DM - Direct manipulation, G - Gestural, M - Mixed

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).



Autotelic Experience: Cell mean plot

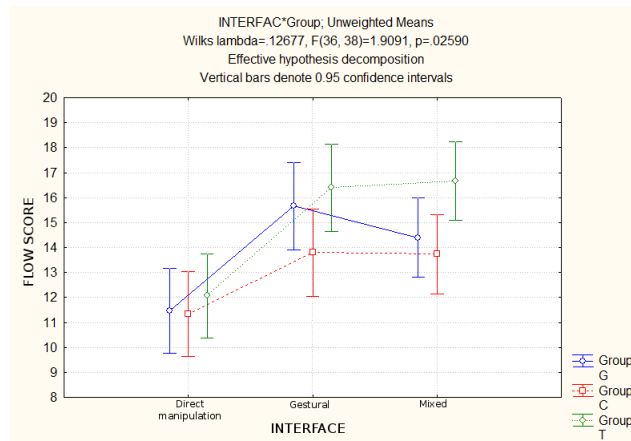
Interaction effect: *Interface*Group*

Challenge-Skill Balance										
Genre		Puzzle			Strategy			Duelling		
		DM	G	M	DM	G	M	DM	G	M
Puzzle	DM		0.001410*	0.073441	1.000000	0.550692	0.589540	0.999876	0.001951*	0.000915*
	G	0.001410*		0.923010	0.010582*	0.804371	0.772556	0.065448	0.999449	0.994700
	M	0.073441	0.923010		0.192342	0.999876	0.999727	0.550692	0.738693	0.589540
Strategy	DM	1.000000	0.010582*	0.192342		0.220025	0.251707	0.999449	0.001332*	0.000632*
	G	0.550692	0.804371	0.999876	0.220025		1.000000	0.860821	0.400057	0.270118
	M	0.589540	0.772556	0.999727	0.251707	1.000000		0.885127	0.365108	0.242218
Duelling	DM	0.999876	0.065448	0.550692	0.999449	0.860821	0.885127		0.000908*	0.000405*
	G	0.001951*	0.999449	0.738693	0.001332*	0.400057	0.365108	0.000908*		0.999999
	M	0.000915*	0.994700	0.589540	0.000632*	0.270118	0.242218	0.000405*	0.999999	

Error: Between; Within; Pooled MSE = 10.256, df = 89.300

DM - Direct manipulation, G - Gestural, M - Mixed

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).



Challenge-Skill Balance: Cell mean plot

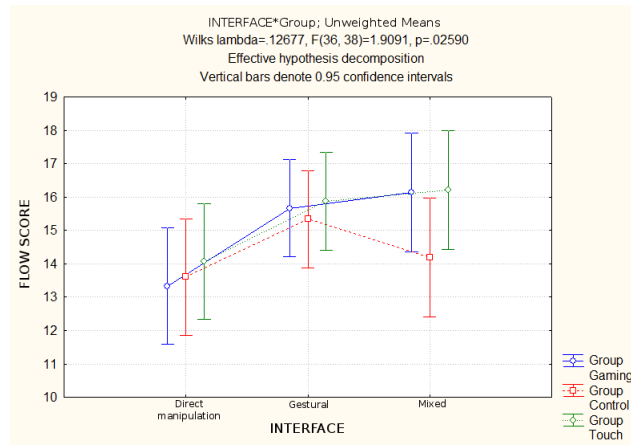
Merging of Action and Awareness

Genre		Puzzle			Strategy			Duelling		
		DM	G	M	DM	G	M	DM	G	M
Puzzle	DM		0.219195	0.068070	1.000000	0.731314	0.997937	0.999414	0.427308	0.262606
	G	0.219195		0.999871	0.695217	0.999999	0.939109	0.903059	1.000000	0.999946
	M	0.068070	0.999871		0.427308	0.998862	0.765733	0.695217	1.000000	1.000000
Strategy	DM	1.000000	0.695217	0.427308		0.611501	0.999148	0.999981	0.580610	0.391191
	G	0.731314	0.999999	0.998862	0.611501		0.942941	0.974118	0.999946	0.997937
	M	0.997937	0.939109	0.765733	0.999148	0.942941		1.000000	0.880870	0.731314
Duelling	DM	0.999414	0.903059	0.695217	0.999981	0.974118	1.000000		0.562018	0.329186
	G	0.427308	1.000000	1.000000	0.580610	0.999946	0.880870	0.562018		0.999990
	M	0.262606	0.999946	1.000000	0.391191	0.997937	0.731314	0.329186	0.999990	

Error: Between; Within; Pooled MSE = 10.115, df = 83.085

DM - Direct manipulation, G - Gestural, M - Mixed

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).



Merging of Action and Awareness: Cell mean plot

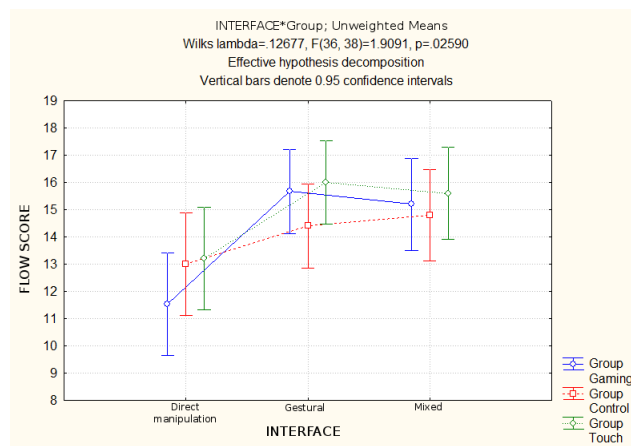
Clear Goals

Genre		Puzzle			Strategy			Duelling		
		DM	G	M	DM	G	M	DM	G	M
Puzzle	DM		0.000855*	0.004454*	0.947441	0.296323	0.150919	0.895625	0.009769*	0.026932*
	G	0.000855*		0.999876	0.393231	0.977970	0.998312	0.501836	0.999999	1.000000
	M	0.004454*	0.999876		0.652699	0.999057	0.999995	0.758465	0.999057	0.999995
Strategy	DM	0.947441	0.393231	0.652699		0.836915	0.569355	1.000000	0.240400	0.428405
	G	0.296323	0.977970	0.999057	0.836915		0.999962	0.984313	0.915488	0.984313
	M	0.150919	0.998312	0.999995	0.569355	0.999962		0.915488	0.984313	0.999057
Duelling	DM	0.895625	0.501836	0.758465	1.000000	0.984313	0.915488		0.071049	0.194563
	G	0.009769*	0.999999	0.999057	0.240400	0.915488	0.984313	0.071049		0.999962
	M	0.026932*	1.000000	0.999995	0.428405	0.984313	0.999057	0.194563	0.999962	

Error: Between; Within; Pooled MSE = 10.667, df = 80.578

DM - Direct manipulation, G - Gestural, M - Mixed

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).



Clear Goals: Cell mean plot

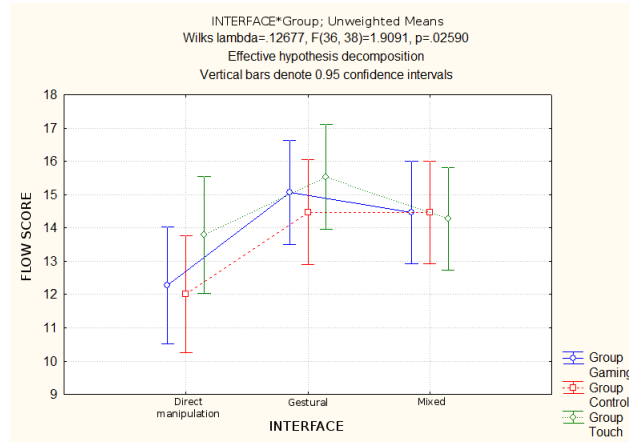
Unambiguous Feedback

Genre		Puzzle			Strategy			Duelling		
		DM	G	M	DM	G	M	DM	G	M
Puzzle	DM		0.115204	0.385606	1.000000	0.587720	0.587720	0.912301	0.106377	0.705055
	G	0.115204		0.999514	0.160014	0.999845	0.999845	0.970262	0.999977	0.998656
	M	0.385606	0.999514		0.429608	1.000000	1.000000	0.999659	0.989960	1.000000
Strategy	DM	1.000000	0.160014	0.429608		0.238615	0.238615	0.809040	0.058412	0.547553
	G	0.587720	0.999845	1.000000	0.238615		1.000000	0.999659	0.989960	1.000000
	M	0.587720	0.999845	1.000000	0.238615	1.000000		0.999659	0.989960	1.000000
Duelling	DM	0.912301	0.970262	0.999659	0.809040	0.999659	0.999659		0.699209	0.999926
	G	0.106377	0.999977	0.989960	0.058412	0.989960	0.989960	0.699209		0.929454
	M	0.705055	0.998656	1.000000	0.547553	1.000000	1.000000	0.999926	0.929454	

Error: Between; Within; Pooled MSE = 9.6370, df = 95.455

DM - Direct manipulation, G - Gestural, M - Mixed

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).



Unambiguous Feedback: Cell mean plot

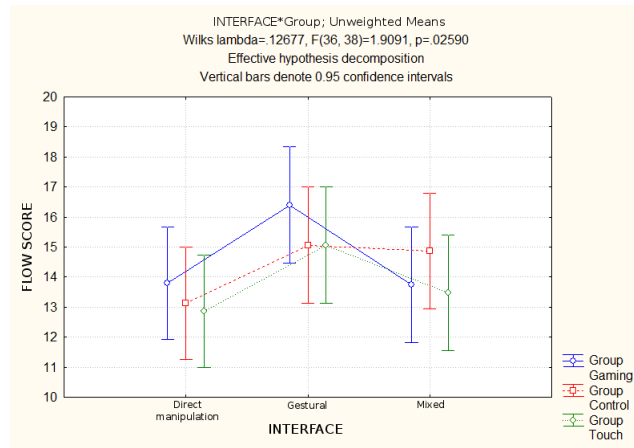
Concentration on the Task at Hand

Genre		Puzzle			Strategy			Duelling		
		DM	G	M	DM	G	M	DM	G	M
Puzzle	DM		0.435262	1.000000	0.999899	0.989254	0.996656	0.998728	0.989254	1.000000
	G	0.435262		0.400196	0.267801	0.985030	0.964498	0.178498	0.985030	0.412069
	M	1.000000	0.400196		0.999955	0.985030	0.994925	0.999262	0.985030	1.000000
Strategy	DM	0.999899	0.267801	0.999955		0.793870	0.875239	1.000000	0.874518	1.000000
	G	0.989254	0.985030	0.985030	0.793870		1.000000	0.773544	1.000000	0.954467
	M	0.996656	0.964498	0.994925	0.875239	1.000000		0.852215	1.000000	0.979618
Duelling	DM	0.998728	0.178498	0.999262	1.000000	0.773544	0.852215		0.657811	0.999893
	G	0.989254	0.985030	0.985030	0.874518	1.000000	1.000000	0.657811		0.916928
	M	1.000000	0.412069	1.000000	1.000000	0.954467	0.979618	0.999893	0.916928	

Error: Between; Within; Pooled MSE = 13.300, df = 100.63

DM - Direct manipulation, G - Gestural, M - Mixed

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).



Concentration on the Task at Hand: Cell mean plot

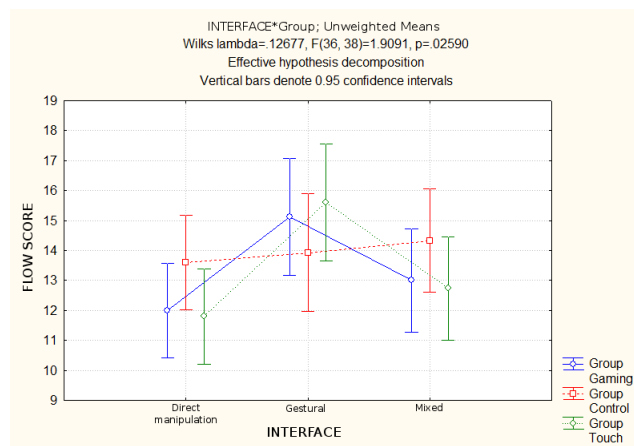
Sense of Control

Genre		Puzzle			Strategy			Duelling		
		DM	G	M	DM	G	M	DM	G	M
Puzzle	DM		0.039190*	0.979364	0.927585	0.815385	0.615103	1.000000	0.095950	0.999609
	G	0.039190*		0.392543	0.942670	0.986972	0.999236	0.158671	0.999987	0.578317
	M	0.979364	0.392543		0.999913	0.997644	0.974738	0.986972	0.468555	1.000000
Strategy	DM	0.927585	0.942670	0.999913		0.999993	0.997396	0.867568	0.785966	0.998629
	G	0.815385	0.986972	0.997644	0.999993		0.999972	0.721512	0.910076	0.986972
	M	0.615103	0.999236	0.974738	0.997396	0.999972		0.504734	0.981625	0.927585
Duelling	DM	1.000000	0.158671	0.986972	0.867568	0.721512	0.504734		0.004790*	0.986654
	G	0.095950	0.999987	0.468555	0.785966	0.910076	0.981625	0.004790*		0.081376
	M	0.999609	0.578317	1.000000	0.998629	0.986972	0.927585	0.986654	0.081376	

Error: Between; Within; Pooled MSE = 11.300, df = 82.059

DM - Direct manipulation, G - Gestural, M - Mixed

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).



Sense of Control: Cell mean plot

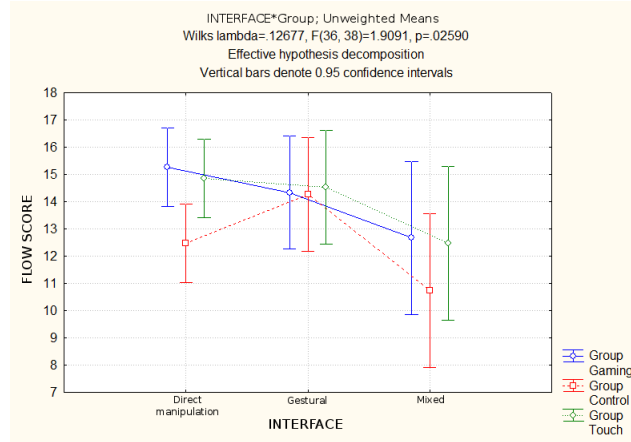
Loss of Self-Consciousness

Genre		Puzzle			Strategy			Duelling		
		DM	G	M	DM	G	M	DM	G	M
Puzzle	DM		0.997037	0.426844	0.657736	0.999199	0.086526	0.999999	0.999923	0.657736
	G	0.997037		0.894187	0.948731	1.000000	0.318717	0.999993	1.000000	0.948731
	M	0.426844	0.894187		1.000000	0.979549	0.937607	0.877426	0.948731	1.000000
Strategy	DM	0.657736	0.948731	1.000000		0.846225	0.871502	0.815617	0.910742	1.000000
	G	0.999199	1.000000	0.979549	0.846225		0.089924	0.999983	1.000000	0.958390
	M	0.086526	0.318717	0.937607	0.871502	0.089924		0.159815	0.250575	0.966669
Duelling	DM	0.999999	0.999993	0.877426	0.815617	0.999983	0.159815		0.999999	0.537515
	G	0.999923	1.000000	0.948731	0.910742	1.000000	0.250575	0.999999		0.722399
	M	0.657736	0.948731	1.000000	1.000000	0.958390	0.966669	0.537515	0.722399	

Error: Between; Within; Pooled MSE = 17.422, df = 82.800

DM - Direct manipulation, G - Gestural, M - Mixed

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).



Loss of Self-Consciousness: Cell mean plot

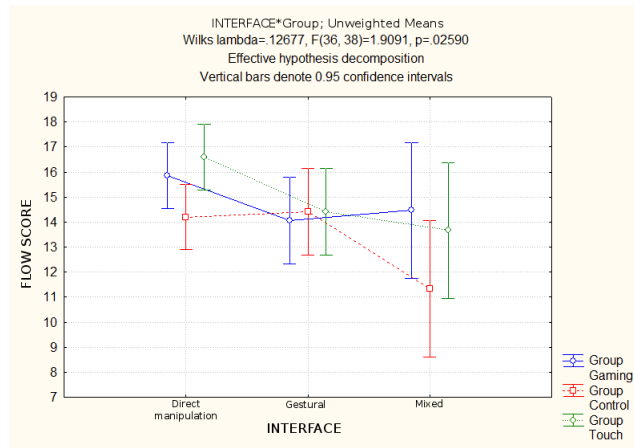
Transformation of Time

Genre		Puzzle			Strategy			Duelling		
		DM	G	M	DM	G	M	DM	G	M
Puzzle	DM		0.815305	0.948404	0.956755	0.979996	0.042998*	0.999856	0.979996	0.818120
	G	0.815305		0.999993	1.000000	1.000000	0.581046	0.676752	1.000000	0.999999
	M	0.948404	0.999993		1.000000	1.000000	0.392091	0.841962	1.000000	0.999723
Strategy	DM	0.956755	1.000000	1.000000		1.000000	0.246087	0.737052	1.000000	0.999987
	G	0.979996	1.000000	1.000000	1.000000		0.172815	0.818120	1.000000	0.999856
	M	0.042998*	0.581046	0.392091	0.246087	0.172815		0.008976*	0.422052	0.765501
Duelling	DM	0.999856	0.676752	0.841962	0.737052	0.818120	0.008976*		0.599703	0.219657
	G	0.979996	1.000000	1.000000	1.000000	1.000000	0.422052	0.599703		0.999329
	M	0.818120	0.999999	0.999723	0.999987	0.999856	0.765501	0.219657	0.999329	

Error: Between; Within; Pooled MSE = 14.722, df = 88.291

DM - Direct manipulation, G - Gestural, M - Mixed

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).



Transformation of Time: Cell mean plot

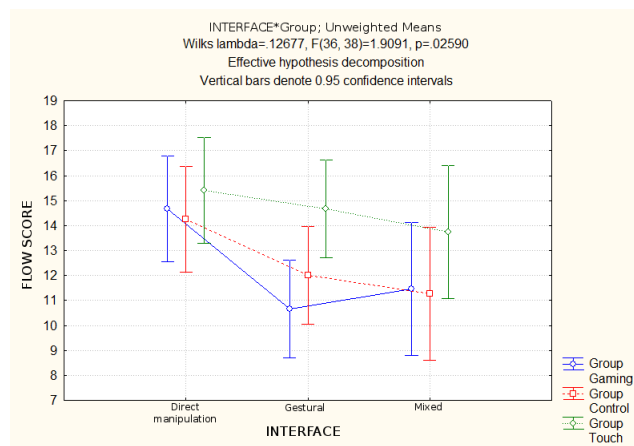
Autotelic Experience

Genre		Puzzle			Strategy			Duelling		
		DM	G	M	DM	G	M	DM	G	M
Puzzle	DM		0.030455*	0.165052	0.999999	0.750029	0.444092	0.999940	1.000000	0.999632
	G	0.030455*		0.998988	0.365053	0.995031	0.999987	0.081621	0.231134	0.585550
	M	0.165052	0.998988		0.697668	0.999995	1.000000	0.250866	0.528253	0.879701
Strategy	DM	0.999999	0.365053	0.697668		0.605801	0.233392	0.998434	0.999999	0.999995
	G	0.750029	0.995031	0.999995	0.605801		0.999479	0.444092	0.750029	0.972756
	M	0.444092	0.999987	1.000000	0.233392	0.999479		0.194865	0.444092	0.820797
Duelling	DM	0.999940	0.081621	0.250866	0.998434	0.444092	0.194865		0.999479	0.891110
	G	1.000000	0.231134	0.528253	0.999999	0.750029	0.444092	0.999479		0.996918
	M	0.999632	0.585550	0.879701	0.999995	0.972756	0.820797	0.891110	0.996918	

Error: Between; Within; Pooled MSE = 18.648, df = 78.186

DM - Direct manipulation, G - Gestural, M - Mixed

* denotes a significant result at a 95% confidence level ($\alpha = 0.05$).



Autotelic Experience: Cell mean plot